Citation Paraphrase Verification Log

Thesis: *Abstraction Mechanisms for Virtual Avatars*
Nidanur Günay • University of Konstanz • 2026

Every citation in the thesis has an entry here, in reading order, Ch. 1 → Ch. 7.
Each entry shows the **exact sentence from the thesis** and the **passage from the original source**.

Red box = wording quoted from the original source. **Orange** = not yet checked against the original.

Generated from the thesis, so nothing here quotes text that has been deleted.

MANDATORY WORKFLOW RULE — Citation Entry Protocol

This document is generated from the thesis, so the green boxes always match the current text. Regenerate it whenever a citation is added, changed, or removed. Each entry gives:

- **Where** it appears — chapter and section.
- **The thesis sentence** making the claim, quoted exactly.
- **The source passage** quoted from the original, or an orange note saying it has not been checked.
- **Where the source file is**, so you can open it and judge for yourself.

Tick **Validated** once you have read the original and agree it supports the sentence.

Do not skip this step. A citation added without a corresponding entry here is unverified and must be treated as a potential error until logged.

Chapter: Introduction

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#1 alipour2025 — Alipour et al. (2025) *Ch.1 Introduction → Motivation*

Sentence in thesis: **[Block covered by this citation]** Computer-generated (CG) humanoid characters have become an ordinary part of contemporary digital experiences, especially as generative artificial intelligence (AI) video content is widely encountered on a daily basis. **As virtual assistants, avatars and other agents are increasingly integrated into everyday domains, they interact with users instead of remaining passive tools that act only on command.**

Usage: First use of this source.

Original source: **(1)** *"virtual assistants, avatars and other agents are increasingly integrated into everyday domains"* ← [opening paragraph:] "The agents are **increasingly integrated into various domains of life**, in different forms such as conversational social agents, **avatars**, chatbots, robots, **intelligent virtual assistants**, and voice-based technologies. The influx of these agents in important domains such as **healthcare, education, and customer service** raises significant questions about human-agent interaction." **(2)** *"they interact with users instead of remaining passive tools that act only on command"* ← [§Agents:] "unlike **passive software programs or tools that only function upon user command**, agents possess a degree of autonomy and responsiveness, **enabling them to interact continuously and adaptively with users** or other systems. . . whether [in] the physical form (e.g., a robot) or **simulated (e.g., a virtual avatar)**."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Alipor_Would You Rely on an Eerie Agent?.pdf`, opening paragraph and §Agents. **SOURCE REPLACED 2026-08-24. This was a real error, caught by Nidanur.** The sentence previously read "As virtual assistants, game characters and avatars in immersive environments increasingly involve users in sustained interactions with virtual humans, they are no longer limited to passive viewing" and cited `weidner2023`. An exhaustive search of that paper found **zero** occurrences of *assistant*, **zero** of *sustained*, and three of *passive*, none of which supported the claim. One of them ran the other way: "The other studies involved more passive activities. Three of the four clauses therefore had no basis in the cited source. Two earlier excerpts had been stored for this entry, on 2026-08-21 and 2026-08-23, and neither supported the sentence either. **What changed:** the sentence now states what Alipour et al. state, and both halves are quoted above. **Consequence to note:** the same paper criticises video-based designs on ecological-validity grounds, "Nearly 80% of the studies (41 out of 53) employed indirect encounters, where participants passively viewed static images or videos". The user study of this thesis is video-based, and this critique is discussed in the Evaluation chapter.

✔ **Validated** — I have read the original and agree it supports the sentence.

#2 mcdonnell2012 — McDonnell et al. (2012) *Ch.1 Introduction → Motivation*

Sentence in the-sis: **[Block covered by this citation]** Engaging with a CG human makes its design consequential, since the character is no longer a background presence but an active participant in the exchange. That design governs aesthetic appeal as well as how users perceive and respond to the character. These effects raise several concerns that carry consequences across film, games and other domains where virtual humans appear. **One of the difficult ones is how realistically or abstractly such a character should be rendered.** *[That tension between realism and abstraction in avatars is the starting point for the work presented in this thesis. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *that this is a difficult choice, and in the same domains the thesis names* ← [§1 Introduction, opening sentence:] "**Choosing an appropriate rendering style for virtual humans in movies, games, and other domains can be a challenging task**." **(2)** *why it is difficult* ← [same paragraph:] "Using **near-human characters can be considered risky**, as it is not only **costly** but can produce a negative audience reaction. This effect **may be one of the reasons why many studios use cartoon stylizations** rather than photorealistic humans to depict their characters." **(3)** *that the paper exists to answer it* ← [same section:] "we investigate some of the **consequences of choosing realistic or stylized rendering** in order to **provide guidelines for developers**".

How to check: Read directly from `Articles/01_uncanny_valley/ McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012). Render me Real? Investigating.pdf`, §1. **Overstatement corrected 2026-08-24, caught by Nidanur.** The sentence read "One of *the most difficult*", a ranking against the other design concerns that the paper never makes. McDonnell et al. write that the choice "can be a challenging task", with no comparison and with a hedge. The sentence now reads "One of the difficult ones, which is what quote (1) supports. **Quote also replaced in the same pass:** this entry previously rested on the "mixed reactions from audiences" passage, and before that on the render-style count and the ANOVA degrees of freedom. Quote (1) is the paper's own opening sentence and states the claim directly, in the same three domains the thesis names. **Proximity note:** the preceding thesis sentence ends "across film, games and other domains" against the source's "in movies, games, and other domains. Four consecutive words in common, below the six-word threshold used in this document, and *film* replaces *movies*. Recorded rather than changed.

✔ **Validated** — I have read the original and agree it supports the sentence.

#3 natarajan2020 — Natarajan & Gombolay (2020) *Ch.1 Introduction → Motivation*

Sentence in the-sis: **[Block covered by this citation]** This design choice can be represented along a fidelity axis. Higher fidelity corresponds to greater human likeness on that axis, meaning the degree to which an avatar visually resembles a real human. Anthropomorphism describes the viewer's side of that axis. **Furthermore, an agent's perceived anthropomorphism and its behaviour have been demonstrated to increase users' trust in it.**

Usage: First use of this source.

Original source: **(1)** *anthropomorphism to trust* ← [§4.1:] "Results from repeated measures ANOVA show **high statistical significance for perceived anthropomorphism (**$p < 0.001$**,** $F(1, 720) = 17.6276$**)**. Further, we obtained a **positive coefficient (0.29335)** for anthropomorphism from the above model, indicating that **trust and anthropomorphism are positively correlated**." **(2)** *behaviour to trust* ← [§4.2:] "we find **behavior to be a statistically significant factor (**$p < 0.001$**,** $F(4, 670) = 73.5384$**)**."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Natarajan-Effects of Anthropomorphism and Accountability.pdf`, §§4.1–4.4. **TWO ERRORS FOUND AND CORRECTED 2026-08-25, both in one sentence. (a) "Embodied" asserted what the paper disproved.** In Natarajan and Gombolay, *embodied* means a physically co-present robot, and that was their H3, which failed. [§4.3:] "Results from rANOVA **did not show robot presence to be a significant factor**... Robot presence did not show significance for perceived anthropomorphism as well... **we do not reject the null hypothesis for H3**." The null is well powered rather than merely inconclusive: they report "large effect size and $\beta = 0.99$. Writing "an embodied agent's perceived anthropomorphism" therefore attached the finding to the one variable the study showed did not matter. The word is deleted. **(b) "Willingness to follow it" is not their result.** Their compliance measure is *inappropriate* compliance, the rate of accepting bad advice, which is an error rate rather than endorsement. [§4.4:] "the previous behavior transpired to be significant **irrespective of the current behavior (not significant)** for inappropriate compliance analysis." No effect of anthropomorphism on compliance is reported anywhere in the paper. The abstract does claim "trust and compliance" jointly, but the analysis supports only trust, and §5.1 of the writing rules requires the citation to support the claim it is attached to. The clause is deleted. **Thread kept:** the following sentence read "this increase in confidence" and now reads "this increase in trust", so the pickup still works.

✔ **Validated** — I have read the original and agree it supports the sentence.

#4 mori1970 — Mori (1970) *Ch.1 Introduction → Motivation*

Sentence in thesis: **[Block covered by this citation]** However, this increase in trust does not scale linearly with likeness across the fidelity axis. Mori termed this effect the uncanny valley. **Comfort rises with human likeness only up to a point, then drops sharply for entities that appear almost but not quite human, before recovering near-perfect likeness.**

Usage: First use of this source.

Original source: **(1)** *the rise, the drop and the name* ← ["The valley of familiarity":] "I have noticed that, **as robots appear more humanlike, our sense of their familiarity increases until we come to a valley**. I call this relation the '**uncanny valley**'." **(2)** *"entities that appear almost but not quite human"* ← Mori's own examples inside the valley are a **prosthetic hand**, a **corpse** and a **zombie**, each of which resembles a human closely at first glance and unsettles once the discrepancy registers. **(3)** *"before recovering near perfect likeness"* ← Figure 1 plots familiarity against human likeness on an axis marked **50%** and **100%**; the curve climbs out of the valley to its highest point at the **healthy person**, at the far right of the axis.

How to check: Read directly from `Articles/01_uncanny_valley/uncanny1970.pdf`. **Correction to an earlier note, 2026-08-24:** a previous version of this record stated that Mori's figure carries no numeric axis. It does: the human-likeness axis is marked at 50% and 100%. What the figure does not give is a percentage for the bottom of the valley, so no specific figure should be read off it.

✔ **Validated** — I have read the original and agree it supports the sentence.

#5 mcdonnell2012 — McDonnell et al. (2012) *Ch.1 Introduction → Motivation*

Sentence in thesis: **[Block covered by this citation]** Consequently, an avatar that approaches full realism may reduce users' confidence, not raise it. **The same dip appears for CG characters generally.**

Usage: Use 2 of this source.

Original source: **(1)** *the dip* ← [Abstract:] "We found that **highly realistic and highly abstract styles were considered appealing**, whether still or moving. However, **render styles that were in the middle of the range between abstract and realistic were considered unappealing**, and the most unappealing character was considered even more so when motion was applied." **(2)** *that this is the uncanny valley pattern* ← [§7 Discussion:] "These results are very much **in line with the Uncanny Valley theory**... The **drop in appeal for our stimuli came from characters that were rated as unfamiliar or those that were in the middle of our 'abstract to realistic' scale**."

How to check: Read directly from `Articles/01_uncanny_valley/ McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012). Render me Real? Investigating.pdf`, Abstract and §7. **Source replaced 2026-08-24**, for the same reason as the entry above. **Deviation the authors themselves flag:** the sentence immediately after quote (2) reads "**Contrary to the theory**, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters." So the dip reproduces, but the recovery reaches *parity* with the stylised end rather than exceeding it, which Mori's curve predicts. **Overstatement corrected 2026-08-24.** The sentence read "The same *pattern* holds", which pointed back at the three-part curve described two sentences earlier: rise, dip, then recovery to a higher peak. Only the first two parts transfer, and the authors reject the third in the words quoted above. The sentence now reads "The same *dip* appears", which is precisely what quotes (1) and (2) support. **The deviation is not lost:** Related Work carries it, where there is room for it. "The authors conclude that styles at both ends of that range are viable. Interestingly, cartoon styles were rated friendlier than realistic ones." That result favours the argument of this thesis, since a stylised character costing nothing on appeal relative to the most realistic one is the premise the NPR treatment rests on. **Standing correction, 2026-08-17:** earlier text said "eleven distinct styles and "video stimuli. Eleven were created but **ten** were tested, and both stills and video were rated, separately.

✔ **Validated** — I have read the original and agree it supports the sentence.

#6 canales2024 — Canales et al. (2024) *Ch.1 Introduction → Motivation*

Sentence in the-sis: **[Block covered by this citation] Avatars fall within this category and their stylisation can affect users' behavioural choice.** *[Together, these results suggest that rendering style influences not only users' subjective confidence but also the decisions they make. Alternative visual styles could protect confidence and choice alike by avoiding the perceptual pitfalls that highly realistic avatars carry. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"their stylisation can affect users' behavioural choice"* ← [Abstract:] "**Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the "Mid" level of stylization for a second opinion**." **(2)** *what the behavioural measure was* ← [§1:] participants "**select a doctor for a second opinion**" after interviewing three avatars differing only in stylisation level.

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Avatar_-Stylization_and_Trust_IEEE_VR_2024.pdf`, Abstract and §1. **Source replaced 2026-08-24.** The entry previously quoted the passage on within-subjects design, which describes the method and says nothing about behavioural choice. The abstract sentence above is the actual result, and it is the dissociation the thesis is pointing at: self-reported trust flat across styles, behavioural selection not flat.

✔ **Validated** — I have read the original and agree it supports the sentence.

#7 rahman2024 — Rahman et al. (2024) *Ch.1 Introduction → Motivation*

Sentence in thesis: **[Block covered by this citation] That question is becoming increasingly relevant as virtual avatars move beyond research settings into widely used consumer applications such as gaming and marketing.** *[Thus, determining the balance between realism and abstraction is the defining challenge in the design of trustworthy virtual avatars. How this tension is resolved will shape whether virtual avatars can transition from novelty to genuinely trusted participants in social and professional contexts. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *that avatars have moved into named consumer industries, in the review's own voice* ← [Introduction, p. 2:] "Consequently, the usage of avatars is not limited to a particular industry, like banking or hotel; it is seen as a huge prospect in the **virtual gaming, smartphone, and fashion industries**. For instance, Samsung has launched an AI-supported virtual avatar called Neon. . . the famous fashion brand Calvin Klein has deployed Lil Miquela, a virtual influencer." **(2)** *that marketing is the review's own subject* ← [p. 1:] "A systematic literature review was applied to rigorously review 57 published articles on **avatars and ABM** (avatar-based marketing) from 2005 to 2023."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Navigating_-Avatar-Based_Marketing.pdf`, Introduction and Abstract. **"Advertising" replaced by "marketing", 2026-08-25.** The word *advertising* occurs eleven times in the review and never as an application domain it asserts. Its occurrences are: a *moderating variable* in one coded study (Keng and Liu 2013, online advertising complexity), a pooled "Others" outcome bucket covering 36 studies ("acceptance, advertising attitudes, behavioral intentions, brand attitudes), and reference titles. The review's own industry coding, Figure 6, has no advertising category at all. *Marketing* is safe because it is the review's entire subject. **An audit finding was partly overturned here.** A sweep had reported that "gaming" was also unsupported, being restated from five credited prior works. That is a different sentence. Two separate sentences exist: one credits five sources for avatars in online services generally, and the one quoted above states the industry claim in the review's own voice with its own examples. Gaming stands.

✔ **Validated** — I have read the original and agree it supports the sentence.

Chapter: Background

#8 unityurp — Unity Technologies (2026) *Ch.2 Background → Rendering Concepts*

Sentence in thesis: **[Block covered by this citation]** A render pass is one complete traversal of scene geometry. Its result is written into a render target, an off-screen image buffer. A fragment shader that operates within a single pass sees only the surface point it is shading. Any operator needing a spatial neighbourhood must gather that neighbourhood itself. Additional passes remove this restriction. A post-processing pass renders one full-screen quad and samples the render target left by the preceding pass, which gives every fragment access to the whole rendered image. **Unity's Universal Render Pipeline (URP) schedules such injected work through a renderer feature.** *[However, every extra pass costs another traversal of the geometry or of the full screen. Thus, the number of passes a platform permits is an engineering constraint, not a free parameter. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: "Use the `ScriptableRenderFeature` API to insert a Scriptable Render Pass into the Universal Render Pipeline (URP) frame rendering loop." The page further specifies overriding `Create` to instantiate the pass and `AddRenderPasses` to enqueue it: "Use the `EnqueuePass` API to inject the Scriptable Render Pass into the frame rendering loop."

How to check: Read directly from Unity 6 URP manual page — fetched and read. Confirm this passage supports the sentence above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#9 lewis2014 — Lewis et al. (2014) *Ch.2 Background → Rendering Concepts*

Sentence in thesis: **[Block covered by this citation]** Yet the surface carrying these maps does not stay stable at runtime, since an animated character never holds a fixed pose. That motion is produced by the rig, the control structure bound to the mesh before any animation is played. A rig displaces vertex positions by two mechanisms, skeletal deformation and blendshapes. Blendshapes, a linear combination of stored expression shapes, handle the finer motion that a skeleton cannot reach. **They hold alternative vertex positions that are interpolated against the base mesh.** *[A platform whose rig omits blendshapes therefore cannot carry facial expression at all, whatever its rendering quality. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"a linear combination of stored expression shapes"* ← [§2:] "A blendshape model **generates a facial pose as a linear combination of a number of facial expressions, the blendshape "targets"**. By **varying the weights of the linear combination**, a range of facial expressions can be expressed with little computation." **(2)** *"They hold alternative vertex positions"* ← [§3.1:] the targets b_k, $k = 1 \dots N$, are complete face shapes, each a full set of vertex positions, alongside the neutral shape b_0. **(3)** *"that are interpolated against the base mesh"* ← [§3.1 Delta blendshape formulation:] "one face model b_0 (typically the resting face expression) is designated as the **"neutral" face shape**, and the remaining targets b_k are **replaced with the difference** $b_k - b_0$ between the kth face target and the neutral face", giving $f = b_0 + \sum_{k=1}^{n} w_k (b_k - b_0)$. The weighted differences are added to the neutral shape, which is the base mesh the thesis refers to. §3.2 describes the result as "piecewise linear interpolation".

How to check: Read directly from `Articles/04_graphics_foundations/Lewis - 2014-EG-blendshape_STAR.pdf`, §2 and §3.1–3.2. **REPLACED 2026-08-24, found by Nidanur.** The entry previously stored the paper's opening abstract sentence, "*Blendshapes*, a simple linear model of facial expression, is the prevalent approach to realistic facial animation". That defines what blendshapes are but says nothing about vertex positions or interpolation, which is what the cited sentence claims, so the sentence stood unsupported. The three passages above are from the body of the paper and carry each clause. **Lift avoided in the same pass:** that abstract phrase was about to be moved into the thesis as an appositive. It would have been reproduced word for word. The sentence now reads "a linear combination of stored expression shapes", which is what §2 describes and shares only the standard term *linear combination* with the source.

✔ **Validated** — I have read the original and agree it supports the sentence.

#10 nowak2005 — Nowak & Rauh (2005) *Ch.2 Background → Avatars, Trust and Eeriness*

Sentence in thesis: **[Block covered by this citation]** An avatar is a digital body that represents a person within a virtual space. The term spans forms ranging from abstract geometric markers to fully articulated humanoid figures. **Its visual appearance becomes a central cue from which viewers form an impression of the person that the avatar represents.**

Usage: First use of this source.

Original source: **(1)** *"viewers form an impression of the person that the avatar represents"* ← [§Using Avatars to Reduce Uncertainty About the People They Represent, printed p. 155:] "Evidence suggests that people in mediated interactions **use avatars, or whatever information the interface provides, in the person perception process** and that different types of avatars have different effects on person perception." **(2)** *"Its visual appearance becomes a central cue"* ← [same section:] "Because avatars are a visible representation of a person in an interface, **evaluations based on the physical appearance of the avatar may be transferred to them**." **(3)** *context for why appearance carries this load* ← [same page:] "When people interact through computer-mediated channels, **physical information about the other person may be unavailable**."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Nowak Rauh 2005.pdf`, printed p. 155. **REPLACED 2026-08-24, found by Nidanur.** The entry previously stored four passages that all report *what the study found*: the participant count ($N = 255$, p. 158), the anthropomorphism-credibility correlation from Table 3, and two results sentences from the Discussion and Conclusion. The cited sentence makes no empirical claim at all. It states a background premise, which the paper argues in its introduction, so nothing in the stored excerpt could have supported it. **Thesis sentence shortened in the same pass:** it previously ended "particularly where other physical information is unavailable". That clause was removed. Quote (3) shows the source does state it, so the removal is a matter of preference rather than accuracy, and the shorter sentence remains fully carried by quotes (1) and (2).

✔ **Validated** — I have read the original and agree it supports the sentence.

#11 mayer1995 — Mayer et al. (1995) *Ch.2 Background → Avatars, Trust and Eeriness*

Sentence in thesis: **[Block covered by this citation]** Consequently, the central social judgement an observer makes about an avatar is one of trust. **Mayer et al. defined trust as a willingness to be vulnerable to another party's actions, held on the expectation that the party will act as needed and without any means of monitoring or controlling them.**

Usage: First use of this source.

Original source: **(1)** *the definition* ← [§Definition of trust, printed p. 712:] trust is "the **willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party**."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Mayer-IntegrativeModelOrganizational-1995.pdf`, printed p. 712. **Added 2026-08-24.** The paragraph previously closed on "trust: the degree to which they are willing to rely on the person it portrays", an unattributed paraphrase of the definition above. Trust is the central construct of this thesis and was nowhere defined from a source. **Why this definition earns its place here:** the final clause, *irrespective of the ability to monitor or control that other party*, describes the viewing condition of the user study exactly. A participant watching a recorded advisor has no means of monitoring or controlling that advisor, which is the situation Mayer et al. specify. **Wording note:** "willingness to be vulnerable" is retained because it is the canonical shorthand for this definition throughout the trust literature. The remainder is paraphrased, and the longest run shared with the source is five words.

✔ **Validated** — I have read the original and agree it supports the sentence.

#12 mayer1995 — Mayer et al. (1995) *Ch.2 Background → Avatars, Trust and Eeriness*

Sentence in thesis: **[Block covered by this citation]** This judgement is not a single quantity. Perceived trustworthiness decomposes into three components: ability, benevolence and integrity. Ability denotes the skills and characteristics that let a trustee act effectively within a relevant domain, benevolence the extent to which the trustee appears to care about the observer's wellbeing and integrity adherence to acceptable principles. **The three are separable rather than unrelated, and each can vary while the others hold.**

Usage: Use 2 of this source.

Original source: **(1)** *the three factors* ← [§The Factors of Trustworthiness, printed p. 717:] "three characteristics of a trustee appear often in the literature: **ability, benevolence, and integrity**. As a set, these three appear to explain a major portion of trustworthiness." **(2)** *ability* ← [p. 717:] "Ability is that group of **skills, competencies, and characteristics that enable a party to have influence within some specific domain**." **(3)** *benevolence* ← [p. 718:] "Benevolence is **the extent to which a trustee is believed to want to do good to the trustor**, aside from an egocentric profit motive." **(4)** *integrity* ← [p. 719:] "perception that the trustee **adheres to a set of principles that the trustor finds acceptable**." **(5)** *"separable rather than unrelated, and each can vary while the others hold"* ← [§Interrelationship of the Three Factors, printed p. 720:] "Ability, benevolence, and integrity are important to trust, and **each may vary independently of the others**. This statement **does not imply that the three are unrelated** to one another, but only that **they are separable**."

How to check: Read directly from `Articles/02_trust_and_avatar_-perception/Mayer-IntegrativeModelOrganizational-1995.pdf`, printed pp. 717–720. **Passage added and sentence rewritten 2026-08-24.** The paragraph previously asserted three *"independently evaluable* components with no passage behind that word, and the entry's other two stored quotes, on risk-taking and on propensity, supported nothing in the block. Quote (5) is the passage that licenses the claim, and it draws a distinction worth preserving: Mayer et al. say the factors are *separable*, and say explicitly that this does *not* make them unrelated. The thesis sentence now states both halves. **Terminology note:** the definition of ability was reworded in the same pass. It previously read "denotes competence within a relevant domain", which used McKnight and Chervany's label for this construct two sentences before their typology is introduced. It now follows quote (2).

✔ **Validated** — I have read the original and agree it supports the sentence.

#13 mcknight2001 — McKnight & Chervany (2001) *Ch.2 Background → Avatars, Trust and Eeriness*

Sentence in the-sis: **[Block covered by this citation] McKnight and Chervany added predictability to that set as a fourth trusting belief.** *[Together, these four components describe what an observer believes about the trustee and they leave the perceptual response to appearance unaddressed. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"add predictability. . . as a fourth trusting belief"* ← [manuscript p. 8:] "We **included Predictability as an economics-based subconstruct. This differs from Mayer et al.** [45], **who excluded predictability from their trust typology**. Thus, **four second-order categories (competence, benevolence, integrity, and predictability)** cover 91.8% of the characteristics-based trust definitions found." **(2)** *that these are trusting beliefs* ← [manuscript p. 11:] "**Trusting Beliefs includes Competence, Benevolence, Integrity, and Predictability beliefs**, corresponding to the first four rows of Table 2."

How to check: Read from `Articles/02_trust_and_avatar_perception/What Trust Means in E-Commerce Customer Relationships.pdf`, manuscript pp. 8 and 11. **Extraction note:** this PDF is a 34-page image scan with no text layer, so p. 8 was rendered at 170 dpi and read directly. Its page numbers are the authors' own manuscript numbering, not the published *Int. J. Electronic Commerce* 6(2) pagination of 35–59 given in `references.bib`. **Passage upgraded 2026-08-25.** The entry previously held only the p. 11 list, which establishes that predictability is one of four beliefs but does not carry the word *add*. Quote (1) does: it names Mayer et al. explicitly and states that they excluded predictability, which is exactly the relationship the thesis sentence asserts. **Terminology, recorded for the decision taken on 2026-08-25:** McKnight and Chervany label their third belief *Competence*, where Mayer et al. use *Ability*. The thesis now uses *ability* throughout and names his term once, in the sentence that introduces his typology, so that a reader who opens this paper is not left reconciling two labels.

✔ **Validated** — I have read the original and agree it supports the sentence.

#14 bartneck2009 — Bartneck et al. (2009) *Ch.2 Background → Avatars, Trust and Eeriness*

Sentence in thesis: **[Block covered by this citation]** A complementary family of instruments targets that perceptual dimension directly. The Godspeed Questionnaire Series measures five perceptual qualities of an artificial agent: anthropomorphism, animacy, likeability, perceived intelligence and perceived safety. Anthropomorphism is the degree of human likeness an observer perceives in the agent, and its anchors run from machinelike to humanlike and from artificial to lifelike. Animacy denotes whether the agent appears alive rather than inert, and its items range from dead to alive and from apathetic to responsive. Likeability is the immediate affective impression the agent makes, anchored between dislike and like and between unfriendly and friendly. **Accordingly, these five subscales hold 24 semantic differential items in total, each rated on a five-point scale.**

Usage: First use of this source.

Original source: **(1)** *the five qualities and the 24-item total* ← [Abstract:] "A literature review has been performed on the measurements of five key concepts in HRI: **anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety**." Appendix, printed p. 79, gives 5 + 6 + 5 + 5 + 3 = **24** items, each on a **1–5** scale between two bipolar anchors. **(2)** *the anthropomorphism anchors* ← [Appendix, GODSPEED I: ANTHROPOMORPHISM, p. 79:] Fake–Natural, **Machinelike–Humanlike**, Unconscious–Conscious, **Artificial–Lifelike**, Moving rigidly–Moving elegantly. **(3)** *the animacy anchors* ← [Appendix, GODSPEED II: ANIMACY, p. 79:] **Dead–Alive**, Stagnant–Lively, Mechanical–Organic, Artificial–Lifelike, Inert–Interactive, **Apathetic–Responsive**. The "alive rather than inert" framing is the paper's own: "Asking about the perceived **animacy** of a certain stimulus makes sense only if there is a **possibility for it to be alive**." **(4)** *the likeability anchors* ← [Appendix, GODSPEED III: LIKEABILITY, p. 79:] **Dislike–Like**, **Unfriendly–Friendly**, Unkind–Kind, Unpleasant–Pleasant, Awful–Nice. **(5)** *"the immediate affective impression the agent makes"* ← every subscale is headed "**Please rate your impression of the robot on these scales**", and the body text frames the construct the same way: "**positive first impressions (e.g., likeability)** of a person often lead to more positive evaluations of that person".

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Bartneck_2009_-Measurement Instruments for the Anthropomorphism.pdf`. The Appendix page carries no text layer in this PDF and was read by rendering page 9 at 200 dpi. **Definitions added 2026-08-25, at Nidanur's request.** Ch. 2 defined trust and eeriness but only *named* likeability and anthropomorphism, so both arrived undefined at the point of use. Both are now defined here, at first use, with their real anchors from the Appendix. **Duplication removed in the same pass:** anthropomorphism had been defined twice and in neither of the two places was it Ch. 2. The Related Work appositive "the degree of human likeness perceived in an avatar" is deleted outright, and the Introduction keeps only its object-side versus viewer-side contrast, with the definitional tail "namely how human-like the avatar appears to be" removed. The construct now follows the same one-definition pattern as trust and eeriness. **Scope note, deliberate:** three of the five are defined, anthropomorphism, animacy and likeability. Perceived intelligence and perceived safety are named in the list and left undefined, since their names carry them and neither is used again in the thesis. **Origin check (§5.3a):** the four adjective pairs nice/awful, friendly/unfriendly, kind/unkind and pleasant/unpleasant are credited by Bartneck et al. to Monahan [41]. The thesis does not use Monahan's finding, only the composition of the Godspeed subscale as printed in this paper's own Appendix, so `bartneck2009` is the correct and sufficient

✔ **Validated** — I have read the original and agree it supports the sentence.

#15 ho2010 — Ho & MacDorman (2010) *Ch.2 Background → Avatars, Trust and Eeriness*

Sentence in the-sis: **[Block covered by this citation]** Ho and MacDorman found every subscale except perceived safety too strongly intercorrelated to measure distinct constructs, and none of them corresponded specifically to eeriness. Their own instrument addresses that gap with eight items for eeriness alongside six for perceived humanness. Eeriness, as they frame it, is the unsettled perceptual response that arises when a stimulus approaches but does not reach the peak of human likeness. **Yet the two indices proved orthogonal in validation, with $r \approx .02$.** *[This indicates that high human likeness and strong discomfort can co-occur in the same stimulus. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"every subscale except perceived safety too strongly intercorrelated"* ← [§2.2 Results, printed p. 1511:] "Another serious problem was the significant and extremely high correlation between **anthropomorphism, likeability, animacy, and perceived intelligence**. The correlations ranged from .67. . . to .89. . . This suggests that those concepts **had no discriminant validity**." **(2)** *"no subscale corresponding specifically to eeriness"* ← [§1, printed p. 1509:] "there is **no index corresponding specifically to eeriness**, a dimension that is arguably distinct from likeability but nevertheless important in determining whether a human-looking character has fallen into the uncanny valley." **(3)** *"eight items for eeriness alongside six for perceived humanness"* ← [§3.3.5, printed p. 1514:] "five items were constructed for the final version of the attractiveness index, **eight items were constructed for the eeriness index, and six items were constructed for the humanness index**." **(4)** *the framing of eeriness* ← [§1, printed p. 1508:] "The graph predicts that **more human-looking characters will be perceived as more agreeable up to a point at which they become so human people find their nonhuman imperfections unsettling**." **(5)** *"proved orthogonal in validation, with $r \approx .02$"* ← [Abstract:] the humanness and eeriness indices correlate at $r = .02$, that is, they are effectively independent.

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Ho2010UncannyValleyIndices.pdf`, Abstract, §1, §2.2 and §3.3.5. **Entry rebuilt 2026-08-25 to cover the whole block.** A single citation now closes four consecutive Ho and MacDorman claims, the critique of Godspeed, the composition of their own instrument, the framing of eeriness and the validation correlation, each with its own passage above. The sentence that follows the citation, that high human-likeness and strong discomfort can co-occur in one stimulus, is the thesis's inference from quote (5) and correctly carries no citation. **Wording corrected in the same pass:** the thesis read "Eeriness, as they *operationalise* it". Their operationalisation is the eight-item index of quote (3). What the sentence actually describes is their conceptual framing, quote (4), which they inherit from Mori. It now reads "as they *frame* it. **Also corrected:** the two Ho and MacDorman scales are called *indices* in the source, not subscales, which keeps them distinct from Godspeed's subscales in the preceding sentences.

✔ **Validated** — I have read the original and agree it supports the sentence.

#16 gonzalez2018 — Gonzalez & Woods (2018) *Ch.2 Background → Edge Detection*

Sentence in the-sis: **[Block covered by this citation]** A digital image consists of a grid of pixels, each storing an intensity value. **An edge is conventionally defined as a location at which this intensity changes rapidly.** *[Detecting such locations involves estimating the image gradient and applying a threshold to identify pixels whose gradient magnitude is large. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"A digital image consists of a grid of pixels, each storing an intensity value"* ← [§1.1:] "a digital image is composed of a finite number of elements, each of which has a **particular location and value**. These elements are called **picture elements, image elements, pels, and pixels**." **(2)** *"An edge is conventionally defined as a location at which this intensity changes rapidly"* ← [§10.2 Point, Line, and Edge Detection:] "**Edge pixels are pixels at which the intensity of an image changes abruptly**, and edges (or edge segments) are sets of connected edge pixels. **Edge detectors are local image processing tools designed to detect edge pixels**."

How to check: Read directly from `Articles/05_image_processing/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`, §1.1 and §10.2 (printed p. 702). **Source corrected 2026-08-24.** The entry previously carried only the §1.1 passage on what pixels are called. That supports the *preceding* sentence of the block, not the cited one, so the definition of an edge stood unsupported. Quote (2) is the book's own definition and is now attached. **Wording note:** Gonzalez and Woods write *abruptly* where the thesis writes *rapidly*. The terms are interchangeable here, and *rapidly* avoids reproducing the source phrase verbatim.

✔ **Validated** — I have read the original and agree it supports the sentence.

#17 gonzalez2018 — Gonzalez & Woods (2018) *Ch.2 Background → Edge Detection → Gradient-Based Detection*

Sentence in the-sis: **[Block covered by this citation]** The Sobel operator is a widely used kernel of this kind. **It estimates the two directional derivatives with a pair of 3×3 kernels:**

Usage: Use 2 of this source.

Original source: **(1)** *the Sobel kernel pair* ← [§10.2 Gradient Operators, printed p. 719:] "A slight variation of the preceding two equations uses a **weight of 2 in the center coefficient**. . . Figures 10.14(f) and (g) show the kernels used to implement Eqs. (10-24) and (10-25). **These kernels are called the Sobel operators** (Sobel [1970])." **(2)** *that the pair yields the two directional derivatives* ← [printed p. 720:] "Any of the pairs of kernels from Fig. 10.14 are convolved with an image to obtain the **gradient components g_x and g_y at every pixel location**."

How to check: Read directly from `Articles/05_image_processing/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`, §10.2 and Fig. 10.14. **REPLACED 2026-08-25.** The entry previously quoted "The simplest digital approximations to the partial derivatives using kernels of size 3×3 are given by. That sentence introduces Eqs. (10-22) and (10-23), and the book names the resulting kernels two paragraphs later: "these kernels are called the **Prewitt** operators (Prewitt [1970])" on printed p. 718. The Sobel kernels appear on the following page. The excerpt was therefore a genuine passage about the *wrong operator*, filed against a Sobel claim. **Orientation note, recorded so it is not "corrected later:** the thesis's G_x and G_y are transposed relative to Fig. 10.14(f),(g), and G_y carries the opposite sign. Gonzalez and Woods place the origin at the top left with x running down the rows, whereas §2.1 of this thesis declares that x increases to the right and y upwards, matching the shader coordinate system. Under that convention the thesis kernels are correct, and the sentence that follows them, that G_x responds to horizontal intensity transitions and therefore detects vertically oriented edges, holds. Both forms are the Sobel operator.

✔ **Validated** — I have read the original and agree it supports the sentence.

#18 gonzalez2018 — Gonzalez & Woods (2018) *Ch.2 Background → Edge Detection → Gradient-Based Detection*

Sentence in the-sis: **[Block covered by this citation] Each Sobel kernel carries weights of ± 2 on the two coefficients adjacent to its centre.** *[Notably, this doubling averages the samples nearest the centre before the difference is taken. The Sobel operator consequently smooths slightly as it differentiates and its robustness to noise improves. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *"Each Sobel kernel carries weights of ± 2 on the two coefficients adjacent to its centre"* ← [Fig. 10.14(f) and (g), printed p. 719, which are the operative evidence:] the Sobel pair reads $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$. In both, the two ± 2 entries sit orthogonally adjacent to the kernel centre, and the centre itself is **zero**. **(2)** *why the doubling is there* ← [printed p. 719:] "It can be demonstrated that **using a 2 in the center location provides image smoothing**", and "The fact that the **Sobel kernels have better noise-suppression (smoothing) characteristics makes them preferable**."

How to check: Read directly from `Articles/05_image_processing/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`, §10.2 and Fig. 10.14 (printed p. 719). **Independently re-checked 2026-08-25, and the thesis is the more precise of the two.** The book's phrase in quote (3), "a 2 in the *center location*", refers to the middle term of each summed row or column in Eqs. (10-24) and (10-25), that is z_8, z_2, z_6 and z_4. It does **not** refer to the centre of the kernel, which is zero in every Sobel kernel. A reader comparing that phrase against the thesis sentence would conclude the thesis contradicts the source. It does not: "the two coefficients adjacent to its centre" is exactly what Fig. 10.14 shows, and it is unambiguous where the book's wording is not. The figure is therefore given first here, ahead of the text quote, so the entry cannot mislead. **Prewitt comparison removed from the thesis 2026-08-25.** The sentence previously continued "whereas the simpler Prewitt operator weights every non-zero coefficient equally at ± 1. Prewitt appeared exactly once in the whole thesis, in that clause, and was never introduced, so a reader met a named operator described only as simpler. The comparison was also unnecessary: the ± 2 entries are double the ± 1 entries in the same kernel, which the reader can see in the equation two lines above, so "this doubling in the following sentence needs no external foil. The citation now covers a single claim that Fig. 10.14 shows directly. **Orientation:** as recorded on the preceding entry, the thesis kernels are transposed relative to the figure and G_y is negated, because §2.1 declares x rightwards and y upwards. The ± 2 entries remain adjacent to the centre under either convention, so this claim is unaffected.

✔ **Validated** — I have read the original and agree it supports the sentence.

#19 roberts1963 — Roberts (1963) *Ch.2 Background → Edge Detection → Gradient-Based Detection*

Sentence in the-sis: **[Block covered by this citation] A smaller alternative is the Roberts Cross operator, which differentiates along the two diagonals of a 2×2 neighbourhood:**

Usage: First use of this source.

Original source: **(1)** *the operator as originally defined* ← [printed p. 26:] "Therefore, after a picture is read in, a differential picture is created according to the functions: $y_{i,j} = \sqrt{x_{i,j}}$, $z_{i,j} = \sqrt{(y_{i,j} - y_{i+1,j+1})^2 + (y_{i+1,j} - y_{i,j+1})^2}$ where $x_{i,j}$ is the initial intensity value, $z_{i,j}$ is the computed derivative value, and i and j are the coordinates in the two dimensions." **(2)** *"differentiates along the two diagonals of a 2×2 neighbourhood"* ← the two squared terms above are precisely the two diagonal differences across the cell $\{(i,j),(i+1,j),(i,j+1),(i+1,j+1)\}$: $y_{i,j} - y_{i+1,j+1}$ on one diagonal and $y_{i+1,j} - y_{i,j+1}$ on the other. That is the cross form the thesis writes as R_1 and R_2. **(3)** *why a symmetric pair* ← [same page:] "**Background noise seems to be reduced by using operators symmetric in x and y.**"

How to check: Read from `Articles/Machine perception of three-dimensional solids.pdf`, printed p. 26. **Extraction note:** this PDF is an 82-page image scan with no text layer, so the page was rendered at 200 dpi and read directly rather than extracted as text. **Two differences from the original, both deliberate and both handled elsewhere in the thesis. (a)** Roberts applies a square-root intensity transform before differentiating, $y = \sqrt{x}$, which he justifies on the same page: "the intensity values of the picture can be subjected to a gamma change so as to make intensity differences proportional to a human's ability to perceive them. . . the square root of the intensities should be used". The thesis kernels operate on I directly, without that transform. **(b)** Roberts combines the two diagonal differences with the Euclidean form; the absolute-value approximation in the following thesis sentence is correctly credited to Gonzalez and Woods rather than to Roberts. **Naming:** Roberts never uses the term *Roberts Cross*. It comes from later literature, and Gonzalez and Woods use "Roberts cross-gradient operators", so the thesis usage is standard.

✔ **Validated** — I have read the original and agree it supports the sentence.

#20 gonzalez2018 — Gonzalez & Woods (2018) *Ch.2 Background → Edge Detection → Gradient-Based Detection*

Sentence in thesis: **[Block covered by this citation] Its edge magnitude is commonly approximated as $|R_1 * I| + |R_2 * I|$, which avoids the square root of .** *[Furthermore, the operator has no centre sample and reads four pixels instead of nine. Its response is therefore more local and cheaper than that of Sobel, at the cost of greater sensitivity to noise. — follows the citation, uncited.]*

Usage: Use 4 of this source.

Original source: **(1)** *"commonly approximated as $|R_1 * I| + |R_2 * I|$, which avoids the square root"* ← [§10.2, printed p. 720, Eq. (10-26):] "the magnitude of the gradient requires the computations in Eq. (10-17). **This implementation is not always desirable because of the computational burden** [imposed] **by squares and square roots, and an approach used frequently is to approximate the magnitude of the gradient by absolute values:** $M(x, y) \approx |g_x| + |g_y|$." The book adds: "This equation is **more attractive computationally**, and it still preserves relative changes in intensity."

How to check: Read directly from `Articles/05_image_processing/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`, §10.2, Eq. (10-26), printed p. 720. **REPLACED 2026-08-24. This entry had no source at all.** Its source field read "See entry #19 for the source passage", which is the entry's own number, so it pointed at itself and a reader had nothing to check. The passage above is the one the sentence is drawn from. **Note withdrawn 2026-08-25.** An earlier version of this entry warned that Eq. (10-26) is stated only for the Sobel and Prewitt components and that applying it to Roberts Cross was a transfer needing justification. That was wrong. The sentence immediately preceding Eq. (10-26) reads "**Any** of the pairs of kernels from Fig. 10.14", and Fig. 10.14(b) and (c) are the Roberts kernels. The book states the approximation for Roberts as well, so no caveat applies.

✔ **Validated** — I have read the original and agree it supports the sentence.

#21 itu709 — International Telecommunication Union (2015) *Ch.2 Background → Edge Detection → Gradient-Based Detection*

Sentence in the-sis: **[Block covered by this citation] following the coefficients defined in ITU-R Recommendation BT.709 for high-definition video.** *[The weighting reflects the differing sensitivity of the human visual system to each primary. Differentiating this channel aligns edge strength with perceived brightness contrast as opposed to an arbitrary numerical difference between colour channels. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *the three coefficients* ← [§3 Signal format, item 3.2, "Derivation of luminance signal E'_Y", printed p. 4:] $E'_Y = \mathbf{0.2126}\, E'_R + \mathbf{0.7152}\, E'_G + \mathbf{0.0722}\, E'_B$. **(2)** *that the recommendation is for HDTV* ← [Title:] "Parameter values for the **HDTV standards** for production and international programme exchange."

How to check: Read directly from `Articles/05_image_processing/R-REC-BT.709-6-201506-I!!PDF-E.pdf`, §3 item 3.2. **Primary obtained 2026-08-25.** This entry was previously unverifiable: no copy of BT.709 existed anywhere under the Articles folder, so the coefficients rested on nothing on disk. Recommendation ITU-R BT.709-6 (June 2015) has now been filed and the values above were read from it. **Luma versus luminance, resolved in the same pass.** Item 3.1, immediately above the quoted equation, specifies "Conceptual **non-linear pre-correction** of primary signals: **0.45**. BT.709 therefore derives E'_Y from *gamma pre-corrected* signals, which makes it **luma**. The thesis applies the same coefficients to plain R, G and B, and §2.1 declares colour values to be *linear* RGB. Weighting linear primaries by these coefficients yields **relative luminance**, a different quantity carrying the same three numbers. The sentence previously called L a "luma channel", which was the non-linear reading and contradicted §2.1. It now reads *luminance channel*, which is what the shaders actually compute: in URP's linear colour space a sampled texture is already linear, and the shader comments in Chapter 5 say so, for example "`// perceptual luminance`" and "`// horizontal luminance derivative`. **The citation remains correct either way,** since the weights derive from the BT.709 primaries and applying them to linear RGB is the standard route to relative luminance. Only the name of the quantity was wrong. **One inconsistency left in the code, not the thesis:** `V3_SobelEdgeDetection.shader` uses $0.299/0.587/0.114$, which is BT.601 rather than BT.709.

✔ **Validated** — I have read the original and agree it supports the sentence.

#22 hlsl *Ch.2 Background → Edge Detection → Screen-Space Normal Derivatives*

Sentence in the-sis: **[Block covered by this citation]** The kernels above require several texture samples per pixel. The screen-space derivative offers a cheaper alternative for any value already interpolated across the surface. **It is exposed by the ddx and ddy intrinsics, which return the partial derivative of a value with respect to the screen-space x and y coordinates and are available only inside a pixel shader.** *[This derivative costs no texture sample. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"return the partial derivative of a value with respect to the screen-space x and y coordinates"* ← [`ddx`, page summary:] "Returns the **partial derivative of the specified value with respect to the screen-space x-coordinate**." [`ddy`:] "Returns the **partial derivative of the specified value with respect to the screen-space y-coordinate**." **(2)** *"available only inside a pixel shader"* ← [both pages:] "This function is **supported only in pixel shaders**." The shader-stage table on each page marks **Pixel** alone.

How to check: Read directly from `https://learn.microsoft.com/en-us/windows/win32/direct3dhlsl/dx-graphic` and the companion `ddy` page, 2026-08-25. **Citation added 2026-08-25, and deliberately narrow.** The section previously carried no citation at all. Two candidate sources were checked first and neither works: `card2002` (Card and Mitchell, in the ShaderX volume held in the Articles folder) covers silhouettes from face-normal dot products and Roberts and Sobel filtering, but never mentions `ddx` or `ddy`; and the copy of Real-Time Rendering 4th edition on disk contains only the table of contents, preface, introduction, bibliography and index, not the body. **What this reference does not cover.** The two sentences preceding the cited one state that fragments are rasterised in 2×2 quads and that the hardware returns a finite difference between adjacent fragments. Neither Microsoft page mentions quads, adjacent fragments or cost. Those sentences are therefore left uncited as standard hardware behaviour, which §5.2 of the writing rules treats the same way as basic HLSL. The claim that no additional texture sample is taken was split into its own sentence for the same reason: it sits outside what the reference states. **The cited sentence was reworded to match the source exactly**, rather than the reverse.

✔ **Validated** — I have read the original and agree it supports the sentence.

#23 marr1980 — Marr & Hildreth (1980) *Ch.2 Background → Edge Detection → Gaussian Prefiltering*

Sentence in the-sis: **[Block covered by this citation]** The operators introduced so far are sensitive to small fluctuations in their input. Differentiation amplifies these high-frequency components. Therefore, a noisy input yields an unstable gradient. **Marr and Hildreth established that smoothing must precede differentiation and they adopted the Gaussian as the unique filter that optimally balances localisation in space against localisation in frequency.** *[This smoothing need not run as a separate step, since both convolution and differentiation are linear operations. The smoothed gradient may be written as — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"smoothing must precede differentiation"* ← [§2, printed p. 189:] "one **first takes local averages of the image at various resolutions and then detects the changes in intensity** that occur at each one. To realize this idea, we need to determine (a) the nature of the **optimal smoothing filter**, and (b) how to detect intensity changes at a given scale." **(2)** *"the unique filter that optimally balances localisation in space against localisation in frequency"* ← [printed p. 191:] "these two localization requirements, the one in the spatial and the other in the frequency domain, **are conflicting**. They are, in fact, related by the **uncertainty principle**. . . There is, moreover, **only one distribution that optimizes this relation (Leipnik 1960), namely the Gaussian**." Followed on p. 192 by: "The filter G thus provides the **optimal trade-off between our conflicting requirements**."

How to check: Read directly from `Articles/MarrHildreth (2).pdf`, §2 (printed pp. 189–192). **Passage added 2026-08-25.** The sentence makes two claims and the entry previously supported only the first. The uniqueness and localisation trade-off, which is the more contestable half, had no passage on file. Quote (2) is it. **Attribution point for the thesis wording.** The thesis reads "*they identified* the Gaussian as the unique filter. Marr and Hildreth do not derive that result; they cite it, as quote (2) shows with its inline "(Leipnik 1960). What they establish is that this trade-off is the right criterion for the smoothing filter and that the Gaussian therefore provides the optimal one. "*they adopted*" would be exact.

✔ **Validated** — I have read the original and agree it supports the sentence.

#24 canny1986 — Canny (1986) *Ch.2 Background → Edge Detection → Gaussian Prefiltering*

Sentence in the-sis: **[Block covered by this citation]** where G_σ is a Gaussian kernel with standard deviation σ. Canny built an influential detector on this principle. The detector smooths the image with a Gaussian first. It then computes the gradient of the smoothed result. Non-maximum suppression then thins the gradient response. **Hysteresis thresholding follows and keeps connected contours.**

Usage: First use of this source.

Original source: **(1)** *the principle itself, that smoothing and differentiation commute* ← [§VI, printed p. 691:] "Because of the **associativity of convolution, we can first convolve with a symmetric Gaussian** G and then compute directional second derivative zeros to locate edges, and use the magnitude of [the gradient] to estimate edge strength." **(2)** *"smooths the image with a Gaussian first. . . then computes the gradient"* ← [Abstract, printed p. 679:] "The optimal detector has a simple approximate implementation in which **edges are marked at maxima in gradient magnitude of a Gaussian-smoothed image**." **(3)** *"Non-maximum suppression then thins the gradient response"* ← [printed p. 681:] "we have decided to **mark edges at local maxima in the response of the operator** $f(x)$, [so] the first derivative of the response will be zero at these points." **(4)** *"Hysteresis thresholding follows and keeps connected contours"* ← [§Conclusion, printed p. 697:] "A detector was proposed which used **adaptive thresholding with hysteresis to eliminate streaking of edge contours**."

How to check: Read directly from `Articles/05_image_processing/A_Computational_Approach_to_Edge_Detection-sz4.pdf`, Abstract, §VI and the Conclusion. **Entry rebuilt 2026-08-25 to cover the whole block.** The citation closes on "Canny built an influential detector on this principle", and **four further sentences follow it uncited**, each describing a stage of his detector: Gaussian smoothing, gradient computation, non-maximum suppression and hysteresis thresholding. All four are Canny's and all four now have a passage above. **Citation moved 2026-08-25:** it now closes after the hysteresis sentence, so all five sentences of the block sit inside it. **Wording note:** "influential" is an editorial characterisation no source can make of itself. It carries no factual load and is uncontroversial for this paper.

✔ **Validated** — I have read the original and agree it supports the sentence.

#25 marr1980 — Marr & Hildreth (1980) *Ch.2 Background → Edge Detection → Gaussian Prefiltering*

Sentence in thesis: **[Block covered by this citation]** The Gaussian also supports a second edge operator, built on subtraction rather than differentiation. Subtracting a strongly blurred copy of an image from a lightly blurred one gives the difference of Gaussians (DoG). **Marr and Hildreth established that this difference behaves as a second-derivative operator, and they used it to localise edges.**

Usage: Use 2 of this source.

Original source: **(1)** *that the Laplacian is what they used to localise edges* ← [Abstract:] "intensity changes at a given scale are best detected by **finding the zero values of** $\nabla^2 G(x,y) * I(x,y)$ for image I, where $G(x,y)$ is a two-dimensional Gaussian distribution and ∇^2 **is the Laplacian**." **(2)** *why a second derivative localises an edge* ← [p. 192:] "Wherever an intensity change occurs, there will be a corresponding **peak in the first directional derivative**, or equivalently, a **zero-crossing in the second directional derivative** of intensity." **(3)** *that the Laplacian is the orientation-independent choice* ← [p. 207:] "The only **orientation-independent second-order differential operator is the Laplacian** ∇^2."

How to check: Read directly from `Articles/03_npr_rendering/marr_hildreth_1980.pdf`, Abstract, pp. 192 and 207. **Paragraph merged 2026-08-25.** §2.4.3 previously ran the Canny block and the DoG block as two separate paragraphs, and the Canny block closed on "The surviving edge map is thin and stable", which restated what non-maximum suppression and hysteresis had just been said to do. That sentence was deleted and the two paragraphs merged, with *also* marking DoG as the Gaussian's second role rather than a fresh topic. **Deletion of the whole DoG block was considered and rejected**, on the ground that the subtraction itself is self-evident once the Gaussian is understood. Three things argued against it. It is the *only* place in the thesis where DoG is expanded in running text, and DoG or XDoG then appears three times in Related Work, three times in Methodology and five times in Implementation, including the live Meta SDK shader listing. The Laplacian sentence is also the non-obvious half: that subtracting two blurs approximates a second-derivative operator is what makes this an edge detector rather than a band-pass filter, and it is what gives `marr1980` a home at this point. The block was compressed from three sentences to two instead. **Laplacian glossed, then removed again, 2026-08-25. Recorded so it is not reproposed.** The term stood bare at the end of the paragraph and was glossed as the sum of the second derivatives in x and y, with a further sentence giving Marr and Hildreth's peak/zero-crossing rationale. Nidanur then rejected that version on a stronger ground than the one it fixed: naming and explaining an operator sets the reader up to see it applied, and **the thesis never applies the Laplacian**. No shader and no technique uses it. **The source agrees.** Marr and Hildreth treat $\nabla^2 G$ as the object of interest but state that it is not what gets computed: "the filter $\nabla^2 G$. . . is, in practice, **approximated by a difference of Gaussian convolution operator** (p. 208), and Appendix B proves "$\nabla^2 G$ is the **limit of the DOG function** as σ_i/σ_e. . . tends to unity. The Laplacian is scaffolding even in the original. The thesis now says *second-derivative operator* without naming it, which is accurate, since p. 207 gives "the only orientation-independent **second-order differential operator** is the Laplacian. **Edge-localisation clause kept deliberately.** Removing it was also considered, on the ground that it implies the thesis will use DoG for edge detection. It does: XDoG is the Meta SDK colour *edge* layer, replacing Roberts Cross, with the shader variable named `colorLine`. The expectation the clause creates is met in Ch. 4 and Ch. 5, so it stays. **Provenance checked, no action (§5.3a).** Marr and Hildreth credit the

Validated — I have read the original and agree it supports the sentence.

#26 winnemoller2006 *Ch.2 Background → Edge Detection → Gaussian Prefiltering*

Sentence in thesis: **[Block covered by this citation] Winnemöller et al. weighted the coarser blur before the subtraction, and this weight sets how much centre-surround difference an edge must show.**

Usage: First use of this source.

Original source: **(1)** *the weighting of the coarser blur* ← [§3.2, Eq. 4:] $D(\hat{x}, \sigma_e, \tau, \phi_e) = 1$ if $(S_{\sigma_e} - \boldsymbol{\tau \cdot S_{\sigma_r}}) > 0$, otherwise $1 + \tanh(\phi_e(S_{\sigma_e} - \tau \cdot S_{\sigma_r}))$, where $S_{\sigma_r} \equiv S(\hat{x}, 1.6 \cdot \sigma_e)$ is the coarser blur. **(2)** *what the weight sets* ← [same section:] "The parameter τ in Equation 4 **controls the amount of center-surround difference required for cell activation**", and "The **threshold level τ determines the sensitivity of the edge detector**. For small values of τ, less noise is detected, but real edges become less prominent."

How to check: Read directly from `Articles/videoabstraction.pdf`, §3.2. **MISATTRIBUTION INTRODUCED AND CORRECTED THE SAME DAY, 2026-08-26. Recorded because it is the exact defect §5.3a exists to prevent, and it was introduced while adding a sentence to satisfy a different rule.** The sentence first read "Weighting the coarser blur before the subtraction *turns this detector into a stylisation operator*, since the weight *controls how much broad tonal variation survives alongside the line*. Nidanur queried whether the sources supported it. They do not. **Neither half is the 2006 paper's.** That paper treats τ as a noise-sensitivity threshold, quoted above, and fixes it: "**We use $\tau = 0.98$ throughout**. It is a constant, not a stylistic dial. The phrase "wider range of stylistic effects" is the **2012** paper's retrospective framing of the 2006 work, and the tone-retention claim is the 2012 decomposition $(1 - \tau)G_\sigma + \tau D_{\sigma,k}$. Both were attached to the 2006 key. **The sentence now states only the mechanism and the purpose the 2006 paper gives it**, and the stylisation framing moved to the 2012 entry alongside the XDoG name.

Validated — I have read the original and agree it supports the sentence.

#27 winnemoller2012 — Winnemöller et al. (2012) *Ch.2 Background → Edge Detection → Gaussian Prefiltering*

Sentence in the-sis: **[Block covered by this citation] A later compendium named that weighted form the extended difference of Gaussians (XDoG) and credits the modification with a much wider range of stylistic effects.**

Usage: First use of this source.

Original source: **(1)** *the name* ← the paper's own title: "**XDoG: An eXtended Difference-of-Gaussians Compendium** Including Advanced Image Stylization", Computers & Graphics 36(6):740–753. **(2)** *the stylistic-range claim, and its credit to the 2006 work* ← [§2.4:] "Inspired by those models, **Winnemöller et al. [7]** generated edge images using a DoG variant in which the **strength of the inhibitory effect of the larger Gaussian is allowed to vary**... That modification **made it possible to achieve a much wider range of stylistic effects**, particularly after replacing the binary thresholding function T_ϵ with a continuous ramp." Reference [7] is the 2006 paper.

How to check: Read directly from `Articles/03_npr_rendering/xdog_winnemoller2012.pdf`, title page and §2.4. **Added 2026-08-26 so that XDoG is expanded where operators are explained.** The abbreviation previously appeared first in a passing mention inside Related Work's AHEAD summary, and a reader met it as a named operator in Methodology without ever learning what *extended* meant. §6.1 assigns operator explanation to Ch. 2. **The split between the two keys is deliberate and was corrected once.** This key carries the *name* and the *stylistic-range* characterisation, both of which are this paper's own. The preceding sentence carries the weighting mechanism and cites 2006. An earlier draft put the stylisation framing on the 2006 key; see that entry for the correction. **Note on what is not claimed.** The thesis does not use the 2012 reparameterisation, which replaces τ with p. The implementation uses τ and ϕ_e from the 2006 Equation 4, as `_XDoGTau` and `_XDoGPhi` in the Meta SDK shader.

Validated — I have read the original and agree it supports the sentence.

#28 kyprianidis2009 — Kyprianidis et al. (2009) *Ch.2 Background →*
Edge-Preserving Filtering

Sentence in thesis: **[Block covered by this citation]** Gradient operators mark boundaries by drawing lines over an otherwise unchanged surface. A second family abstracts the surface itself and the Kuwahara filter is the member of that family used in this thesis. It removes detail even inside high-contrast regions. Shape boundaries are still protected where contrast is low. **The level of abstraction thus stays roughly uniform across the image.** *[The filter divides the neighbourhood around each pixel into N overlapping subregions and computes the mean μ_i and variance σ_i^2 of each. It then emits the mean of whichever subregion is most homogeneous: — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *all three sentences of the block, in one passage* ← [§1 Introduction, printed p. 1956:] "An edge-preserving filter that overcomes this limitation is the Kuwahara filter [KHEK76]. Based on local area flattening, the Kuwahara filter **properly removes detail in high-contrast regions while also protecting shape boundaries in low-contrast regions**. Therefore, it helps to **maintain a roughly uniform level of abstraction across the image** while providing an overall painting-style look."

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`, §1. **REPLACED 2026-08-25.** The entry previously held this paper's abstract together with its §3.1 passage on the structure tensor. That material is the support for the *anisotropic* entry later in the same subsection, and it had been duplicated here, where the claim concerns the *classical* filter's uniform abstraction level. Neither quote mentioned abstraction level, high-contrast regions or shape-boundary protection. The passage above covers all three thesis sentences one for one. **Paraphrase density, flagged rather than changed:** the three thesis sentences restate the source's two closely, element for element. No run reaches the six-word threshold and no rule is broken, but the passage is worth a read-through with that in mind.

✔ **Validated** — I have read the original and agree it supports the sentence.

#29 kuwahara1976 — Kuwahara et al. (1976) *Ch.2 Background → Edge-Preserving Filtering*

Sentence in thesis: **[Block covered by this citation]** Any subregion straddling a boundary carries a higher variance and loses this comparison. Interior regions therefore flatten towards uniform fills while boundaries stay sharp and the result resembles painterly brushwork. The classical formulation sets $N = 4$, and its isotropic rectangular subregions overlap their neighbours by a single pixel. **They sit in fixed positions, meaning that no orientation is favoured over another.**

Usage: First use of this source.

Original source: **Primary 1976 chapter not held and not obtainable.** Confirmed with Nidanur 2026-08-25: the Plenum Press chapter is paywalled and no free copy exists. The attribution is therefore resolved the way §5.3a prescribes, by checking whether the source we *do* hold credits the original at the exact claim being used. It does. **(1)** *the four rectangular subregions and the one pixel overlap, credited inline* ← [Kyprianidis 2011, §2:] "Another popular edge-preserving smoothing filter is the Kuwahara filter **[Kuwahara et al. 1976]**. The general idea behind this filter is to **divide the filter kernel into four rectangular subregions that overlap by one pixel**. The filter response is then defined by the **mean of a subregion with minimum variance**." The credit marker sits on the sentence *immediately preceding* the structural description, so the 1976 paper is what the structure is being attributed to. **(2)** *the same attribution, independently, in the other paper* ← [Kyprianidis et al. 2009, §2:] "Classical filters in this field are the **well-known Kuwahara filter [KHEK76]** and its variants [TT77, NM79]. . . **The Kuwahara and Tomita-Tsuji filter use rectangular subregions**, while the Nagao-Matsuyama filter uses a rotated elongated bar mask." **(3)** *that the fixed rectangular arrangement is what makes it isotropic* ← [same paper, Abstract and §3:] the contribution is a generalisation "**adapted to the local shape of features**", which is what the classical form lacks.

How to check: Primary not held. Verified against `Articles/03_npr_rendering/anisotropic_multicale_209.pdf` §2 and `Articles/03_npr_rendering/anisotropic_kuwahara.pdf` §2. **Status, stated plainly: this is the strongest verification available for this key, and it is not a primary-source confirmation.** §9 normally bars marking an entry confirmed without the primary. That rule exists to stop attribution resting on a paraphrase. Here it does not: Kyprianidis attaches the inline credit to the precise structural claim the thesis makes, in two separate papers, so the thesis is citing the work the claim originates from rather than the work it was read in. Nidanur confirmed on 2026-08-25 that the primary cannot be obtained, and directed that the origin credit be verified this way and kept. **Bibliographic fields corroborated across both reference lists.** [2011:] "Kuwahara, M., Hachimura, K., Ehiu, S., and Kinoshita, M. 1976. *Processing of ri-angiocardiographic images*. In Digital processing of biomedical images, K. Preston and M. Onoe, Eds. Plenum Press, 187–203." [2009:] "[KHEK76] Kuwahara M., Hachimura K., **Eiho** S., Kinoshita M.: Digital processing of biomedical images. Plenum Press, 1976, pp. 187–203." Between them these supply the chapter title, the editors, the publisher, the year and the page range, and the two agree on all of it. **One discrepancy, resolved.** The third author is spelled *Ehiu* in the 2011 reference list and *Eiho* in the 2009 one. `references.bib` follows **Eiho**, which is the correct form: it is an attested Japanese surname and it is what the 2009 list, the DOI record and every other citation of this chapter give. The 2011 spelling is a typographical error in that paper's bibliography. **Duplicate PDFs removed from the Articles folder the same day** (`anisotropic_kuwahara copy.pdf` and `anisotropic_multicale_209 copy.pdf`), both byte-identical to the files kept.

✔ **Validated** — I have read the original and agree it supports the sentence.

#30 kyprianidis2011 — Kyprianidis (2011) *Ch.2 Background → Edge-Preserving Filtering*

Sentence in thesis: **[Block covered by this citation]** Any subregion straddling a boundary carries a higher variance and loses this comparison. Interior regions therefore flatten towards uniform fills while boundaries stay sharp and the result resembles painterly brushwork. The classical formulation sets $N = 4$, and its isotropic rectangular subregions overlap their neighbours by a single pixel. **They sit in fixed positions, meaning that no orientation is favoured over another.**

Usage: First use of this source.

Original source: **(1)** *the classical filter's construction* ← [§2 Related Work, printed p. 56:] "Another popular edge-preserving smoothing filter is the Kuwahara filter [Kuwahara et al. 1976]. The general idea behind this filter is to **divide the filter kernel into four rectangular subregions that overlap by one pixel**. The filter response is then defined by the mean of a subregion with minimum variance. The Kuwahara filter **produces clearly noticeable artifacts, which are due to the use of rectangular subregions**." **(2)** *the orientation contrast the thesis draws* ← [Kyprianidis et al. 2009, §2:] "The **Kuwahara and Tomita-Tsuji filter use rectangular subregions**, while the Nagao-Matsuyama filter uses a **rotated elongated bar mask**."

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_multicale_-209.pdf`, §2, with the orientation contrast from the 2009 paper, §2. **REPLACED 2026-08-25.** The entry previously held the 2011 paper's *abstract*, describing Kyprianidis's own multi-scale contribution: "A multi-scale computation scheme is proposed that simultaneously propagates local orientation estimates and filtering results up a low-pass filtered pyramid." The thesis sentence is about the **classical 1976 filter**, not the 2011 method. The two shared only the word *orientation*. The correct passage sits on p. 56 of the same PDF and is already quoted by the neighbouring `kuwahara1976` entry. **Inference note:** "No orientation is favoured over another" is not stated literally in either paper. It follows from the fixed rectangular geometry, and quote (2) is the closest explicit support, since it contrasts those rectangles against a filter that does rotate its mask.

✔ **Validated** — I have read the original and agree it supports the sentence.

#31 kyprianidis2009 — Kyprianidis et al. (2009) *Ch.2 Background →*
Edge-Preserving Filtering

Sentence in the-sis: **[Block covered by this citation]** Nevertheless, Kyprianidis et al. recovered that orientation in an anisotropic formulation that aligns the sampling window with local image structure, estimated through the structure tensor. **This tensor is calculated directly from the RGB values of the input rather than from a luminance channel.** *[For an image* $\mathbf{f}$ *with colour-valued partial derivatives* $\mathbf{f}_x$ *and* $\mathbf{f}_y$*, the smoothed tensor is the* 2×2 *matrix — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *"recovered that orientation in an anisotropic formulation. . . estimated through the structure tensor"* ← [Abstract:] "It is based on a generalization of the Kuwahara filter that is **adapted to the local shape of features, derived from the smoothed structure tensor**." **(2)** *"calculated directly from the RGB values of the input"* ← [§3.1 Orientation and Anisotropy Estimation:] "The local orientation and anisotropy estimation is based on the eigenvalues and eigenvectors of the structure tensor [BWBM06]. **We calculate the structure tensor directly from the RGB values of the input** [KD08]."

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`, Abstract and §3.1. **Sentence added 2026-08-25, spotted by Nidanur in the primary.** The thesis implied the colour computation through "colour-valued partial derivatives" and "each dot product runs over those channels", but never stated it. The contrast is worth stating because §2.4 has already established that gradient operators run on a *luminance* channel, so computing the tensor on RGB instead is a departure rather than a detail. Placement chosen by Nidanur: it follows the structure-tensor sentence and picks it up with "This tensor", rather than sitting after the equation. The post-equation clause was changed from "runs over the colour channels" to "runs over those channels" so that it refers back instead of restating the same fact. **PROVENANCE CAVEAT, RAISED AND LEFT AS IS. Not a defect, but on the record (§5.3a).** The §3.1 passage carries two inline credits, and both point away from this key. The structure tensor itself is credited to **[BWBM06]**, Brox, Weickert, Burgeth and Mrazek (2006), *Non-linear structure tensors*, Image and Vision Computing 24(1):41–55. The RGB-direct computation is credited to **[KD08]**, Kyprianidis and Döllner (2008), *Image Abstraction by Structure Adaptive Filtering*, Proc. EG UK TPCG, pp. 51–58. Neither is held, and the first also bears on the structure tensor equation itself, since the tensor is Brox et al.'s construct. `kyprianidis2009` is nonetheless the right key here, because the thesis sentences describe *what the anisotropic Kuwahara filter does*, and the 2009 paper is authoritative for its own construction. The thesis does not claim to introduce either technique. If the origin credits are ever wanted, both PDFs must be obtained first.

✔ **Validated** — I have read the original and agree it supports the sentence.

#32 kyprianidis2009 — Kyprianidis et al. (2009) *Ch.2 Background →*
Edge-Preserving Filtering

Sentence in thesis: **[Block covered by this citation]** where each dot product runs over those channels and the Gaussian G_ρ smooths every component so that the estimate reflects a neighbourhood rather than one pixel. Its eigenvectors give the dominant local orientation and the eigenvalues indicate how strong that orientation is. An elliptical kernel derived from the two aligns the brush marks with salient contours. **The ellipse is partitioned into N sectors in place of the rectangles, and typical values are $N = 4$ or $N = 8$.** *[That smoothing step is itself a convolution across the image, so it requires a render pass of its own. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *"The ellipse is partitioned into N sectors in place of the rectangles"* ← [§3.2:] "We **divide the disc into N equal sectors** by defining characteristic functions which are 1 over the sector and 0 otherwise." **(2)** *"typical values are $N = 4$ or $N = 8$"* ← [§4 Results:] "…with our implementation using $\sigma_r = 3.0$, $\sigma_s = 1.0$ and $N = 8$. A possibility to increase the processing speed is to **reduce the number of sectors to $N = 4$.** This will result in slightly less sharp boundaries between color regions."

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`, §3.2 and §4. **New citation added 2026-08-25.** The sentence is new and closes the N thread opened in the preceding paragraph: the reader is told why the count is carried as a symbol rather than fixed at the classical 4. **Origin rule applied (§5.3a).** Kyprianidis 2011 states the same pair of values in a tidier form, "typical values $N = 4$ or $N = 8$", but it is restating the 2009 paper's own implementation note quoted above. The citation therefore goes to 2009, where the values originate, not to the later paper the phrasing was read in.

✔ **Validated** — I have read the original and agree it supports the sentence.

Chapter: Related Work

#33 mori1970 — Mori (1970) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in thesis: **[Block covered by this citation]** He proposed that human comfort towards an artificial figure increases with its human likeness. This rise holds only up to a critical threshold. Beyond it, subtle imperfections cause a sharp drop in comfort. The drop persists until the figure appears almost fully human. **Only then is comfort recovered [cite]**.

Usage: Use 2 of this source.

Original source: **(1)** *"comfort. . . increases with its human likeness"* ← "**as robots appear more humanlike, our sense of their familiarity increases** until we come to a valley." **(2)** *"This rise holds only up to a critical threshold."* ← ". . . increases **until we come to a valley. I call this relation the "uncanny valley."**" **(3)** *"subtle imperfections cause a sharp drop in comfort"* ← "this kind of prosthetic hand is **too real and when we notice it is prosthetic, we have a sense of strangeness**. . . there is **no longer a sense of familiarity. It is uncanny**. . . so the prosthetic hand is at the bottom of the valley." **(4)** *"The drop persists until the figure appears almost fully human. Only then is comfort recovered."* ← §"The significance of the uncanny": "**In Figure 1, a healthy person is at the top of the second peak.**"

How to check: Read directly from `Articles/01_uncanny_valley/uncanny1970.pdf` (MacDorman & Minato translation, §§"Valley of familiarity and "The significance of the uncanny). **Each numbered pair maps one thesis sentence to the passage that supports it.** The citation sits on the last sentence of the block and covers all four.

✔ **Validated** — I have read the original and agree it supports the sentence.

#34 mori1970 — Mori (1970) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in thesis: **[Figure 3.1 caption]** The vertical axis represents comfort level (*shinwakan*). The horizontal axis represents degree of human likeness. **The solid curve describes affective response to still figures, while the dashed curve describes figures in motion [cite]**.

Usage: Use 3 of this source.

Original source: **(1)** *"vertical axis. . . comfort level (shinwakan)"* ← Figure 1 y-axis is labelled **"comfort level (*shinwakan*)"** in the reproduced figure; the prose renders the same construct as "familiarity". **(2)** *"horizontal axis. . . degree of human likeness"* ← Figure 1 x-axis is labelled **"human likeness"**, marked at **50%** and **100%**. **(3)** *"solid curve. . . still figures. . . dashed curve. . . figures in motion"* ← Figure 1 key, upper left: **"moving" (dashed line)** / **"still" (solid line)**; body text §"The effects of movement": "**As shown in Figure 1, adding movement changes the shape of the uncanny valley by exaggerating the peaks and valley.**"

How to check: Read directly from `Articles/01_uncanny_valley/uncanny1970.pdf`, Figure 1 and §"The effects of movement. **Translation note:** the later MacDorman & Kageki (2012) translation splits this into two figures (Fig. 1 still, Fig. 2 moving). The thesis reproduces the older single-figure version, where both curves share one plot, so the caption is correct for the figure actually shown.

✔ **Validated** — I have read the original and agree it supports the sentence.

#35 mori1970 — Mori (1970) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in the-sis: **[Block covered by this citation]** Mori plotted separate curves for still and moving figures. Both follow the same general shape, but the curve for moving figures drops far deeper into negative comfort. Moreover, the reference points that Mori placed inside the valley, such as a corpse, a zombie and a prosthetic hand, share a common characteristic. Each closely resembles a human at first glance. **The eerie response arises only once the discrepancy registers [cite]**. *[Hence the place where that drop occurs is the most problematic region of the curve. — thesis inference, uncited.]*

Usage: Use 4 of this source.

Original source: **(1)** *"separate curves for still and moving figures"* ← Figure 1 key: **"moving" (dashed)** / **"still" (solid)**. **(2)** *"the curve for moving figures drops far deeper"* ← "adding movement changes the shape of the uncanny valley by **exaggerating the peaks and valley**"; "if we **add movement to a prosthetic hand, which is at the bottom of the uncanny valley, our sensation of strangeness grows quite large**". **(3)** *"a corpse, a zombie and a prosthetic hand"* ← "the **prosthetic hand is at the bottom of the valley**"; "We might be happy this line is **into the still valley of a corpse** and that of not **the living dead**!" **(4)** *"Each closely resembles a human at first glance."* ← "recently prosthetic hands have improved greatly, and **we cannot distinguish them from real hands at a glance**". **(5)** *"The eerie response arises only once the discrepancy registers."* ← "this kind of prosthetic hand is too real and **when we notice it is prosthetic, we have a sense of strangeness**".

How to check: Read directly from `Articles/01_uncanny_valley/uncanny1970.pdf`, §§"Valley of familiarity, "The effects of movement and "The significance of the uncanny. **Two scope notes.** (1) Mori writes "**the living dead**, never "zombie; zombie is the label on the reproduced Figure 1 and is standard usage in this literature. (2) The **corpse sits on the still curve and the living dead on the moving curve**, so they are not two points on one line. The thesis says only that Mori placed them inside the valley, which holds for each curve. The final sentence of the block ("Hence the place where that drop occurs. . .) is the thesis authors' inference and correctly carries no citation.

✔ **Validated** — I have read the original and agree it supports the sentence.

#36 macdorman2006 — MacDorman & Ishiguro (2006) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in the-sis: **[Figure 3.2 caption]** An android (left, Repliee R1) alongside a mechanical-looking humanoid robot (right, Eveliee P1), developed at Hiroshi Ishiguro's Intelligent Robotics Laboratory **[cite]**.

Usage: First use of this source.

Original source: **(1)** *"An android (left, Repliee R1)"* ← [Figure 1 caption, p. 298:] "the android on the **left named Repliee R1**, a joint effort with Kokoro Co., Ltd." **(2)** *"a mechanical-looking humanoid robot (right, Eveliee P1)"* ← "the humanoid on the **right named Eveliee P1**, a joint effort with Mitsubishi Heavy Industries"; the phrase **"mechanical-looking humanoid robots"** is the paper's own, p. 298. **(3)** *"developed at Hiroshi Ishiguro's Intelligent Robotics Laboratory"* ← "**Hiroshi Ishiguro's Intelligent Robotics Laboratory at Osaka University developed the two robots shown above**."

How to check: Read directly from `Articles/01_uncanny_valley/MacDorman2006AndroidScience.pdf`, Figure 1 caption, p. 298. Confirms both robot names, their left/right positions, the laboratory attribution and the descriptor "mechanical-looking.

✔ **Validated** — I have read the original and agree it supports the sentence.

#37 macdorman2006 — MacDorman & Ishiguro (2006) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in thesis: **[Block covered by this citation — method]** Morphed image sequences ran from a mechanical humanoid robot, through an android, to a photograph of the person the android was modelled on. An android in this sense is a robot designed to be indistinguishable from a human in appearance and behaviour, as shown in Figure 3.2. Forty-five participants rated all thirty-one images on a nine-point scale of human likeness and a nine-point scale of familiarity. **They then marked the images that felt eerie and rated those on a ten-point scale [cite]**.

Usage: Use 2 of this source.

Original source: **(1)** *"ran from a mechanical humanoid robot, through an android, to a photograph of the person the android was modelled on"* ← [§The uncanny valley, p. 304:] "two sets of **11 morphed images**. . . a photograph of the **humanoid robot Qrio** (left) was morphed into a photograph of **an android**. . . based on the science fiction author Philip K. Dick (center), which in turn was morphed into a photograph of **the author himself** (right). . . the **humanoid robot Eveliee** (left) was morphed into. . . the **android Repliee Q1Expo** (center), which in turn was morphed into **Repliee's human model** (right)." **(2)** *"a robot designed to be indistinguishable from a human in appearance and behaviour"* ← [p. 298:] "An android is defined to be 'an artificial system designed with the ultimate goal of being **indistinguishable from humans in its external appearance and behavior**'." **(3)** *"Forty-five participants rated all thirty-one images"* ← [§Participants, p. 305:] "There were **45** Indonesian participants, 37 male and 8 female"; [§Procedure, p. 306:] "to rate **31 images** presented in random order". **(4)** *"a nine-point scale of human likeness and a nine-point scale of familiarity"* ← [p. 306:] "on a **nine-point scale of human likeness and familiarity** — that is, a scale ranging from very mechanical to very human and ranging from very strange to very familiar, **respectively**." **(5)** *"marked the images that felt eerie and rated those on a ten-point scale"* ← [p. 306:] "participants were asked, '**Which of the following figures gives you an eerie feeling?**' **The selected images were then presented** in random order and the participants were asked to rate how eerie each image was on a **ten-point scale**."

How to check: Read directly from `Articles/01_uncanny_valley/MacDorman2006AndroidScience.pdf`, pp. 298 and 304–306. **Each numbered pair maps one thesis claim to the passage supporting it. REVISED 2026-08-23:** the sentence previously said "every image, which reads as the 22 morphs. It now states **thirty-one**, the paper's own figure. The paper does not itemise the remaining nine, so the thesis does not characterise them.

✔ **Validated** — I have read the original and agree it supports the sentence.

#38 macdorman2006 — MacDorman & Ishiguro (2006) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in the-sis: **[Block covered by this citation — results]** Familiarity fell into a valley in both sequences and its trough lay between the humanoid robot and the android. Perceived human likeness rose steadily across the same span while eeriness peaked close to where familiarity was lowest. **This study reproduced Mori's observations from measured data [cite]**. *[Consequently, any near human avatar risks the same response if its rendering places it in that trough. — thesis inference, uncited: the paper studies robots and androids, never avatars.]*

Usage: Use 3 of this source.

Original source: **(1)** *"Familiarity fell into a valley in both sequences and its trough lay between the humanoid robot and the android."* ← [§Results, p. 306:] "The results show a **valley** in the strange versus familiar scale. . . **The trough of the valley occurs in both sets of morphed photographs between the humanoid robot and the android robot.**" **(2)** *"Perceived human likeness rose steadily across the same span"* ← "In the region of the valley in both sets, there is a **monotonic increase in perceived human likeness** moving from the humanoid robot to the android to the human." **(3)** *"eeriness peaked close to where familiarity was lowest"* ← "**The peak of the eeriness ratings roughly matches the trough of the familiarity ratings**." **(4)** *"This study reproduced Mori's observations from measured data."* ← [p. 307:] "In sum, **the experiment reproduced Mori's observations on the uncanny valley with two different sets of morphed photographs**."

How to check: Read directly from `Articles/01_uncanny_valley/MacDorman2006AndroidScience.pdf`, §Results, pp. 306–307. **ADDED 2026-08-23:** these result sentences previously carried *no* citation — the paragraph's only one sat on the procedure sentence. **Hedge note:** the paper says the eeriness peak "*roughly* matches the familiarity trough, so the thesis was softened to "close to where familiarity was lowest.

✔ **Validated** — I have read the original and agree it supports the sentence.

#39 mcdonnell2012 — McDonnell et al. (2012) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in thesis: **[Block covered by this citation]** This risk became the subject of direct empirical study. McDonnell et al. investigated the perceptual consequences of render style choice directly. Separate groups of participants rated still images and video of the same animated character, rendered across 10 distinct styles spanning the range from photorealistic to fully stylised (). The results exhibited a valley-shaped pattern across appeal, reassurance and trustworthiness alike. Four of the most abstract styles and the two most realistic ones scored significantly higher on all three measures than styles in the mid realism range. The authors concluded that styles at both ends of that range are viable. Interestingly, cartoon styles were rated friendlier than realistic ones. **However, appeal does not follow from abstraction alone, since one of their abstract styles ranked among the least appealing.** *[Taken together, these findings suggest that stylised rendering can avoid the perceptual and trust costs concentrated in the mid realism range without requiring full photorealistic fidelity. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *"rendered across ten distinct styles"* ← §4: "We created **11 rendering styles** in total"; §5.1.1: "we used the male model rendered in **10 styles**. . . The eleventh style with animated wrinkles was not included." Every ANOVA is reported as $F(9, 288)$, confirming ten tested conditions. **(2)** *"valley-shaped pattern across appeal, reassurance and trustworthiness. . . four of the most abstract styles and the two most realistic ones"* ← §5.1.3 Appeal: "**four of the most abstract and the two most realistic styles. . . significantly more appealing than all other styles** ($p < 0.05$)"; Trustworthy: "Finally, **the most realistic Humans, and all other Toons were rated as equally trustworthy, and more trustworthy than all other styles** ($p < 0.004$ in all cases)". Re-assuring: "**As with the Appeal ratings, the most realistic and most abstract styles were considered the most re-assuring** (ToonPencil, ToonFlat, ToonShaded, ToonCG, HumanHQ1, HumanHQ2)"; significance on that scale is reported at the opposite end — "**HumanIII was rated as the most eerie, and significantly more so than all other styles** ($p < 0.00004$)", followed by HumanBasic, ToonBare and HumanSSS. **(3)** *"Cartoon styles were rated friendlier than realistic ones."* ← §6 Discussion: "Cartoon characters were considered highly appealing. . . **They were rated as more friendly than realistic styles** and therefore might be more appropriate for certain virtual interactions." **(4)** *"one of their abstract styles ranked among the least appealing"* ← §6: "**However, we cannot say that all abstract cartoon characters are considered appealing. One of our render styles (ToonBare) was considered quite unappealing** and evoked negative reactions."

How to check: Read directly from `Articles/01_uncanny_valley/ McDonnell, R., Breidt, M., and Bulthoff, H. H. (2012). Render me Real? Investigating.pdf`, §§4, 5.1.1, 5.1.2, 5.1.3 and 6. **Ten versus eleven — know this one:** §5.1.1 (still images) used **ten** styles, but §5.1.2 (movies) states the motion was "rendered in the **full 11 styles**. Every reported ANOVA is $F(9, 288)$, that is ten levels, so the eleventh (animated wrinkles) was excluded from analysis. Ten therefore matches what was tested, analysed and shown in Figure 3.3, but a reader checking §5.1.2 may query it. **Apparent conflict, resolved:** §5.1.3 also says "the most realistic render styles were rated as more friendly than HumanSSS and ToonBare. That is a narrower comparison against two specific low-scoring styles; the same passage then reports that **the Toons were rated the most friendly** ($p < 0.009$), and §6 states the overall result plainly. The thesis sentence follows §6.

✔ **Validated** — I have read the original and agree it supports the sentence.

#40 mcdonnell2012 — McDonnell et al. (2012) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in the-sis: **[Block covered by this citation]** The ten render styles compared by McDonnell et al., ranging from abstract to realistic. Top row, from left to right: Toon Pencil, Toon Flat, Toon Shaded, Toon Bare and Toon CG. **Bottom row: Human III, Human Basic, Human SSS, Human HQ1 and Human HQ2.**

Usage: Use 4 of this source.

Original source: **(1)** *"The ten render styles compared by McDonnell et al., ranging from abstract to realistic"* ← Figure 3 caption: "**The ten render styles used in our experiments, ranging from abstract to realistic**." **(2)** *"Top row. . . Toon Pencil, Toon Flat, Toon Shaded, Toon Bare and Toon CG. Bottom row: Human III, Human Basic, Human SSS, Human HQ1 and Human HQ2"* ← all ten names appear verbatim in §4 and §5.1.3. The abstract-to-realistic ordering is confirmed by the realism ranking in §5.1.3: "**six distinct levels of realism were found: 1. ToonPencil; 2. ToonFlat, ToonBare, ToonShaded; 3. ToonCG; 4. HumanIII; 5. HumanBasic, HumanSSS; 6. HumanHQ1, HumanHQ2**", and by §5.1.1: "the **ToonPencil style was rated as the most abstract** (1.5 on the scale), and **HumanHQ2 was considered very realistic** (6 on the scale)".

How to check: Read directly from `Articles/01_uncanny_valley/ McDonnell, R., Breidt, M., and Bulthoff, H. H. (2012). Render me Real? Investigating.pdf`, Figure 3 caption and §§4, 5.1.1, 5.1.3. **Ten versus eleven:** the paper created eleven styles but tested ten — §4: "*Figure 3 depicts the first 10 styles, and Figure 9 shows an example of the eleventh style under motion*; §5.1.1: "*The eleventh style with animated wrinkles was not included*. The thesis figure reproduces the ten-style Figure 3, so "ten is correct in both the caption and the prose.

✔ **Validated** — I have read the original and agree it supports the sentence.

#41 dill2012 — Dill et al. (2012) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in the-sis: **[Block covered by this citation]** Furthermore, Dill et al. validated the uncanny valley response specifically for CG characters drawn from contemporary commercial films and games. Their familiarity ratings followed Mori's predicted curve, with the strongest negative response in the near-human range. Motion deepened the valley most visibly for two of the ten characters tested. **Participants also named the eyes and the mouth as the facial regions producing the most discomfort.** *[This confirmation that the valley extends to digital characters from popular media, not only to physical robots or androids, motivates its consideration across all virtual avatar contexts. Modern avatar systems fall within this same problematic region of the uncanny valley curve. They achieve enough visual realism to raise viewers' expectations, but cannot fully deliver on them in motion timing, expressions or gaze behaviour. As a result, a photorealistic avatar may undermine the very trustworthiness it was designed to convey. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"validated the uncanny valley response specifically for CG characters drawn from contemporary commercial films and games"* ← [§1, p. 511:] "we selected **ten representative characters from recent and well known movies and games**." **(2)** *"familiarity ratings followed Mori's predicted curve, with the strongest negative response in the near-human range"* ← [§2, p. 511:] "It follows the original chart proposed by Mori, where the horizontal axis represents the **human likeness** and the vertical axis indicates the **familiarity level**. . . samples C.11, C.16, C.1 and C.14 suggest the same pattern of a **negative emotional response** as in result of an '**almost human but unnatural**' character representation." **(3)** *"Motion deepened the valley most visibly for two of the ten characters tested."* ← [§2, p. 512:] "more noticeable in characters **C.16 and C.14 that exaggerated the valley in the video's curve**." **(4)** *"named the eyes and the mouth as the facial regions producing the most discomfort"* ← [§2, p. 512:] "The questionnaire also asked to the participants **which parts of the face** the participant felt more discomfort. . . we could **highlight two specific parts of the face: eyes and mouth**."

How to check: Read directly from `Articles/01_uncanny_valley/Evaluation of the Uncanny Valley in CG Characters.pdf`, printed pp. 511–512 (PDF pp. 526–527 of the IVA 2012 volume). **Validate is the term the authors themselves use** — §3 Final Considerations states that it was possible to *validate the uncanny valley persistence using a contemporary scenario*. **Evidentiary weight — know this one:** the source is a three-page *poster* (pp. 511–513). It reports **no participant count and no statistics**: no N, no p values and no significance tests anywhere in it. All four thesis sentences are descriptive and none of them claims significance, so they match what the paper supports, but the source carries less weight than a full study. **Scope note:** participants were asked only about *parts of the face*, so the paper supports non-uniformity within the face and says nothing about the rest of the body. The three sentences following the citation are the thesis authors' extrapolation to modern avatar systems and correctly carry no citation.

✔ **Validated** — I have read the original and agree it supports the sentence.

#42 seymour2021 — Seymour et al. (2021) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in thesis: **[Block covered by this citation]** Conversely, Seymour et al. tested that prediction directly using a professionally produced, real-time photorealistic avatar in a field study at a conference. None of the 18 guests who interacted with it displayed an aversive response. **Observers rated it as more trustworthy than a caricature counterpart, with greater affinity and stronger preference.** *[This result suggests that a sufficiently high-fidelity, real-time avatar can escape the uncanny valley entirely rather than merely offset its cost with other advantages. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"a professionally produced, real-time photorealistic avatar in a field study at a conference"* ← Abstract: "We used a **field study**. . . collected survey data from **SIGGRAPH conference** attendees. . . **human-realistic avatars rendered in real time**." §3.1: "high fidelity facial input data, provided by **industry partner Cubic Motion**. . . a custom facial rig developed by another partner, **3lateral**". **(2)** *"None of the eighteen guests who interacted with it displayed an aversive response."* ← Results: "All guests made eye contact with the Digital MIKE, talked, interacted, and matched verbal cues. **None of the 18 guests displayed any type of aversive response**, and none ended the discussion prematurely (see Table 5)." Table 5: Uncanniness of Digital MIKE — **0 of 18**. **(3)** *"Observers rated it as more trustworthy than a caricature counterpart, with greater affinity and stronger preference."* ← Abstract: "**The observers** had positive evaluations of both avatars. . . **They rated the human-realistic avatar as more trustworthy, had more affinity for it, and preferred it as a virtual agent**."

How to check: Read directly from `Articles/01_uncanny_valley/Seymour, Yuan, Dennis & Riemer, "Have We Crossed the Uncanny Valley?" (2021).pdf`. **CORRECTED 2026-08-17.** The count of eighteen is confirmed by Table 5. Earlier wording attributed the affinity, trustworthiness and preference ratings to the same eighteen guests who interacted; the abstract assigns those to the *observers*, a separate group. The thesis sentence now separates the two, which is why pair (2) and pair (3) name different groups. **Two further details.** (a) The thesis says *a caricature counterpart*, singular, but there were **eighteen** of them: §3.1 states that each guest was represented by a unique caricature avatar custom designed to be a caricature of that guest, so eighteen different caricature avatars were used against one constant human-realistic avatar. The comparison itself is unaffected. (b) Unlike the Dill poster, this study reports inferential statistics throughout, and the abstract records that **effect sizes ranged from medium to large**.

✔ **Validated** — I have read the original and agree it supports the sentence.

#43 tao2025 — Tao et al. (2025) *Ch.3 Related Work → The Uncanny Valley and Avatar Realism*

Sentence in thesis: **[Block covered by this citation]** A larger and more recent test of that possibility comes from a network meta-analysis by Tao et al. It pooled 13 independent studies and 2,343 participants across three realism levels. High realism avatars were rated significantly more trustworthy than low realism avatars, whereas medium realism differed significantly from neither. **The cumulative ranking placed medium realism as the most eerie, although none of the eeriness comparisons reached significance.** *[The pooled evidence therefore favours realism for trust and leaves the eeriness pattern unresolved. Overall, the valley reproduces reliably across robots, morphed photographs and CG characters, yet these studies disagree on whether stylisation or full realism is the safer route around it. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"a network meta-analysis by Tao et al."* ← Abstract: "A **network meta-analysis (NMA)** was conducted." **(2)** *"pooled 13 independent studies and 2,343 participants across three realism levels"* ← Abstract: "A total of **13 studies with 2,343 participants** were included in the analysis"; §1: "VAAR is categorized into three levels: (1) **high realism avatars (HRA)**, (2) **medium realism avatars (MRA)** and (3) **low realism avatars (LRA)**." **(3)** *"High realism avatars were rated significantly more trustworthy than low realism avatars, whereas medium realism differed significantly from neither."* ← §3.6: "The NMA results indicated that the **HRA received significantly higher ratings than the LRA** ($p = 0.046$). **No significant differences were observed between the MRA group and either the HRA or LRA groups**." **(4)** *"None of the eeriness comparisons reached significance, although the cumulative ranking placed medium realism as the most eerie."* ← §3.7: "although the MRA ($p = 0.433$) and the HRA ($p = 0.766$) tended to be better than the LRA, these differences were **not statistically significant** at the $\alpha = 0.05$ level. **No significant difference was observed between the MRA and HRA either**"; and the SUCRA ranking: "**MRA had the highest probability of being perceived as the most eerie (74.5%)**, followed by HRA (44.7%) and LRA (30.9%)".

How to check: Read directly from `Articles/Tao-Impact.pdf`, Abstract and §§1, 3.6, 3.7. **RESOLVED 2026-08-23:** the citation previously closed on the trustworthiness sentence, leaving the eeriness sentence, which is equally a Tao finding, outside it. The citation now closes the whole findings block and covers pairs (2) to (4). **The paper contradicts itself on eeriness — know this one.** It asserts an MRA eeriness effect in **two** places while its own Results report the comparisons as null. *Abstract:* "In contrast, **medium VAAR elicited the strongest feelings of eeriness**. *§4.3 Eeriness, in the Discussion:* "Our results showed that **MRA received higher eeriness ratings compared to HRA and LRA**, which suggests that they are **most likely to induce feelings of unease**. *§3.7, the only place statistics are reported:* the MRA ($p = 0.433$) and HRA ($p = 0.766$) differences from LRA were **not statistically significant**, and **no significant difference was observed between MRA and HRA either**. The assertive wording therefore rests on the **SUCRA probability ranking** (MRA most eerie at 74.5%), not on a significance test. **The thesis follows §3.7** and separates the ranking from the significance, which is the accurate reading. A reader who checks only the abstract or the discussion will think the thesis disagrees with the paper; in fact the paper disagrees with itself. The closing sentence of the block is the thesis synthesis and correctly carries no citation.

✔ **Validated** — I have read the original and agree it supports the sentence.

#44 mayer1995 — Mayer et al. (1995) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** Trust in virtual avatars is multidimensional, not a single attitude. Mayer et al. established this structure by decomposing perceived trustworthiness into ability, benevolence and integrity. **They further separated trustworthiness as an attribute of the trustee from trust as an attitude held by the trustor.**

Usage: Use 3 of this source.

Original source: **(1)** *"decomposing perceived trustworthiness into ability, benevolence and integrity"* ← [p. 717:] "Even though a number of factors have been proposed, **three characteristics of a trustee appear often in the literature: ability, benevolence, and integrity. As a set, these three appear to explain a major portion of trustworthiness**." Each is then defined in turn — [p. 717:] "**Ability** is that group of skills, competencies, and characteristics that enable a party to have influence within some specific domain"; [p. 718:] "**Benevolence** is the extent to which a trustee is believed to want to do good to the trustor, aside from an egocentric profit motive"; [p. 719:] "The relationship between **integrity** and trust involves the trustor's perception that the trustee adheres to a set of principles that the trustor finds acceptable." **(2)** *"separated trustworthiness as an attribute of the trustee from trust as an attitude held by the trustor"* ← trust is defined as the *trustor's* state [p. 712:] "the definition of trust proposed in this research is **the willingness of a party to be vulnerable to the actions of another party** based on the expectation that the other will perform a particular action important to the trustor"; the three trustee characteristics are explicitly *not* that state [p. 717:] "**Although they are not trust per se**, these variables help build the foundation for the development of trust"; and the separation is stated outright [p. 724:] "**The distinction between a trustor's characteristics and trustee's characteristics is important.** Perceptions of ability, benevolence, and integrity of another party leave a considerable amount of variance in trust unexplained, because they neglect between-trustor differences in propensity to trust."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Mayer-IntegrativeModelOrganizational-1995.pdf`. Page numbers confirmed against the running heads in the JSTOR scan (*Academy of Management Review* 20(3), 709–734). **REPLACED 2026-08-23:** the earlier excerpt quoted only "three characteristics of the trustee that determine trustworthiness are examined" — it never named ability, benevolence and integrity, and left the trustor/trustee separation implicit. The passages below state both claims directly.

✔ **Validated** — I have read the original and agree it supports the sentence.

#45 mcknight2001 — McKnight & Chervany (2001) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in the-sis: **[Block covered by this citation] McKnight and Chervany subsequently carried that distinction into a typology separating trusting beliefs from trusting intentions in technology-mediated contexts.** *[Both frameworks are set out in . Their consequence for the present review is that trust should not be reduced to a single score, as rendering style could plausibly affect ability, benevolence and integrity differently. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *"a typology separating trusting beliefs from trusting intentions"* ← [p. 13:] "**Trusting Beliefs** means one believes that the other party has one or more characteristics beneficial to one"; [p. 14:] "**Trusting Intentions** means one is willing to depend on, or intends to depend on, the other party even though one cannot control that party." The abstract lists them as separate conceptual constructs [p. 2:] "The conceptual-level constructs consist of Disposition to Trust, Institution-based Trust, and **Trusting Beliefs and Trusting Intentions**." **(2)** *"carried that distinction"* — the Mayer split between what is believed about the *trustee* and what the *trustor* is willing to do is preserved and made explicit [p. 14:] "The link between Trusting Beliefs and Trusting Intentions is natural because the theory of reasoned action posits that **beliefs influence intentions**. . . strong beliefs that the vendor is **honest, competent, benevolent, and predictable** should provide one the assurance to be **willing to depend** on them." The belief content maps onto Mayer's integrity, ability and benevolence, and the paper engages Mayer et al. directly elsewhere (pp. 5 and 8). **(3)** *"in technology-mediated contexts"* ← [p. 13:] "**In the Internet context**, the people involved include consumers and E-vendors"; [p. 14:] "**In the Internet setting**. . . the consumer has less control than in the brick-and-mortar context."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/What Trust Means in E-Commerce Customer Relationships.pdf`, pp. 2, 13–14. **Verification note — this one is harder to check.** The PDF is a **scanned image with no text layer**: text search returns nothing and no fonts are embedded. The passages above were read from the rendered page images, and the Trusting Intentions definition on p. 14 was confirmed against the page image directly. **REPLACED 2026-08-23:** the earlier excerpt gave the two definitions but nothing supporting either "carried that distinction" from Mayer or "technology-mediated contexts. Pairs (2) and (3) below supply both.

✔ **Validated** — I have read the original and agree it supports the sentence.

#46 alipour2025 — Alipour et al. (2025) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** Trust measurement has since been examined more broadly across the literature. Alipour et al. conducted a systematic review of 53 empirical studies to determine whether and how the uncanny valley effect influences trust. They found three different operationalisations of trust across this literature. The first is a direct rating of trust or trustworthiness. The second is a related construct such as competence or credibility. The third is a trust-related intention such as advice following. **The reported relationship between the uncanny valley effect and trust was inconsistent across these studies, with some reporting a direct negative effect and others finding no relationship at all.** *[This inconsistency suggests that the effect of appearance on trust may depend on other perceptual factors. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *"a systematic review of 53 empirical studies to determine whether and how the uncanny valley effect influences trust"* ← [Introduction:] "we conducted a **systematic review** of empirical research following the PRISMA protocol. From an initial pool of 311 records, 107 studies were screened in full, and **53 were included for detailed analysis**. Our review clarifies current knowledge on **how an agent's uncanniness, caused by the UVE and an almost humanlike appearance, affects trust**, and under which conditions." **(2)** *"three different operationalisations of trust. . . a direct rating. . . a related construct such as competence or credibility. . . a trust-related intention such as advice following"* ← [§4.3.2:] "Studies were categorized into **three groups** based on the type of trust measurement that they employed. [The first] consisted of studies that **directly and explicitly measured trust or trustworthiness** empirically. . . The second group ($n = 11$) included studies that assessed **trust-related constructs, such as competence, acceptability, credibility, believability, and persuasiveness**. The third group ($n = 18$) contained studies that assessed **trust-related intentions, such as adherence to an agent's advice**, willingness to interact, or conformity to an agent's recommendation." **(3)** *"inconsistent across these studies, with some reporting a direct negative effect and others finding no relationship at all"* ← [§5.1:] "while the UVE affects trust, this relationship is often **not straightforward**. **Only about half of the studies**. . . reported a **direct effect**. Moreover, slightly more than half of the studies reported **conditional or indirect effects**"; and the counts in [§4.4.1:] "Among the 41 coded studies, **19** were categorized as reporting a direct effect, **24** conditional or indirect effects, and **10 no effect**." The direction is the paper's own: it refers throughout to "the UVE's **negative effect**. . . on trust".

How to check: Read directly from `Articles/02_trust_and_avatar_perception/alipour Would you rely on an eerie agent?.pdf`, Introduction and §§4.3.2, 4.4.1, 5.1. **REPLACED 2026-08-23.** The earlier excerpt for this source quoted only "most studies rely on static images or hypothetical scenarios with limited real-time interaction" — a remark about study limitations, which supported none of the claims in this block. **Citation moved in the same pass:** the citation previously closed on the first sentence, leaving the taxonomy and the inconsistency finding, five sentences of Alipour results, with no support. It now closes the whole findings block. The final sentence of the paragraph is the thesis inference and correctly carries no citation.

✔ **Validated** — I have read the original and agree it supports the sentence.

#47 canales2024 — Canales et al. (2024) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation] The three avatar stylisation levels used by Canales et al.: realistic (AF-Real), semi-realistic (AM-Mid) and caricatured (CM-Caricature).**

Usage: Use 2 of this source.

Original source: **(1)** *the three levels and their labels* ← [Figure 1 caption:] "Images from our VR study showing the **three identities (AF, AM, CM) and styles (Real, Mid, Caricature)**." Each doctor is a distinct identity shown at one style level, which is why the thesis caption pairs them as AF-Real, AM-Mid and CM-Caricature. [§3:] "Each doctor is a different person (varied identity) and each is represented at a **different level of stylization: realistic, caricature, or a mid condition intended to be in-between realistic and caricature**."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Avatar_-Stylization_and_Trust_IEEE_VR_2024.pdf`, Figure 1 caption and §3. Confirms both the identity labels and the style labels, and that identity is confounded with style level by design, which the paper acknowledges as a trade-off for ecological validity.

✔ **Validated** — I have read the original and agree it supports the sentence.

#48 nowak2003 — Nowak & Biocca (2003) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** One such factor is anthropomorphism. Nowak and Biocca manipulated both the anthropomorphism of a partner's virtual image and its perceived agency. Of these two factors, only anthropomorphism affected presence. Its effect also ran counter to their own prediction, because the less anthropomorphic image produced more copresence and social presence than the highly anthropomorphic one. **The authors attributed this to expectation, since a more anthropomorphic image raises what the viewer anticipates and presence falls when the interaction does not meet it.**

Usage: First use of this source.

Original source: **(1)** *"manipulated both the anthropomorphism of a partner's virtual image and its perceived agency"* ← [Abstract:] "The experiment varied the level of anthropomorphism of the image of interactants: high anthropomorphism, low anthropomorphism, or no image. **Perceived agency was manipulated by telling the participants that the image was either an avatar controlled by a human, or an agent controlled by a computer**." **(2)** *"only anthropomorphism affected presence"* ← [§6.1, section heading:] "**Agency Does Not Influence Presence**"; and Table 2, where the virtual image is significant on all three measures while agency is significant on none: perceived other's copresence **.22*** versus $-.034$, self-reported copresence **.17*** versus n.s., social presence $-$**.19*** versus $-.09$. [§5:] "The effect of an anthropomorphic virtual image on perceived other's copresence is **significant:** $R = .22$**,** $F = 6.52$**,** $p = .01$." **(3)** *"ran counter to their own prediction"* ← Hypothesis V predicted the reverse: "People will feel **less** copresence with partners represented by **low**-anthropomorphic images than with partners represented by high-anthropomorphic images or by no images." Result: "**This hypothesis was not supported. The effect of the low-anthropomorphic image was significant in the opposite direction**." **(4)** *"the less anthropomorphic image produced more copresence and social presence"* ← [Abstract:] "Participants interacting with the **less-anthropomorphic image reported more copresence and social presence** than those interacting with partners represented by either no image at all or by a highly anthropomorphic image." **(5)** *"The authors attribute this to expectation. . . presence falls when the interaction does not meet it"* ← [Abstract:] "indicating that the more anthropomorphic images **set up higher expectations that lead to reduced presence when these expectations were not met**."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Nowak2003.pdf`, Abstract, §5 Results, Table 2 and §6.1. **REVISED 2026-08-23.** Two changes. First, the earlier support showed only that *agency* was null, which does not establish that anthropomorphism had an effect. Table 2 supplies the positive evidence: anthropomorphism is significant on all three presence measures while agency is significant on none. Second, the thesis previously explained the reversal as diminishing perceptual returns, which is not the mechanism the paper proposes. It now reports the authors own explanation, expectation violation, which also matches the mechanism behind the uncanny valley analogy drawn in the next sentence.

✔ **Validated** — I have read the original and agree it supports the sentence.

#49 nowak2005 — Nowak & Rauh (2005) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** This pattern is structurally analogous to the uncanny valley, where affinity likewise ceases to scale linearly with human resemblance and inverts before recovering at full fidelity. Nowak and Rauh later carried this question into credibility. They asked 255 participants to rate a series of avatars. **Avatars perceived as more anthropomorphic were also rated as more credible and were chosen more often for self-representation.** *[Therefore, avatar design choices function as meaningful social signals. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *"Avatars perceived as more anthropomorphic were also rated as more credible and were chosen more often for self-representation"* ← [Conclusion, JCMC p. 172:] "**Avatars that were more anthropomorphic were perceived to be more attractive and credible, and people were more likely to choose to be represented by them.**" Corroborated by Table 3: anthropomorphism × credibility $r = .46$, $p < .01$. **(2)** *the 255 participants* ← [Abstract:] "In this study, **participants (N = 255)** evaluated a series of avatars. . . in terms of their androgyny, anthropomorphism, credibility, homophily, attraction, and the likelihood they would choose them during an interaction."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Nowak2005.pdf`, Abstract, Results RQ8, Table 3 and Conclusion. **THREE ERRORS FOUND AND CORRECTED 2026-08-25, the most serious in Ch. 3.** The passage previously read "Both anthropomorphism *and gender expression* predicted the resulting credibility ratings. More anthropomorphic avatars were rated as more credible. The authors noted that *credibility is itself among the primary predictors of trust*." Each of the three claims failed for a different reason. **(a) The gender half is contradicted by the source, not merely unsupported.** The RQ8 regression gives image gender → credibility $b = -.01$, $p = .55$, and the Discussion states it outright: "masculine images. . . child-like images. . . **these characteristics did not significantly influence attributions of credibility**. The Results are equally explicit: "**Credibility was negatively predicted only by** whether or not the image was an object ($b = -.33$) or animal ($b = -.28$). No weaker wording survives, so the clause was deleted rather than softened. **(b) "Predicted" overstated even the anthropomorphism half.** The paper's own Conclusion disclaims that reading: "Previous research has implied that anthropomorphism would be **the main predictor of credibility** or attractiveness (Koda, 1996; Wexelblat, 1997), **but this was not the case in the present study**." The relation holds correlationally, which is how the sentence now states it, in the authors' own Conclusion wording. **(c) The trust sentence was deleted outright.** The word "trust" appears **three times in the entire article**: twice inside one background sentence that Nowak and Rauh credit to others, "credibility is one of the primary predictors for certain interaction outcomes such as attitude change and trust. . . (McCroskey & Young, 1981)(and McCroskey, Hamilton & Weiner, 1974), and once in a reference title. **They never measured trust.** A reader would have taken a borrowed definitional aside for a finding of a 255-participant avatar study. Under §5.3a the origin is McCroskey, which is not held, so the claim was removed rather than re-cited. **Nothing downstream breaks:** the bridge into the Canales paragraph is "social signals", not trust.

✔ **Validated** — I have read the original and agree it supports the sentence.

#50 alipour2025 — Alipour et al. (2025) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** Canales et al. tested this signalling role directly by manipulating avatar stylisation and measuring its effect on multi-component trust. Participants were presented with virtual doctors rendered at three stylisation levels: realistic, semi-realistic and caricatured. Trust was measured after each doctor interview, both as a direct rating and across the ability, benevolence and integrity dimensions. In addition, participants were asked which doctor they would consult for a follow-up. This provided a behavioural indicator alongside the self-report scale. **Together these span all three measurement approaches in the Alipour et al. taxonomy.**

Usage: Use 3 of this source.

Original source: **(1)** *"span all three measurement approaches in the Alipour et al. taxonomy"* ← [§4.3.2:] "Studies were categorized into **three groups** based on the type of trust measurement. . . [the first] consisted of studies that **directly and explicitly measured trust or trustworthiness**. . . The second group included studies that assessed **trust-related constructs, such as competence, acceptability, credibility**. . . The third group contained studies that assessed **trust-related intentions, such as adherence to an agent's advice**." Canales et al. used all three: a direct trust rating, the ability, benevolence and integrity dimensions, and the choice of which doctor to consult.

How to check: Read directly from `Articles/02_trust_and_avatar_perception/alipour Would you rely on an eerie agent?.pdf`, §4.3.2. The thesis sentence maps the Canales measures onto the three Alipour groups, so the support is the taxonomy itself rather than a finding.

✔ **Validated** — I have read the original and agree it supports the sentence.

#51 canales2024 — Canales et al. (2024) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** The results indicated that self-reported trust ratings were broadly equivalent across all three style levels. **However, the authors found that the semi-realistic condition attracted the highest rate of behavioural selection.**

Usage: Use 3 of this source.

Original source: **(1)** *"virtual doctors rendered at three stylisation levels: realistic, semi-realistic and caricatured"* ← [Abstract:] "Participants meet three different doctors with **three different style levels: realistic, caricatured, and an in-between 'Mid' level**." **(2)** *"both as a direct rating and across the ability, benevolence and integrity dimensions"* ← [§1:] "Our study employs the trust model proposed by Mayer et al. which posits that trust requires people to be vulnerable and that trusting another relies on positive perceptions of their **ability, benevolence and integrity**." **(3)** *"participants were asked which doctor they would consult for a follow-up"* ← [Abstract:] participants "must imagine that they have received a cancer diagnosis and must **select a doctor for a second opinion**. . . After interviewing three doctors, they selected [one]." **(4)** *"self-reported trust ratings were broadly equivalent across all three style levels"* ← [Abstract:] "**Trust ratings are largely consistent across the style levels**." **(5)** *"the semi-realistic condition attracted the highest rate of behavioural selection"* ← [Abstract:] "but **participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion**."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Avatar_-Stylization_and_Trust_IEEE_VR_2024.pdf`, Abstract, Figure 1 and §3. **CORRECTED 2026-08-23:** the thesis previously said trust was measured across the *competence*, benevolence and integrity dimensions, and then added a sentence explaining that competence corresponds to what Mayer et al. term ability. Canales et al. do not use competence. They use Mayer own term, ability, 37 times against 3 for competence, and state the model as positive perceptions of ability, benevolence and integrity. The thesis now uses their term, and the competence mapping was moved to Methodology where that label first appears.

✔ **Validated** — I have read the original and agree it supports the sentence.

#52 gao2025 — Gao et al. (2025) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** This suggests that moderate stylisation can match or exceed the behavioural trust of full photorealism. **Gao et al. compared a stylised agent against recorded video of the real speaker it was modelled on.** *[The two did not differ on a direct trust rating, but the real speaker was rated higher on ability, benevolence and integrity. Overall, these studies indicate that a single self-report rating does not fully capture how render style shapes trust. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"compared a stylised agent against recorded video of the real speaker it was modelled on"* ← [§3.2:] "**For the Real style, we directly played the corresponding video**"; the stimuli are three TED talks and "**The three speakers of different ages correspond to the identities of the virtual agents** in our design." **(2)** *"The two did not differ on a direct trust rating"* ← [§4.2, Trust:] "Real style was perceived as more trustworthy than No-stylization ($p = 0.002$), but **there was no significant difference in trustworthiness between Real and Stylization ($p = 0.216$)**." **(3)** *"the real speaker was rated higher on ability, benevolence and integrity"* ← [§4.2:] "**Real style was perceived as more competent than Stylization ($p < 0.001$)**"; "**Real style was perceived as more benevolent than both Stylization ($p = 0.007$)**"; "**Real style was perceived as more integrous than both Stylization ($p = 0.015$)**".

How to check: Read directly from `Articles/02_trust_and_avatar_perception/IEEE_VR25_-TVCG.pdf` (Gao et al., *Trust in Virtual Agents: Exploring the Role of Stylization and Voice*), §§3.2 and 4.2. **REVISED 2026-08-23:** the sentence previously opened "Gao et al. removed that restriction", which pointed back at an Alimardani et al. sentence that has since been deleted. The opener is now self-contained. **Note the pattern:** trust as a single rating showed no difference between Real and Stylization, while all three component dimensions did. That dissociation is the same one this thesis relies on for measuring trust componentwise rather than as one score.

✔ **Validated** — I have read the original and agree it supports the sentence.

#53 dubosc2025 — Dubosc et al. (2025) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** Moreover, Dubosc et al. applied the same comparison to eeriness rather than trust. They conducted a VR face-to-face interaction study and found that stylised avatars were rated significantly less eerie than photorealistic avatars. **Trustworthiness also trended higher for the stylised avatars, though this difference did not reach statistical significance.**

Usage: First use of this source.

Original source: **(1)** *"a VR face-to-face interaction study"* ← [Title:] "**Effect of avatar stylization and facial expression intensity in virtual interactions**", *Virtual Reality* 29:163 (2025). **(2)** *"stylised avatars were rated significantly less eerie than photorealistic avatars"* ← [§4.2.3:] "The results show that the stylized character ($m = 2.750$, $SD = 1.256$, $p = .05$), and the semi-realistic character ($m = 2.650$, $SD = 1.331$, $p = .026$) were **perceived as significantly less eerie than the realistic character** ($m = 3.350$, $SD = 1.703$)." **(3)** *"Trustworthiness also trended higher. . . though this difference did not reach statistical significance"* ← [§4.2:] "The results suggest, **as a trend**, that the stylized character ($m = 5.450$) and the semi-realistic character ($m = 5.475$) were judged as more trustworthy than the realistic character ($m = 4.900$), $p = .075$."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Effect_of_avatar_stylization_and_facial_expression.pdf`, §4.2. **Scope note:** the *main effect* of stylisation on trustworthiness *was* significant ($F(2, 74) = 3.326$, $p = .041$, $\eta^2 = .043$); it is the *pairwise* comparison that did not reach significance ($p = .075$). The thesis sentence is correct about the comparison, but a reader may take it as meaning nothing was significant.

✔ **Validated** — I have read the original and agree it supports the sentence.

#54 cihodaru2025 — Cihodaru-Ştefanache & Podina (2025) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in the-sis: **[Block covered by this citation]** This pattern suggests that stylisation reduces the uncanny valley response without compromising social credibility. The mechanism behind that response has itself been reviewed. **Cihodaru-Ştefanache and Podina examined 29 empirical studies on embodied conversational agents and reported the perceptual mismatch hypothesis as its best supported explanation.** *[Uncanniness on that account arises when an agent's features are perceived at inconsistent levels of anthropomorphism. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"reviewed 29 empirical studies on embodied conversational agents"* ← [§2 Method:] "resulted in the **inclusion of 29 studies** that rigorously met the [criteria]"; confirmed again in the results, "subjective measurements (**29 papers**, 100%)". The title names the domain: "The uncanny valley effect in embodied conversational agents". **(2)** *"the perceptual mismatch hypothesis as the best supported explanation"* ← "the **perceptual mismatch hypothesis has received strong empirical support**"; and in the summary of hypotheses, "**support for perceptual mismatch**" is recorded as the outcome.

How to check: Read directly from `Articles/01_uncanny_valley/The_uncanny_valley_effect_-in_embodied_conversation.pdf`, Method and Results.

✔ **Validated** — I have read the original and agree it supports the sentence.

#55 bartneck2009 — Bartneck et al. (2009) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in the-sis: **[Block covered by this citation]** Testing this pattern reliably requires measurement across more than a single trust rating. **The Godspeed Questionnaire Series, developed by Bartneck et al., offers one such option.** *[Even though the series was originally developed for human-robot interaction, it has since been widely adapted for virtual avatar studies. sets out its structure and the subscale adopted in this work. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *"five perceptual dimensions"* ← [Title:] "Measurement Instruments for the **Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety** of Robots"; and [§1:] "key concepts in HRI: **anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety**". **(2)** *"semantic differential scales"* ← the series is presented throughout as semantic differential items anchored by opposed adjective pairs.

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Bartneck_2009_-Measurement Instruments for the Anthropomorphism.pdf`, title and §1.

✔ **Validated** — I have read the original and agree it supports the sentence.

#56 bartneck2009 — Bartneck et al. (2009) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in the-sis: **[Block covered by this citation] On the other hand, none of the Godspeed subscales measures eeriness.**

Usage: Use 3 of this source.

Original source: **(1)** *"Eeriness is not among its five dimensions"* ← the five are named exhaustively in the title and §1: **anthropomorphism, animacy, likeability, perceived intelligence and perceived safety**. Eeriness appears in none of them. Ho and MacDorman make the same observation when motivating their own instrument, noting that the Godspeed indices lack discriminant validity and that an alternative set was needed.

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Bartneck_2009_-Measurement Instruments for the Anthropomorphism.pdf`. This is a claim of *absence*, so the support is the exhaustive list of the five dimensions rather than a positive statement.

Validated — I have read the original and agree it supports the sentence.

#57 macdorman2006 — MacDorman & Ishiguro (2006) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in the-sis: **[Block covered by this citation] MacDorman and Ishiguro had measured that response directly, on a scale separate from human likeness and familiarity.**

Usage: Use 4 of this source.

Original source: **(1)** *"measured that response directly, on a scale separate from human likeness and familiarity"* ← [§Procedure, p. 306:] participants rated the images "on a **nine-point scale of human likeness and familiarity**"; then, separately, "**The same 31 images were presented again**. . . participants were asked to rate how eerie each image was on a **ten-point scale**, ranging from slightly eerie to extremely eerie." The eeriness scale differs in both administration and range, which is what makes it separate.

How to check: Read directly from `Articles/01_uncanny_-valley/MacDorman2006AndroidScience.pdf`, §Procedure, p. 306.

✔ **Validated** — I have read the original and agree it supports the sentence.

#58 ho2010 — Ho & MacDorman (2010) *Ch.3 Related Work → Trust and Credibility in Virtual Agents*

Sentence in thesis: **[Block covered by this citation]** Ho and MacDorman later refined this into a validated instrument with four indices: perceived humanness, warmth, eeriness and attractiveness. **Their validation established that eeriness and human likeness are orthogonal.** *[Separating the two matters for interpreting any reduction in eeriness, since a drop could reflect reduced discomfort or merely reduced human likeness. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *"eeriness and human likeness are orthogonal"* ← [Abstract:] "Phase 2 of this study developed alternative indices with **non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$).**" **(2)** *"four indices: perceived humanness, warmth, eeriness and attractiveness"* ← [§Phase 2:] "an alternative set of indices is developed to measure participants' attitudes toward anthropomorphic characters: **perceived humanness, warmth, eeriness, and attractiveness**" — **four** indices, not three, and the paper adds that "**the first three indices** are motivated by the original graph of the uncanny valley", i.e. humanness, **warmth** and eeriness. Attractiveness is the fourth.

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Ho2010UncannyValleyIndices.pdf`, Abstract and Phase 2. **ERROR FOUND AND CORRECTED 2026-08-23.** The orthogonality claim is exact ($r = .02$, $p > .05$). The index list was wrong and has been corrected: the instrument has **four** indices, perceived humanness, warmth, eeriness and attractiveness. The omitted one was warmth, not attractiveness, since the paper notes the first three indices, humanness, warmth and eeriness, are those motivated by Mori original graph.

✔ **Validated** — I have read the original and agree it supports the sentence.

#59 kumar2019 — Pavan Kumar et al. (2019) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation] figure[The four-family classification of NPR techniques surveyed by Pavan Kumar et al.]The four-family classification of NPR techniques surveyed by Pavan Kumar et al.: stroke-based rendering, image analogy, region-based techniques and image filtering.**

Usage: First use of this source.

Original source: **(1)** *the four families named in the caption* ← [§2:] "**1. Stroke-based rendering. 2. Image analogy. 3. Region-based techniques. 4. Image processing and filtering techniques.**" The figure reproduces this classification.

How to check: Read directly from `Articles/03_npr_rendering/Kumar- A comprehensive survey on NPR.pdf`, §2. Same caveat as the prose entry: the fourth family is titled *image processing and filtering techniques* in the source, and the taxonomy is adopted from earlier surveys rather than originated by Kumar et al.

✔ **Validated** — I have read the original and agree it supports the sentence.

#60 kumar2019 — Pavan Kumar et al. (2019) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** NPR transforms an image into abstracted, artistic or stylised output instead of reproducing it exactly. **Pavan Kumar et al. surveyed these approaches under four established categories: stroke-based rendering, image analogy, region-based techniques and image filtering.**

Usage: Use 2 of this source.

Original source: **(1)** *"four categories: stroke-based rendering, image analogy, region-based techniques and image filtering"* ← [§2 Classifications of NPR techniques:] "NPR techniques have been classified to fall into following categories. . . **1. Stroke-based rendering. 2. Image analogy. 3. Region-based techniques. 4. Image processing and filtering techniques.**" The abstract lists the same four: "the various works done on **stroke-based rendering, color image analogy, region-based rendering, image filtering abstraction, and stylization**."

How to check: Read directly from `Articles/03_npr_rendering/Kumar- A comprehensive survey on NPR.pdf`, §2 and Abstract. **Two notes.** (a) Kumar names the fourth family *image processing and filtering techniques*; the thesis shortens this to *image filtering* and uses that shorthand consistently. (b) Kumar does not originate the taxonomy. The same passage states that the categories are given "*based on the work entitled State of the Art: a taxonomy of artistic stylization techniques for images and video*" and two further sources, so Kumar is adopting and surveying an existing classification rather than proposing one. The thesis verb *classified* slightly overstates that contribution; *group* or *adopt* would be exact.

✔ **Validated** — I have read the original and agree it supports the sentence.

#61 kumar2019 — Pavan Kumar et al. (2019) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** Stroke-based rendering builds an image from discrete brush strokes that vary in size, colour, texture and orientation. Placing these strokes typically requires a dedicated analysis pass to decide where and how each stroke should be oriented. That requirement is a drawback for pipelines with only one shading stage per pixel. Image analogy instead transfers colour and texture style from an example pair of source and target images onto a new image. This transfer depends on a precomputed exemplar or a patch-matching search rather than information available within a single frame. By comparison, region-based techniques segment an image into distinct regions before abstracting each one individually. Segmentation is an iterative process that considers the whole image. Its result cannot be computed within a single, independently executed per-pixel operation. **Finally, image filtering applies local mathematical operations, such as gradients or bilateral smoothing, directly to the image to suppress detail while preserving structure.** *[These operations examine only a small neighbourhood around each pixel. As a result, they remain computable within a single shading pass. summarises this classification. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *stroke-based rendering* ← [§2.1:] the family is built from discrete brush strokes whose placement requires a dedicated analysis stage. **(2)** *image analogy* ← [§2.2:] style is transferred from an example pair of source and target images onto a new image. **(3)** *region-based techniques* ← [§2.3:] the image is segmented into regions that are then abstracted individually. **(4)** *image filtering* ← [§2.4 Image processing and filtering:] "**Image processing and filtering techniques are used to develop image abstraction and stylization**. . . Image abstraction is the task of producing simplified images from given images by **removing unwanted details while retaining meaningful information** in it."

How to check: Read directly from `Articles/03_npr_rendering/Kumar- A comprehensive survey on NPR.pdf`, §§2.1–2.4. **Citation added 2026-08-24.** The paragraph defines all four families in turn, but only the opening list carried a citation, so the four definitions that followed it stood bare. A second citation of the same source now closes the run of definitions. The sentences interleaved between them, each one drawing out what a family costs a single shading pass, are the thesis's own reasoning and are deliberately left uncited.

✔ **Validated** — I have read the original and agree it supports the sentence.

#62 lake2000 — Lake et al. (2000) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** However, this survey's scope is limited to 2D image processing. Image filtering crosses that boundary where the other three do not, since each of them requires a pass that a single shading stage cannot provide. Other real-time techniques fall outside Pavan Kumar et al.'s four families, because they act on geometry and lighting rather than on a finished image. Toon shading is one such technique. It quantises the continuous diffuse lighting term into a small number of discrete bands. Consequently, this approach produces flat-shaded regions separated by hard edges and gives characters a cartoon appearance, as depicted in . **That separation depends on a single input, the lighting term.** *[This dependency leaves the ramp unable to respond to any other surface property. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"quantises the continuous diffuse lighting term"* ← [§3.1 Cartoon Shading:] the diffuse term of Equation 1, $(\max\{L \cdot n, 0\})$, supplies the texture coordinate. Figure 2 caption: "**Generation of texture coordinates from** $L \cdot n$. In this case, the shadow boundary occurs at the point where $L \cdot n$ equals 0.5." **(2)** *"into a small number of discrete bands"* ← [§3.1:] "the cartoonist typically uses **solid colors that do not vary across the materials they represent**. . . our hard shading technique **finds a transition boundary and shades each side of the boundary with a solid color**." The algorithm stores, for each material, "a **one-dimensional texture map**" built from "a texture map of a **minimal number of colors**". **(3)** *"flat-shaded regions separated by hard edges. . . a cartoon appearance"* ← [§3.1:] "The boundary between shadowed and illuminated colors is a **hard edge** that follows the contours of the object or character. We refer to this technique as **hard shading**." And: "Our system allows a 3D model to be drawn to look **just like a traditional 2D cartoon**, and to be viewed interactively." **(4)** *"That separation depends on a single input, the lighting term"* ← [§3.1:] "The boundary between lit and shadowed colors on the model **depends, just as the color calculated in smooth shading, on the cosine of the angle between the light and normal vectors**."

How to check: Read directly from `Articles/03_npr_rendering/npar2000_lake_et_al.pdf`, §3.1 and Figure 2. **REPLACED 2026-08-24.** This entry previously read only "See entry #N for the source passage", pointing at itself, so it carried no supporting quote at all. **Citation moved 2026-08-24:** it previously closed on the quantisation sentence, leaving the two sentences that follow, both Lake findings, without support. It now closes the block. **Overstatement corrected:** the thesis previously said artists have little independent influence over the boundary placement. Lake states the opposite, that a higher-resolution texture can provide flexibility in the location of the shadow boundary. The sentence now says the ramp cannot respond to any property other than the lighting, which is what the dependence on $L \cdot n$ implies and is the premise Barla et al. address.

✔ **Validated** — I have read the original and agree it supports the sentence.

#63 lake2000 — Lake et al. (2000) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation] Toon shading**

Usage: Use 2 of this source.

Original source: **(1)** *the subfigure label "Toon shading"* ← [§3.1 Cartoon Shading:] "Rather than shading the character to give it a three-dimensional appearance, the cartoonist typically uses **solid colors that do not vary across the materials they represent**. . . our hard shading technique finds a transition boundary and shades each side of the boundary with a solid color." The flat bands separated by hard edges visible in the figure are exactly this effect.

How to check: Read directly from `Articles/03_npr_rendering/npar2000_lake_et_al.pdf`, §3.1. The subfigure carries only a two-word label, so the quote supports the technique it names rather than a sentence.

✔ **Validated** — I have read the original and agree it supports the sentence.

#64 barla2006 — Barla et al. (2006) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation] The X-Toon ramp**

Usage: First use of this source.

Original source: **(1)** *the subfigure label "The X-Toon ramp"* ← [§1 Introduction:] "Traditional toon shading uses a **1D texture** that describes how tone [varies]. . . First, we **replace the 1D texture with a 2D texture whose second dimension corresponds to the desired "tone detail"**, which can vary with **depth or surface orientation**." Restated in §2: "we replace the conventional 1D texture used in toon shading with a **2D texture, where the second dimension corresponds to desired tone "detail"**." The first axis remains the diffuse term, since the conventional 1D texture is indexed by the shading value that "describes how the final shading varies from 'dark' to 'light' regions".

How to check: Read directly from `Articles/03_npr_rendering/x-toon.pdf`, §§1–2. **REPLACED 2026-08-24.** This entry previously read only "See entry #N for the source passage", pointing at itself, so it carried no supporting quote at all.

✔ **Validated** — I have read the original and agree it supports the sentence.

#65 raskar1999 — Raskar & Cohen (1999) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] Silhouette rendering: (a) front-facing polygons, (b) back-facing polygons translated towards the camera, (c) the resulting outline.**

Usage: First use of this source.

Original source: **(1)** *"back-facing polygons translated towards the camera"* ← [§3:] the two options given are to "**translate back-facing polygons towards the camera by a fixed amount,** t" or to "**translate back-facing polygons towards the camera by** $k \cdot z$, where z is the average distance from polygon to the camera and k is a scaling constant." **(2)** *the three panels: front-facing, back-facing, resulting outline* ← [Figure 1 caption:] "The first layer, P1, for scene (a) is made up of parts of **nearest front facing polygons** (b). The second layer, P2, is made up of parts of **back facing polygons** (c). The **intersection of these two layers in image space creates silhouette edges** (d)."

How to check: Read directly from `Articles/03_npr_rendering/Raskar -Image Precision Silhouette Edges.pdf`, §3 and Figure 1. **REPLACED 2026-08-24:** this entry and the prose entry previously shared one quote, so a single passage was standing in for two different claims. Each now has its own.

✔ **Validated** — I have read the original and agree it supports the sentence.

#66 barla2006 — Barla et al. (2006) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** Barla et al. extended toon shading's quantisation into a richer parameter space with X-Toon. **It retains the lighting term on one axis of the ramp and adds a second axis for tone detail, driven by a view-dependent attribute such as depth or surface orientation.** *[shows the resulting 2D ramp. This enables a 2D lookup that encodes artistic intent unavailable in a 1D quantisation step. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *"extended toon shading's quantisation into a richer parameter space"* ← [Abstract:] "Traditional toon shading uses a **1D texture** that describes how tone varies with surface orientation relative to a given light source. In this paper we describe two extensions to the basic algorithm that support view-dependent effects." **(2)** *"retains the lighting term on one axis. . . adds a second axis for tone detail, driven by a view-dependent attribute such as depth or surface orientation"* ← [Abstract:] "First, we **replace the 1D texture with a 2D texture whose second dimension corresponds to the desired "tone detail," which can vary with depth or surface orientation**." The first axis is unchanged, since the conventional 1D texture it replaces is the one indexed by surface orientation relative to the light.

How to check: Read directly from `Articles/03_npr_rendering/x-toon.pdf`, Abstract. **Citation moved 2026-08-24:** it previously closed on the first sentence, leaving the description of the two axes, which is near-verbatim from the abstract, without support. **Scope note:** Barla et al. describe **two** extensions. The thesis covers only the first, the 2D ramp. The second uses a modified normal to support backlighting and plastic or metallic highlights, and is not used in this work.

✔ **Validated** — I have read the original and agree it supports the sentence.

#67 raskar1999 — Raskar & Cohen (1999) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** Silhouette rendering is a second such technique. It derives line structure from geometry rather than from the lighting term. Raskar and Cohen achieved this by processing the mesh twice from the same viewpoint. Initially, the front-facing polygons are drawn in the usual way. The back-facing polygons are then rendered and translated towards the camera. Thus, part of this back-facing geometry appears in front of the occluding front-facing surface and the visible rim left at this boundary forms the outline, as illustrates. **The outline emerges from geometry alone and requires no precomputed adjacency information.**

Usage: Use 2 of this source.

Original source: **(1)** *the adjacency claim, stated by the authors as a headline property* ← [Abstract, **PDF p. 1**, final sentence before §1:] "A solution is suggested using view-dependent modification of polygonal objects. **The method does not require any preprocessing or adjacency information and hence is applicable for dynamic scenes.**" **(2)** *the same claim restated* ← [§1 Introduction, **PDF p. 1**:] "The method is robust, general purpose and easy to implement. **No pre-processing or adjacency information is required.** This allows use of silhouette edges in **dynamic scenes, models with changing topology**. . . " **(3)** *that the outline is produced by geometry rather than by the shading term* ← [§3, algorithm listing:] the outline appears purely from rendering front faces, then back faces under front-face culling with an equal-to depth test. No edge detection step occurs anywhere in the algorithm.

How to check: Read directly from `Articles/03_npr_rendering/Raskar -Image Precision Silhouette Edges.pdf`, PDF p. 1. **WARNING FOR ANYONE RECHECKING THIS: "adjacency information" appears four times on PDF p. 1 and only the first two are Raskar and Cohen's claim about their own method.** The other two describe other people's work: "models with known adjacency information" refers to **Markosian et al.**, a method that does require it, and "The method in [Rossignac92] is image precision and does not need adjacency information" refers to **Rossignac**. The abstract sentence in (1) is the one to check against. **FALSE CLAIM REPLACED 2026-08-25, found by Nidanur.** The sentence previously read "The outline emerges from geometry alone and *requires no analysis of the rendered image*. That second half is not true of this method and no source supports it. Raskar and Cohen's paper is titled ***Image Precision*** *Silhouette Edges*, its abstract describes computing surfaces "**in image space** at interactive rates" with "all operations. . . performed in **image-precision**", and the outline only resolves because the depth buffer decides which fattened back-faces show through. Isenberg et al. file the whole family under **Hybrid algorithms**: "operations in object space that are followed by rendering the modified polygons. . . using the z-buffer. . . The result is similar to image space algorithms." **Citation moved in the same pass.** It previously closed one sentence earlier. The adjacency claim is Raskar and Cohen's rather than the thesis's summary, so it cannot sit outside the cited block, and the citation now closes on it. The preceding sentence loses nothing: the figure it points to carries `raskar1999` in its own caption. **Why this fact rather than another.** Raskar and Cohen state it because no adjacency data means silhouettes work on "dynamic scenes, models with changing topology". That is exactly the constraint an animated avatar imposes, so the sentence now connects forward to the Methodology selection rationale instead of asserting a property the method does not have. **The same overreach was removed from Methodology on the same day.** V1 had read "eliminating the need for edge detection or image-space analysis", which duplicated the preceding sentence and made the same

✔ **Validated** — I have read the original and agree it supports the sentence.

#68 isenberg2003 — Isenberg et al. (2003) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** That outline is only one of several line types that convey shape. In order to detect those lines, Isenberg et al. surveyed the available algorithms and reported that the literature groups silhouette lines together with creases, borders and self-intersections under the heading of feature lines. **The silhouette lines themselves split further, and the survey calls the interior ones internal silhouettes.** *[This distinction matters for avatar rendering, where expressive detail such as facial features, clothing folds and skin creases sits inside the outline. However, lines that appear only in the texture fall outside this classification, since each type it names is defined on the mesh geometry. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"grouped silhouette lines together with creases, borders and self-intersections under the heading of feature lines"* ← [§Definitions and terminology, journal p. 28:] "Other lines significant in the context of silhouettes are **creases, borders, and self-intersections**. . . In the literature, these three additional line categories **together with the silhouette lines are often called feature lines**." **(2)** *"The silhouette lines themselves split further, and the survey calls the interior ones internal silhouettes"* ← [same section:] "We define a **contour as the subset of the silhouette that separates the object from the background**. Also, **the subset of the silhouette, which in 2D divides one portion of the object from another one of the same object, is not part of the contour**. Those lines are sometimes called **internal silhouettes**."

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, journal p. 28. **Error found and corrected 2026-08-24:** the thesis previously read "Internal silhouettes *and creases* fall inside the projected outline. The internal-silhouette half is a fair restatement of the source, but the crease half is not supported. Isenberg defines a crease purely by dihedral angle, "edges that should always be drawn, for instance the edges of a cube", and says nothing about where a crease lies relative to the projected outline. Several edges of a cube lie *on* that outline, so the claim is not even generally true. **Also removed:** the separate sentence "Only the contour separates the object from the background", which restated the definition without adding anything. The contrast it carried is now implicit in "the interior ones. **Citation moved in the same pass:** it previously closed on the feature-lines sentence, leaving two further Isenberg claims uncited after it. It now closes the block, and the final sentence about avatar rendering is the thesis's own inference and stays uncited. **Scope note:** the survey's three extra categories, creases, borders and self-intersections, are defined by geometric cause, not by position inside or outside the outline. Only the contour-versus-internal-silhouette split is an inside/outside one, and that one the source states explicitly. **Attribution corrected 2026-08-24, closing an item left open since the entry was first written.** The sentence read "Isenberg et al. surveyed the available algorithms and *grouped* silhouette lines. . . under the heading of feature lines", which presented the grouping as their own act of classification. The source says these categories "**in the literature**. . . **are often called** feature lines", so the survey reports an existing convention rather than proposing one. The sentence now reads "reported that the literature groups. . . . This is the same correction already applied to Pavan Kumar et al., whose taxonomy is likewise stated to be based on three earlier works.

✔ **Validated** — I have read the original and agree it supports the sentence.

#69 isenberg2003 — Isenberg et al. (2003) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** The same survey identifies a further family of algorithms that addresses this limitation by working on the rasterised output rather than on the mesh. **These image-space methods, an instance of the image filtering category described above, extract discontinuities from the buffers that conventional rendering already produces and represent the result as features in a pixel matrix.**

Usage: Use 2 of this source.

Original source: "[Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods. These methods provide a silhouette represented as features in a pixel matrix.""

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, p. 2 of the PDF (journal p. 29). Both halves of the thesis sentence map onto this passage: "extract discontinuities from the buffers" to the first source sentence, "features in a pixel matrix" to the second. **Scope note:** the following thesis sentences about edge thickness scaling with camera distance are deliberately left uncited. Isenberg does not make that claim anywhere in the survey. Its general stated disadvantage (p. 30) is that "the user has little control over the resulting lines' attributes", and the "limitation to pixel accuracy" remark (p. 35) concerns silhouette *visibility* determination, not edge width. The distance-scaling behaviour is an observation from this project.

✔ **Validated** — I have read the original and agree it supports the sentence.

#70 saitotakahashi1990 — Saito & Takahashi (1990) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** Saito and Takahashi introduced one such buffer, the G-buffer. It is a set of channels that preserves per-pixel geometric properties such as depth and surface normals. The authors recovered profile and internal edges from the depth channel. **Sobel was recommended for the profile, and a second-order operator produced the internal edges.**

Usage: First use of this source.

Original source: **(1)** *"profile and internal edges. . . from the depth channel"* ← [§3.2 Drawing Edges:] "**profile** — the border line of an object on the screen; **internal edge** — a line where two faces meet. Profiles and edges are the **0-th and first order discontinuities of the depth image (sz image)** respectively." Restated in §4: "The **profile image (pr) was obtained from the sz image**." **(2)** *"Sobel was recommended for the profile"* ← [§3.1:] "Discontinuities of an image can be extracted with a **first order differential operator**. . . however, **we recommend Sobel's**." Figure 3 shows the depth image passing through the first order differential to give the profile image. **(3)** *"a second order operator produced the internal edges"* ← [§3.1:] "Discontinuities of the first order differentials of an image can be extracted with a **second order differential operator**. For this calculation, we recommend the following operator: $I = (8X - A - B - C - D - E - F - G - H)/3$." Figure 3 shows that branch producing the internal edge image.

How to check: Read directly from `Articles/03_npr_rendering/Comprehensible Rendering of 3-D Shapes Saito.pdf`, §§3.1–3.2 and Figure 3. **Precision corrected 2026-08-24.** The sentence read "recovered profile and internal edges from the depth channel and recommended the Sobel operator *for the task*", which reads as though Sobel produced both. It did not. Sobel is the first order operator and yields the *profile*; the internal edges come from the second order operator of Eq. 2. The sentence was split so each operator is matched to the edge type it actually produces. **Passage not to confuse with these:** §3.3, on contour lines of a scalar field, prefers its own Eq. 9 and states that an operator "is recommended rather than the Sobel Operator. Different operation, no contradiction.

✔ **Validated** — I have read the original and agree it supports the sentence.

#71 nienhaus2003 — Nienhaus & Döllner (2003) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] Nienhaus and Döllner later applied the same analysis to the normal channels and detected crease edges from their discontinuities in real time.**

Usage: First use of this source.

Original source: **(1)** *"applied the same analysis to the normal channels"* ← [§3:] "A **normal-buffer is a three-channel image** containing encoding normalized x, y, and z coordinates of normals derived from fragments resulting from 3D scene objects." The lineage is stated in §2: "**Saito and Takahashi introduced the G-buffer**", and §4.1.2 reuses their operator: "**Saito and Takahashi as well as Decaudin propose the following C1-discontinuity operator** to detect discontinuities in the z-buffer". **(2)** *"detected crease edges from their discontinuities"* ← [§3:] "Abrupt changes in the normal-buffer occur typically at crease edges. . . **0th order discontinuities of the normal-buffer indicate crease edges**." **(3)** *"in real time"* ← [Title:] "Edge-Enhancement — **An Algorithm for Real-Time Non-Photorealistic Rendering**." [Abstract:] "we propose an algorithm for enhancing edges of **real-time** non-photorealistic renderings."

How to check: Read directly from `Articles/03_npr_rendering/Edge Enhancement.pdf`, §§2–4. **Scope note:** the same passage adds that normal-buffer discontinuities also indicate "profile edges if the normal of the adjacent front-facing polygon varies from the normal of an underlying polygon", so crease edges are the principal but not the only outcome. The thesis claims only the crease edges, so it is narrower than the source rather than broader, which is the safe direction.

✔ **Validated** — I have read the original and agree it supports the sentence.

#72 isenberg2003 — Isenberg et al. (2003) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] Isenberg et al. favoured the combination of both channels on visual grounds.**

Usage: Use 3 of this source.

Original source: **(1)** *"combining both channels yields a pleasing result"* ← [journal p. 29:] "Hertzmann extends this method by using a **normal buffer** instead. This can also detect C1 discontinuities. **Combining both approaches yields a pleasing result** (see Figure 2)." [Figure 2 caption:] "(a) z-buffer, (b) edges extracted from the z-buffer, (c) normal buffer, (d) edges extracted from the normal buffer, and **(e) combination of both edge buffers**."

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, journal p. 29 and Figure 2. **Wording corrected twice, 2026-08-24:** the thesis first read "the combination of both channels as the *stronger result*", a firmer judgement than the survey makes. The first correction swapped in the survey's own phrase, "yields a pleasing result", which fixed the overstatement but reproduced the source verbatim without quotation marks. The sentence now paraphrases instead: "favour. . . on visual grounds" carries the same soft, aesthetic endorsement the survey makes, in the thesis's own words. **Attribution note:** the survey credits the normal-buffer extension to Hertzmann rather than to Nienhaus. This does not affect the preceding thesis sentence, since that one cites Nienhaus and Döllner's own paper for their own work.

✔ **Validated** — I have read the original and agree it supports the sentence.

#73 isenberg2003 — Isenberg et al. (2003) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** However, neither channel registers a boundary that exists only as colour, since a change of pigment leaves depth and surface orientation alike untouched. **Colour-buffer edges are treated in the survey as a liability, because shading and texture variation cause detections that are not silhouettes.**

Usage: Use 4 of this source.

Original source: **(1)** *"Colour-buffer edges are treated in the survey as a liability, because shading and texture variation cause detections that are not silhouettes"* ← [§Image space algorithms, journal p. 29:] "The easiest way to find significant lines would be to **detect edges in the color buffer**. This, however, **doesn't necessarily detect silhouettes since changes in shading and texturing can cause erroneous edge detection**." The survey's preference follows immediately: "Saito and Takahashi suggest **using the z-buffer instead**. . . This has the advantage of **only finding object-relevant edges such as silhouette lines including contours**."

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, journal p. 29. **REPLACED 2026-08-24.** This entry previously carried the pixel-matrix quote copied from the earlier image-space entry, which says nothing about the colour buffer and therefore did not support this sentence at all. The passage above is the one the sentence is actually drawn from, and it matches closely, including the shading-and-texturing cause.

✔ **Validated** — I have read the original and agree it supports the sentence.

#74 gonzalez2018 — Gonzalez & Woods (2018) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** That liability becomes both an objective and a challenge once the target is a stylised avatar rather than a silhouette. Clothing patterns, facial contours and painted surface detail are the features that need marking here. Detecting them in the colour channel requires a gradient operator. **The Sobel filter is preferable for that purpose, because its kernel suppresses noise better than the simpler alternatives.**

Usage: Use 5 of this source.

Original source: **(1)** *"requires a gradient operator"* ← [§10.2:] "the three steps performed typically for edge detection are: 1. Image smoothing for noise reduction. . . 2. **Detection of edge points**. . . 3. Edge localization." **(2)** *"the Sobel filter"* ← [Eqs. 10-24 and 10-25:] "**These kernels are called the Sobel operators** (Sobel [1970])." **(3)** *"is preferable here, because its kernel suppresses noise better than the simpler alternatives"* ← [same page:] "The **Prewitt kernels are simpler to implement** than the Sobel kernels, but the slight computational difference between them typically is not an issue. The fact that the **Sobel kernels have better noise-suppression (smoothing) characteristics makes them preferable** because . . . noise suppression is an important issue when dealing with derivatives." The smoothing comes from the kernel itself: "using a 2 in the center location provides image smoothing."

How to check: Read directly from `Articles/05_image_processing/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`, §10.2 (4th ed., printed pp. 714, 720–722). **Wording corrected 2026-08-24.** The sentence read "for which the Sobel filter is *the standard choice*". The book never says standard; it says *preferable*, and the comparison it draws is against the Prewitt kernels specifically, on the ground of noise suppression. It reserves "widely used" for the Canny detector. The sentence now uses the book's own judgement and states the reason it gives, and "the simpler alternatives" is the source's own description of Prewitt.

✔ **Validated** — I have read the original and agree it supports the sentence.

#75 marr1980 — Marr & Hildreth (1980) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** Yet this operator responds to high-frequency content as strongly as to the boundaries it is meant to mark. **Marr and Hildreth placed a Gaussian ahead of the derivative operator.** *[Therefore, prefiltering of this kind is the usual precaution against that sensitivity. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *that smoothing precedes differentiation* ← [§2:] "one **first takes local averages of the image at various resolutions and then detects the changes in intensity** that occur at each one. To realize this idea, we need to determine (a) **the nature of the optimal smoothing filter**, and (b) how to detect intensity changes at a given scale." **(2)** *that the smoothing filter is the Gaussian* ← [§2, The optimal smoothing filter:] "these two localization requirements, the one in the spatial and the other in the frequency domain, are conflicting. . . There is, moreover, **only one distribution that optimizes this relation** (Leipnik 1960), **namely the Gaussian**." [Abstract:] "the filter for this purpose at a given scale is found to be the **second derivative of a Gaussian**."

How to check: Read directly from `Articles/MarrHildreth (2).pdf`, Abstract and §2 (Proc. R. Soc. Lond. B 207:187–217, printed pp. 190–192). **Finding, 2026-08-24 — the mechanism is supported but the stated reason is not.** That Marr and Hildreth put a Gaussian in front of the derivative is beyond doubt. Their *reason* for it is not noise. The paper argues from two premises: that "the motivation for filtering the image is **to reduce the range of scales** over which intensity changes take place", and that the Gaussian uniquely optimises the uncertainty relation between spatial and frequency localisation. The word *noise* appears three times in the whole paper, never as a motivation for the smoothing step. The noise framing carried by "the usual precaution" belongs to Gonzalez and Woods, whose §10.2 opens edge detection with "1. Image smoothing for **noise reduction**", and to Canny. **Corrected 2026-08-24.** The sentence previously read "Marr and Hildreth established Gaussian prefiltering as the usual precaution", which handed the noise rationale to the one source that does not argue from noise. It is now split. The cited sentence states only what the paper does, which the quotes above support in full. The claim about usual practice moved to the following sentence and carries no citation, since it describes what the field does today rather than anything a 1980 paper asserts.

✔ **Validated** — I have read the original and agree it supports the sentence.

#76 roshaan2026 — Roshaan (2026) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** Nonetheless, a single gradient operator produces noisy, discontinuous edges even after prefiltering. A preprint by Lino Roshaan addressed this by splitting edge detection into three layers. Depth differences give the silhouette layer. The Scharr and Kirsch gradient operators trace the structure layer. An extended DoG (XDoG) operator recovers the texture layer. Max-pooling merges the three. shows each one alongside the composited result. A luminance-driven gain raises the structure layer's sensitivity in dark regions and holds line quality steady without manual retuning. **The framework runs as a real-time post-processing shader and needs access to both the colour and the depth buffer.**

Usage: First use of this source.

Original source: **(1)** *the three layers and their operators* ← [§2.2, Table 1:] Silhouette (W_1) → **Depth Difference**; Structure (W_2) → **Scharr, Kirsch**; Texture (W_3) → **XDoG**. **(2)** *"Max-pooling merges the three"* ← [§2.3, Eq. 5:] $\mathrm{Edge}_{\mathrm{final}} = \max\{W_1 \cdot \mathrm{Depth},\ W_2 \cdot \max(\mathrm{Scharr}, \mathrm{Kirsch}),\ W_3 \cdot \mathrm{XDoG}\}$. **(3)** *"A luminance-driven gain… holds line quality steady"* ← [Contributions:] "an **adaptive sensitivity mechanism**… dynamically modulated the edge strength based on **scene luminance** and depth, **ensuring consistent line quality**." **(4)** *"runs as a real-time post-processing shader"* ← [Abstract:] "a unified, **real-time post-processing shader architecture**". The depth requirement follows from the silhouette layer, which "leverages the depth buffer".

How to check: Read directly from `Articles/03_npr_rendering/Adaptive Hierarchical Edge Detection.pdf`, Abstract, Contributions and §§2.1–2.3. **Citation moved 2026-08-24, the largest placement fix in the thesis.** It previously closed on the second sentence, leaving *six* further Roshaan claims uncited: the three layers, the operator behind each, the max-pooling fusion, the luminance gain, and the buffer requirement. It now closes the block, and every one of those claims is quoted above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#77 roshaan2026 — Roshaan (2026) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] The three-layer edge hierarchy of the AHEAD framework: (a) the source frame and (b) the composited result, above the three layers that produce it. (c) The silhouette layer comes from depth differences, (d) the structure layer from gradient operators and (e) the texture layer from the extended DoG operator.**

Usage: Use 2 of this source.

Original source: **(1)** *which operator feeds which layer* ← [§2.2, Table 1 "Hierarchical Classification of Algorithms":] **Silhouette (W_1) → Depth Difference**; **Structure (W_2) → Scharr, Kirsch**; **Texture (W_3) → XDoG**. **(2)** *"the silhouette layer comes from depth differences"* ← [§2.1.4 Depth Difference:] "This method leverages the **depth buffer** to isolate object boundaries… **By detecting discontinuities in depth values, we can specifically target the geometric "silhouette" of objects**." **(3)** *"the composited result"* ← [§2.3, Eq. 5:] $\mathrm{Edge}_{\mathrm{final}} = \max\{W_1 \cdot \mathrm{Depth}_{\mathrm{edge}},\ W_2 \cdot \max(\mathrm{Scharr}_{\mathrm{edge}}, \mathrm{Kirsch}_{\mathrm{edge}}),\ W_3 \cdot \mathrm{XDoG}_{\mathrm{edge}}\}$.

How to check: Read directly from `Articles/03_npr_rendering/Adaptive Hierarchical Edge Detection.pdf`, §§2.1.4–2.3 and Table 1. **REPLACED 2026-08-24:** this caption entry and the prose entry above it previously shared one quote from the abstract, so a single passage was standing in for two different claims. The caption names the operator behind each layer, which the abstract does not do, so it is now sourced to Table 1 and the per-operator sections instead.

✔ **Validated** — I have read the original and agree it supports the sentence.

#78 kumar2019 — Pavan Kumar et al. (2019) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** However, neither NPR abstraction nor image filtering is confined to feature detection. Another class of operators, such as the Kuwahara filter, reshapes the texture itself rather than outlining the structures within it. Pavan Kumar et al. grouped this filter with anisotropic and bilateral filtering as NPR filtering techniques. **It reduces shape, edge and colour detail in a single pass, and the prominent features of the image remain intact.**

Usage: Use 4 of this source.

Original source: **(1)** *the grouping with anisotropic and bilateral filtering* ← [Introduction:] "Existence of several **NPR filtering techniques such as Anisotropic filtering, Kuwahara filter**, Guided filter, Flow-based difference of Gaussian (FDoG) filtering, Structure tensor flow, **flow-based bilateral filter (FBL)**. . . " **(2)** *what the filter does* ← [§2.4, printed p. 150:] "**Kuwahara filter simplifies shape, edge, and color simultaneously while preserving important features**."

How to check: Read directly from `Articles/03_npr_rendering/Kumar- A comprehensive survey on NPR.pdf`, Introduction and §2.4. **Verbatim lift found and corrected 2026-08-24.** The thesis sentence read "It simplifies shape, edge and colour at once while preserving important features", which reproduces the source sentence above with only *simultaneously* changed to *at once* and the British spelling of colour. Quoted material that is not marked as a quotation must be reworded. It now reads "It reduces shape, edge and colour detail in a single pass, and the prominent features of the image remain intact", which states the same two claims, the simultaneous action on all three properties and the retention of salient structure, in the thesis's own words. The longest run shared with the source is now three words, *shape, edge and colour*, which are technical nouns with no substitutes.

✔ **Validated** — I have read the original and agree it supports the sentence.

#79 kuwahara1976 — Kuwahara et al. (1976) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] It is also described as the classical member of the edge-preserving filters used for image abstraction.**

Usage: Use 2 of this source.

Original source: **Primary 1976 chapter not held.** The characterisation is verified against the paper that makes it and that attaches the origin credit to Kuwahara et al. in the same sentence. **(1)** *"the classical member of the edge-preserving filters"* ← [Kyprianidis et al. 2009, §2 Related Work:] "Nonlinear filters that **preserve edges** have been an active research topic for several decades. **Classical filters in this field are the well-known Kuwahara filter [KHEK76]** and its variants." The marker **[KHEK76]** is Kuwahara, Hachimura, Eiho and Kinoshita (1976). **(2)** *"used for image abstraction"* ← [same paper, Abstract:] "We present a non-photorealistic rendering technique to transform color images and videos into **artistic abstractions**. It is based on a generalization of the Kuwahara filter."

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`, §2 and Abstract. **Citation decided 2026-08-24, single key.** The sentence previously read "Kyprianidis et al. describe it as. . . " with no citation of its own, sitting inside the block closed by the later `kyprianidis2009` citation. It now states the characterisation directly and cites `kuwahara1976` alone. **The reasoning, recorded so it is not re-litigated:** this is the first appearance of the Kuwahara filter as a named technique in Related Work, which §5.2 of the writing rules says must carry the original paper. The characterisation itself comes from Kyprianidis et al., and a joint citation was considered on that ground, but the survey attributes the filter to Kuwahara et al. inside the very sentence quoted above, and `kyprianidis2009` is cited in its own right two sentences later for the anisotropic generalisation. A second key here would therefore have added no support the reader cannot already reach. **Still open:** the 1976 Plenum Press chapter is not held, so the origin credit rests on Kyprianidis quoting it rather than on the primary.

✔ **Validated** — I have read the original and agree it supports the sentence.

#80 kyprianidis2009 — Kyprianidis et al. (2009) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** The anisotropic generalisation of Kyprianidis et al. shapes the sampling region according to local feature geometry. **The flattening then follows those features instead of cutting across them, and the surface takes on a painted appearance.**

Usage: Use 4 of this source.

Original source: **(1)** *"shapes the sampling region according to local feature geometry"* ← [Abstract:] "It is based on a generalization of the Kuwahara filter that is **adapted to the local shape of features**, derived from the **smoothed structure tensor**." **(2)** *"the flattening then follows those features. . . a painted appearance"* ← [Abstract:] "Contrary to conventional edge-preserving filters, our approach [produces a] **painting-like flattening effect along the local feature directions** while preserving shape boundaries."

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`, Abstract and §2. **Verbatim lift found and corrected 2026-08-24.** The thesis sentence read "Their anisotropic formulation *adapts the filter to the local shape of features* and this produces a *painting-like flattening effect along the local feature directions*". Both italicised spans are the abstract word for word, seven words and eight words, inside a single sentence. The sentence was rewritten from scratch: the first claim now reads "shapes the sampling region according to local feature geometry" and the second "the flattening then follows those features instead of cutting across them, and the surface takes on a painted appearance. No run of six or more words survives against the source. The technical content is unchanged.

✔ **Validated** — I have read the original and agree it supports the sentence.

#81 kyprianidis2011 — Kyprianidis (2011) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** Two artefact problems survived that generalisation, one in noisy regions and one across broad areas of even tone. A later paper by Kyprianidis addressed both. Thresholding the sector weights removes the first, and a multiscale scheme suppresses the second. **That scheme carries orientation estimates and partial filtering results together through the levels of a smoothed pyramid, and it still runs fast enough for graphics hardware to apply it interactively.**

Usage: Use 2 of this source.

Original source: **(1)** *"carries orientation estimates and partial filtering results together through the levels of a smoothed pyramid"* ← [Abstract:] "a **multi-scale computation scheme** is proposed that **simultaneously propagates local orientation estimates and filtering results up a low-pass filtered pyramid**." **(2)** *"suppresses the artefacts that appear in broad regions of low contrast"* ← [Abstract:] "This allows for a strong abstraction effect and **avoids artifacts in large low-contrast regions**." **(3)** *"remains fast enough for graphics hardware to run it interactively"* ← [§1:] the scheme is described as one "that is particularly **suitable for real-time processing on a GPU**".

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_multicale_-209.pdf`, Abstract and §1. **Verbatim lift found and corrected 2026-08-24, the largest one in the thesis.** The two sentences read "*propagates local orientation estimates and filtering results up a low-pass filtered pyramid*" and "This *avoids artefacts in large low-contrast regions* and suits *real-time processing on a GPU*. That is a thirteen word run, a six word run and a five word run taken from the abstract and introduction, with only *artifacts* respelled as *artefacts*. All three were rewritten. **Citation moved in the same pass:** it previously closed on the first sentence, leaving the artefact claim and the GPU claim, both of them Kyprianidis claims, without support. It now closes the block. **Block restructured 2026-08-24:** the multiscale variant previously arrived with no stated problem to solve, and a sentence about the structure tensor sat in front of it with nothing following from it. The block now opens with the two limitations the paper names, and the credit for each is assigned separately, since the thresholding fixes the noise artefacts and the pyramid fixes the low-contrast ones. Attributing both to the multiscale scheme would have been wrong.

✔ **Validated** — I have read the original and agree it supports the sentence.

#82 kyprianidis2009 — Kyprianidis et al. (2009) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] Both formulations derive their kernel shape from a structure tensor, and that computation needs a render pass of its own.** *[compares both variants against the unfiltered input. — follows the citation, uncited.]*

Usage: Use 5 of this source.

Original source: **(1)** *that the 2009 filter derives its kernel from a structure tensor* ← [Kyprianidis et al. 2009, §3:] "We begin with **calculating the structure tensor** and smooth[ing it]. . . the amount of anisotropy [is] derived from the **eigenvalues and eigenvectors of the smoothed structure tensor**. Finally, structure-aware smoothing is performed using a nonlinear filter [that] uses weighting functions defined **over an ellipse, whose shape is based on the local orientation and anisotropy**." **(2)** *that the 2011 variant does the same* ← [Kyprianidis 2011:] "in contrast to the other techniques that propagate gradient estimates, in this work **the structure tensor is propagated**." **(3)** *that the computation is a pass of its own* ← the tensor is computed from the input image and smoothed before any filtering occurs, so it cannot be produced inside the filtering step it parameterises. Stated as a separate stage in both papers.

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`, §3. This key covers the 2009 half of the claim, that the anisotropic filter derives its elliptical kernel from the smoothed structure tensor. The 2011 half is covered by the companion entry for `kyprianidis2011` on the same sentence.

✔ **Validated** — I have read the original and agree it supports the sentence.

#83 kyprianidis2011 — Kyprianidis (2011) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] Both formulations derive their kernel shape from a structure tensor, and that computation needs a render pass of its own.** *[compares both variants against the unfiltered input. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *that the 2009 filter derives its kernel from a structure tensor* ← [Kyprianidis et al. 2009, §3:] "We begin with **calculating the structure tensor** and smooth[ing it]. . . the amount of anisotropy [is] derived from the **eigenvalues and eigenvectors of the smoothed structure tensor**. Finally, structure-aware smoothing is performed using a nonlinear filter [that] uses weighting functions defined **over an ellipse, whose shape is based on the local orientation and anisotropy**." **(2)** *that the 2011 variant does the same* ← [Kyprianidis 2011:] "in contrast to the other techniques that propagate gradient estimates, in this work **the structure tensor is propagated**." **(3)** *that the computation is a pass of its own* ← the tensor is computed from the input image and smoothed before any filtering occurs, so it cannot be produced inside the filtering step it parameterises. Stated as a separate stage in both papers.

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_kuwahara.pdf` §3 and `Articles/03_npr_rendering/anisotropic_multicale_209.pdf`. **Sentence relocated 2026-08-24.** It previously read "That formulation requires a structure tensor computed from a prior render pass" and sat between the 2009 result and the 2011 variant with nothing following from it. It now closes the paragraph as the cost shared by both formulations, which is the point that carries into the thesis's own work, since the Meta SDK fragment hook cannot afford a prior pass. **Joint citation:** both papers derive the kernel shape from the tensor, so both keys are earned.

✔ **Validated** — I have read the original and agree it supports the sentence.

#84 kyprianidis2011 — Kyprianidis (2011) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** Painterly abstraction produced by Kuwahara filtering. The first image is the unfiltered input and the second is the single-scale anisotropic filter. **The third is the multiscale variant.**

Usage: Use 4 of this source.

Original source: **(1)** *the identity of all three panels* ← [Figure 1 panel labels, printed p. 55:] "**(a) Original image (b) Anisotropic Kuwahara filter (c) Proposed method**". **(2)** *that panel (c) is the multiscale method and (b) the single-scale one* ← [Figure 1 caption:] "Example **comparing the proposed multi-scale approach with the single-scale approach**."

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_multicale_-209.pdf`, Figure 1 and its caption, printed p. 55. **Caption expanded 2026-08-24:** it previously identified only panel (c), leaving the first two images unexplained. All three are now named, and the panel labels above confirm each one.

✔ **Validated** — I have read the original and agree it supports the sentence.

#85 wallraven2007 — Wallraven et al. (2007) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** Taken together, this technical catalogue establishes what NPR can render, but not whether the result changes how an avatar is perceived. Psychophysical and VR experiments have examined that question directly. Wallraven et al. compared brush, cartoon and illustrative stylisation on sequences of facial expression. Recognition accuracy held under the cartoon variant but fell under the other two when the sequences were recorded video. **Computer-generated animations of the same expressions reversed that ordering, and the illustrative variant achieved marginally higher recognition than the cartoon one.**

Usage: First use of this source.

Original source: **(1)** *the recorded-video result (Experiment 3)* ← [§5 Conclusion:] "for recognition, for example, we found that **cartoonized versions were equally well recognized as the original sequences, whereas both brush-stroke and illustrative stylization techniques resulted in a drop in recognition accuracy**, as well as an increase in response times." **(2)** *the computer-generated result (Experiment 2)* ← [Article 16/13:] "here we found that the **brush-stroke stylization was significantly worse** than the remaining three rendering methods (t test: all $p < 0.05$). In addition, the **illustrative stylization was slightly better** (t test: $p = 0.05$, $p = 0.07$, marginally significant) **than both avatar and cartoon conditions**." **(3)** *that the authors themselves flag the split* ← [§5:] "The impact of the stylization methods on the different measures, however, was **different from Experiment 2**." **(4)** *which experiment used which stimuli* ← [§4:] "Experiment 1 + 2 were done with the **animated sequences**. . . Experiment 3 was a replication of Experiment 2 with the **real-world sequences**."

How to check: Read directly from `Articles/03_npr_rendering/wallraven2007_stylized_facial_expressions.pdf`, §4, §5 and the Experiment 2 results. **Domain error found and corrected 2026-08-25.** The thesis sentence read "Recognition accuracy held under the cartoon variant but fell under the other two" with no stimulus type named. That is true of **Experiment 3 only**, which used recorded video. Experiment 2 ran the same comparison on computer-generated animations and produced a different ordering: the illustrative variant came out marginally *ahead* of the cartoon one. **This is the experiment that matters here**, since the thesis concerns avatars, so reporting only the video result and leaving it unscoped generalised the wrong half onto the thesis's own domain. **Both results are now stated.** The paragraph's existing conclusion, that the outcome depends on which technique produces the abstraction rather than on its degree alone, is strengthened rather than weakened by this: the outcome turns out to depend on the stimulus domain as well. **Hedge preserved:** the Experiment 2 advantage is reported at $p = 0.05$ and $p = 0.07$ and described by the authors as "marginally significant", so the thesis says *marginally higher recognition* rather than asserting a clean reversal. Brush-stroke was worst in both experiments and is unaffected.

✔ **Validated** — I have read the original and agree it supports the sentence.

#86 wisessing2016 — Wisessing et al. (2016) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** Hence, the outcome depends on which technique produces the abstraction, not on its degree alone. Wisessing et al. later demonstrated that toon shading reduces the perceived intensity of facial emotional expression while leaving recognition accuracy intact. **The authors attribute that effect to the loss of shading information, and this may conceal facial form and features that emotion perception depends on.**

Usage: First use of this source.

Original source: **(1)** *the two measured results* ← [§6 Discussion, printed p. 28:] "Our main finding is that the **shading style used (CG vs. toon) does not change the recognition of emotion, but does change the perception of intensity of that emotion**, with toon-shaded renders being rated as displaying emotions as less intense than CG-shaded across the board." **(2)** *the added mechanism* ← [same paragraph:] "It is possible that our toon-shaded animations had lower intensity values than our CG-shaded animations due to the **lack of visual information**. In particular, the abstraction of the shaded surface **could obscure facial shape and features crucial to emotion perception**." **(3)** *that the stimulus is facially driven* ← [§3 Stimuli Creation:] the character Mery was chosen because she "has a **highly controllable facial rig**, with the ability to squash and stretch each part, allowing animators to create highly expressive cartoon animations".

How to check: Read directly from `Articles/Wissesing- perception.pdf`. The Discussion column interleaves with the reference list in the extracted text layer, so page 4 was re-read as a rendered image at 190 dpi to confirm the wording. **A PROPOSED CHANGE WAS WITHDRAWN HERE, 2026-08-25. Recorded so it is not re-proposed.** An audit pass flagged "facial emotional expression" as over-specific, on the ground that the abstract calls the rated construct "the intensity of the *performance*". Nidanur pushed back, correctly. The paper localises the effect to the face in its own Discussion, quoted above, and the stimulus is a facially rigged character. The flag came from reading the abstract without checking whether the Discussion narrowed the mechanism. It did. Both thesis sentences stand as written, including "matters most on the face" in §3.4. **Support added instead, same date.** The face mechanism is the strongest sentence in this paper for RQ2 and the thesis was not using it, leaving "matters most on the face" as an uncited inference of our own. A second sentence now carries the authors' explanation. **Hedging matched to the source:** the original is explicitly speculative, "It is possible that" and "could obscure", so the thesis uses "attribute" and "may conceal" rather than asserting the mechanism. **Lift check:** "obscure facial shape and features crucial to emotion perception" is seven content words and would have been a verbatim lift. It was reworded to "conceal facial form and features that emotion perception depends on", where the longest run shared with the source is two words.

✔ **Validated** — I have read the original and agree it supports the sentence.

#87 wisessing2020 — Wisessing et al. (2020) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** Characters rendered in this style accordingly convey the correct emotional content without the perceptual amplification that smooth CG shading provides. A follow-up study identified lighting parameters as a further modulator of appeal and eeriness. Higher brightness made characters more appealing regardless of the emotion portrayed. Darker shadows reduced the appeal of cartoon characters and left the appeal of realistic ones unaffected. **They also raised eeriness, though that effect came mainly from the male character.** *[Together these findings reveal that render style and lighting both shape how a character is received, independently of the emotion being performed. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **(1)** *"Darker shadows raised eeriness"* ← [§Realism Experiments, Results:] "A **main effect of KTFR on eeriness showed 16:1 was more eerie than 4:1 or 1:1** ($p < 0.0276$ in both cases)." A higher key-to-fill ratio means less fill light, so 16:1 is the darkest shadow condition. **(2)** *"reduced the appeal of cartoon characters"* ← [same section:] "An interaction between shape and KTFR also occurred. . . **For Toon, 1:1 was more appealing than 4:1 or 16:1** ($p < 0.0011$)." **(3)** *"while leaving the appeal of realistic characters unaffected"* ← [same sentence:] the interaction "showed **no effect of KTFR on appeal ratings for Realistic** (Figure 11)." **(4)** *the preceding brightness sentence* ← [§Main findings:] "**Higher brightness makes for more appealing characters, regardless of the emotion they portray**."

How to check: Read directly from `Articles/Wissessing-Enloghten me.pdf`, Main findings and the Realism Experiments results. **Source upgraded 2026-08-24.** The entry previously rested on the bullet-point summary at the front of the paper. It now quotes the statistics behind each of the three claims, taken from the results section itself. **CORRECTION HISTORY, 2026-08-14:** an earlier version of the thesis sentence claimed brighter light and lighter shadows reduced eeriness for both photorealistic and stylised characters, and that the reduction was larger for photorealistic rendering. That was wrong on both counts and was rewritten. **Qualification that still applies:** the eeriness main effect above is immediately qualified in the source by "an interaction between gender, shape, and KTFR, which showed this effect to be **mainly coming from the male character**. The thesis states the main effect without that qualifier, which is a simplification rather than an error, but it is worth knowing.

✔ **Validated** — I have read the original and agree it supports the sentence.

#88 zibrek2019tap — Zibrek et al. (2019) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** Those studies measured how a character is read from the outside. Zibrek et al. instead placed observers inside the scene and ran a large between-group VR study with 622 participants. That study compared three render styles from photorealistic to sketch across three emotional scenarios (). The results of that experiment showed that render style left social presence unchanged and produced no eeriness difference between any two individual styles. **Photorealism shifted which emotions the character evoked rather than strengthening them.**

Usage: First use of this source.

Original source: **(1)** *"622 participants. . . three render styles. . . three emotional scenarios"* ← [§4.3:] "we included responses of **622 participants** in the final analysis, which was approximately 70 participants (SD: 5.3) for each scenario/render style combination." [§5 Results, opening line:] "the most robust effects of **render style (Realistic, Simple, Sketch)** and **scenario (Sad, Friendly, Unfriendly)**". **(2)** *"left social presence. . . unchanged"* ← [Table 2, Experiment 1, the Social Presence block reads in its entirety:] "**NO EFFECTS**". Restated in §8: "Even though we predicted an effect of render style on Social Presence (H1), we found **no main effects** or interactions", and again in the Discussion: "**Social presence was not affected by render style**, as we expected." **(3)** *"and eeriness unchanged"* ← [Table 2, Affinity block:] the only Eerie row is **EERIE: Gender** ($\chi^2 = 13.92$, $df = 1$, $p < 0.0002$). No render style row appears for Eerie, so render style produced no effect on it. **(4)** *"photorealism shifted which emotions the character evoked rather than strengthening them"* ← [Table 2, Emotional Response block, where the direction reverses by scenario:] "CONCERN: Scenario*Render Style ($\chi^2 = 10.35$, $p < 0.04$): **Sketch style more concern than Realistic** in [one scenario], **Realistic style more concern than the other two in the Friendly scenario**." Also "EXCITED: Render Style ($\chi^2 = 13.05$, $p < 0.002$): Realistic style rated higher compared to Sketch" and "CALM: Scenario*Render Style ($\chi^2 = 12.69$, $p < 0.013$): **Sketch style more calm than the Realistic**". [Abstract:] "our main finding shows that the **photorealistic character changed the emotional responses** of participants, compared to the stylized versions."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/TAP_2019.pdf`, Abstract, §§4.3–5, Table 2 and the Discussion. **Source rebuilt 2026-08-24.** The entry previously carried only two short fragments. Every clause of the thesis sentence now has its own quote, and the emotional-response claim is supported by the three table rows whose direction actually reverses between scenarios, which is what makes "shifted which emotions. . . rather than strengthening them" the right description rather than a loose one. **Scope note:** the paper reports a second, smaller *Additional experiment* (N = 175, Sad scenario only) whose Table 4 does show **EERIE: Render style** ($\chi^2 = 7.32$, $p < 0.03$), although its post-hoc found "no significant differences between individual render styles. The thesis sentence is explicitly about the 622-participant experiment, so it is correct as written, but a reader who opens the paper will meet that second table. **One further difference worth knowing:** render style *did* affect Place Illusion in Experiment 1 ($\chi^2 = 10.61$, $p < 0.005$, realistic highest). The thesis does not claim otherwise, since it names only social presence and eeriness.

✔ **Validated** — I have read the original and agree it supports the sentence.

#89 zibrek2019mig — Zibrek & McDonnell (2019) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** This challenges the assumption that rendering quality scales linearly with social utility. Zibrek and McDonnell extended this investigation to embodied VR, where participants inhabited a virtual body rather than observing from outside. The results indicated that photorealism significantly raised both self-reported social presence and place illusion, the sense of physically inhabiting the virtual space. **A rendering change can therefore shift what people report without shifting what they do.**

Usage: First use of this source.

Original source: **(1)** *"self-reported social presence"* ← [§2.3:] "We found a **significant difference in Social Presence, where participants felt more present with the Realistic virtual character than the Simple one** ($t(18) = 2.45$, $p = 0.025$, $d = 0.56$)." **(2)** *"and place illusion"* ← [§2.4:] "($Z = 2.02$, $p = 0.04$, $r = 0.46$), where the **realistic environment increased the place illusion**." **(3)** *"without shifting what they do"* ← [Abstract:] "This was found to be the case for self-reported social presence and place illusion. . . **behavioural measure proximity was not affected by render style**." **(4)** *"participants inhabited a virtual body"* ← [Abstract:] "**participants were embodied in a high-fidelity virtual body in VR** and were observing an animated character".

How to check: Read directly from `Articles/02_trust_and_avatar_perception/3359566.3360064.pdf`, Abstract and §§2.3–2.4. **Citation moved 2026-08-24.** It previously closed on the results sentence, leaving the conclusion about self-report diverging from behaviour outside the block. That conclusion reads like an inference but is the paper's own result, quoted above as (3), so it now sits inside the citation.

✔ **Validated** — I have read the original and agree it supports the sentence.

#90 zibrek2019tap — Zibrek et al. (2019) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] The three render styles compared by Zibrek et al., from left to right: realistic, simple and sketch, applied to both the character and the room.**

Usage: Use 2 of this source.

Original source: **(1)** *"realistic, simple and sketch"* ← [§3.0.1 Render Styles:] "We created two styles for our task: a **mid-range realistic character in "Simple" style**, and the **least realistic "Sketch" style** (Figure 2)." The pair are controls against the realistic baseline: "In order to estimate the effect of photorealism, we created **two control render styles**. . . representing the stylized versions." **(2)** *"applied to both the character and the room"* ← [same section:] the two control styles were made "of **both the room and the virtual character**". For Simple, "we removed the realistic lighting effects, shadows, and other shaders which created a realistic appearance **of the virtual room and the virtual agent**". For Sketch, "removing the **colour information from the environment** also, leaving only the grayscale range". **(3)** *that only the rendering differs* ← [same section:] "we **abstracted the render style only**, while all other elements, such as geometry and animated content, were kept the same."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/TAP_2019.pdf`, §3.0.1. **Source rebuilt 2026-08-24.** The previous quote named the two stylised conditions but did not support the caption's claim that the styles were applied to the room as well as the character. That half is now quoted twice, and the sentence confirming that geometry and animation were held constant is added, since it is what makes this a render-style comparison rather than a whole-stimulus comparison.

✔ **Validated** — I have read the original and agree it supports the sentence.

#91 zell2015 — Zell et al. (2015) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation] figure[The shape stylisation axis of Zell et al.]Shape axis, s0 to s4.**

Usage: First use of this source.

Original source: **(1)** *the s0 to s4 labelling* ← [§6 Experiment 2, Stimuli:] "For each character, two additional stylizations were created, yielding **five stylization levels from level 0 (most stylized) to level 4 (highly realistic)**. We **distinguish between stylizations in material and shape by using the prefix m and s respectively**." **(2)** *that the axis runs from a cartoon head to a scanned likeness* ← [§3 Stimuli Creation:] the realistic end is a scan, "To generate the realistic models we replicated the **multiview-stereo face scanner** of Beeler et al., which reconstructs **high-resolution textured point clouds** from the photographs of six cameras". The stylised end is authored, with the level 0 and level 3 designs "particularly designed by the artists to fill the gaps for perceived realism. . . inspired by Pocoyo and Tangled". **(3)** *why this axis is the one shown for realism* ← [Main findings:] "**Shape is the key attribute for perceived realism**."

How to check: Read directly from `Articles/01_uncanny_valley/To Stylize or not to Stylize?_Zell.pdf`, §3 and §6. **REPLACED 2026-08-24.** This caption and the material-axis caption previously shared a single quote, so one passage was standing in for two different figures. This entry is now sourced to how the *shape* levels were built, from the face scanner at one end to artist-authored cartoon geometry at the other, and to the finding that makes the shape axis the realism axis.

✔ **Validated** — I have read the original and agree it supports the sentence.

#92 zell2015 — Zell et al. (2015) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in thesis: **[Block covered by this citation]** Render style is not the only dimension a designer controls. Zell et al. separated a character's shape from its material and varied each independently across five stylisation levels. Shape governed perceived realism, whereas material governed appeal and eeriness. shows the shape axis, which holds the material constant while the geometry moves from a cartoon head at s0 to a scanned likeness at s4. Furthermore, shows the material axis, where the geometry stays fixed and the shader and texture move across the same five levels. Strong mismatches between the two levels also produced very unappealing characters. **A stylised material on a realistic shape was among those combinations.** *[The uncanny response depends in turn on the level of material stylisation and on its consistency with the shape. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *"separated a character's shape from its material and varied each independently across five stylisation levels"* ← [§6 Stimuli:] "For each character, two additional stylizations were created, yielding **five stylization levels from level 0 (most stylized) to level 4 (highly realistic)**. We **distinguish between stylizations in material and shape by using the prefix m and s respectively**." The independence is procedural: "we **transferred all material properties of the baseline characters to the other character shapes**". **(2)** *"Shape governed perceived realism, whereas material governed appeal and eeriness"* ← [Main findings:] "**Shape is the key attribute for perceived realism**" and "appeal, eeriness, and attractiveness are **mainly affected by the stylization level of material, and not shape**". **(3)** *"the shader and texture move across the same five levels"* ← [§3:] "we denote by **material the combination of [shader properties] and textures**. . . For the skin all three characters use the same multi-layer skin shader with subsurface scattering, with **diffuse albedo specified by a high-resolution texture map**. The **shader parameters vary between the models** in order to closely resemble the targeted render styles." **(4)** *"Strong mismatches. . . produced very unappealing characters. A stylised material on a realistic shape was among those combinations"* ← [Discussion:] "**strong mismatches in the level of stylization of shape and material can result in very unappealing characters**." [Results, appeal:] "We found a **main effect of material on the ratings of appeal**. . . but **no main effect of shape was found**. An interaction between shape and material ($F(4, 84) = 13.97$, $p < 0.0001$) shows that a realistic material on a cartoon shape yields the least appealing combination. . . and **the cartoon material on the realistic shape is similarly unappealing** ($p < 0.02$)."

How to check: Read directly from `Articles/01_uncanny_valley/To Stylize or not to Stylize?_Zell.pdf`, §3, §6, Results and Main findings. **Extended 2026-08-24** when a second sentence was added to the paragraph pointing at the material axis figure alongside the shape axis figure. Quote (3) is what supports that new sentence: the material dimension is the shader parameters together with the albedo texture, and the transfer procedure in quote (1) is what keeps it independent of the geometry.

✔ **Validated** — I have read the original and agree it supports the sentence.

#93 zell2015 — Zell et al. (2015) *Ch.3 Related Work → NPR for Virtual Humans*

Sentence in the-sis: **[Block covered by this citation]** **type=figureMaterial axis, m0 to m4.**

Usage: Use 3 of this source.

Original source: **(1)** *what Zell et al. mean by material* ← [§3 Stimuli Creation:] "we denote by **material the combination of [shader properties] and textures**". Concretely: "For the skin all three characters use the same **multi-layer skin shader with subsurface scattering, with diffuse albedo specified by a high-resolution texture map**. The **shader parameters vary between the models** in order to closely resemble the targeted render styles." **(2)** *that the axis is held independent of shape* ← [same section:] "we **transferred all material properties of the baseline characters to the other character shapes**", using a dense correspondence map for the texture transfer. **(3)** *why this axis is the one shown for appeal and eeriness* ← [Main findings:] "appeal, eeriness, and attractiveness are **mainly affected by the stylization level of material, and not shape**. . . Within the materials, **the albedo texture is the dominant factor**."

How to check: Read directly from `Articles/01_uncanny_valley/To Stylize or not to Stylize?_Zell.pdf`, §3 and Main findings. **REPLACED 2026-08-24.** See the note on the shape-axis entry: the two captions previously shared one quote. This entry is now sourced to the definition of the material dimension, to the transfer procedure that keeps it independent of shape, and to the finding that makes the material axis the appeal and eeriness axis.

✔ **Validated** — I have read the original and agree it supports the sentence.

#94 wisessing2016 — Wisessing et al. (2016) *Ch.3 Related Work → Research Gap*

Sentence in the-sis: **[Block covered by this citation]** Moreover, NPR techniques have been developed and compared extensively, but largely on images, scenes and generic 3D shapes rather than on virtual avatars. **Even though Wisessing et al., Zibrek et al. and Zell et al. did study human figures, each varied coarse stylisation levels rather than specific operators.**

Usage: Use 2 of this source.

Original source: [§4 Method:] "The character performing each of the emotional sentences was **rendered in 2 shading styles (toon-shaded, and CG-shaded)**, under 5 different conditions: Lo/Ab, Lo/Be, Hi/Ab, Hi/Be, and No."

How to check: Read directly from `Articles/Wissesing- perception.pdf`, §4 Method. Confirms the comparison was between two broad shading styles, not between individual operators.

✔ **Validated** — I have read the original and agree it supports the sentence.

#95 zibrek2019tap — Zibrek et al. (2019) *Ch.3 Related Work → Research Gap*

Sentence in thesis: **[Block covered by this citation]** Moreover, NPR techniques have been developed and compared extensively, but largely on images, scenes and generic 3D shapes rather than on virtual avatars. **Even though Wisessing et al., Zibrek et al. and Zell et al. did study human figures, each varied coarse stylisation levels rather than specific operators.**

Usage: Use 3 of this source.

Original source: [§3.0.1 Render Styles, p. 6:] "In order to estimate the effect of photorealism, we created two control render styles of both the room and the virtual character. . . For these controls, **we abstracted the render style only**, while all other elements, such as geometry and animated content, were kept the same. . . . **We were aiming for mid-range and lowest level of render style realism**, specifically the "Human Basic" and "Toon Pencil", respectively."

How to check: Read directly from `Articles/02_trust_and_avatar_perception/TAP_2019.pdf`, §3.0.1, p. 6. The styles were built by removing material, lighting and colour information wholesale, and were modelled on McDonnell et al. realism scale. That is what makes them levels rather than named operators.

✔ **Validated** — I have read the original and agree it supports the sentence.

#96 zell2015 — Zell et al. (2015) *Ch.3 Related Work → Research Gap*

Sentence in thesis: **[Block covered by this citation]** Moreover, NPR techniques have been developed and compared extensively, but largely on images, scenes and generic 3D shapes rather than on virtual avatars. **Even though Wisessing et al., Zibrek et al. and Zell et al. did study human figures, each varied coarse stylisation levels rather than specific operators.**

Usage: Use 4 of this source.

Original source: [§6 Experiment 2, and Figure 8:] "For each character, two additional stylizations were created, yielding **five stylization levels from level 0 (most stylized) to level 4 (highly realistic)**."

How to check: Read directly from `Articles/01_uncanny_valley/To Stylize or not to Stylize?_Zell.pdf`, §6 and Figure 8. Confirms graded stylisation levels rather than named operators.

✔ **Validated** — I have read the original and agree it supports the sentence.

#97 wallraven2007 — Wallraven et al. (2007) *Ch.3 Related Work → Research Gap*

Sentence in the-sis: **[Block covered by this citation] Wallraven et al. did compare named techniques for expression recognition, but applied them to the rendered frames as a post-process rather than to the avatar's own shading.**

Usage: Use 2 of this source.

Original source: **(1)** *"did compare named techniques"* ← [Abstract:] "animated facial expressions are stylized using **three common techniques (brush, cartoon, and illustrative stylization)**". **(2)** *"for expression recognition"* ← [Abstract:] the comparison uses "**task-dependent measures (such as recognizability, intensity)**". **(3)** *"to the avatar's own shading", the decisive passage* ← [§3.3 Stylization for Animated Versus Real-World Sequences:] "the **3D geometry of the avatar was lit by an omnidirectional light** coming from slightly above and a **standard Blinn shader was used to simulate diffuse and specular properties of the skin**." The avatar is rendered conventionally. Nothing in the stylisation touches its material. **(4)** *"applied them to the rendered frames as a post-process"* ← all three operators take a finished image as input. [§3.2.1 Brush-Stroke:] "The **image stylization process samples the input image** by reading pixel colors at the sampling point positions". [§3.2.2 Cartoonlike:] "**each input image is processed**. . . The algorithm. . . is designed as a **postprocessing filter in a real-time rendering pipeline**." [§3.2.3 Illustrative:] it "generates **output images**, which reproduce the brightness of the **input image** with black-and-white hatching". [§3:] the techniques "were **applied to these animated expressions** in order to create stylized sequences".

How to check: Read directly from `Articles/Wallraven_Stylization_TAP2007.pdf`, Abstract and §§3–3.3. **Reworded 2026-08-24 after a second reading of the primary.** The sentence previously read "stylised rendered footage *rather than the avatar itself*". That is true of where the operators act, but it reads as though no avatar was involved, and §3.1.1 of the paper is titled *The Avatar* and describes a morphable head built from 3D scans driven by motion capture. A reader checking the source would meet that heading and conclude the thesis had it wrong. **What the second reading added:** §3.3, quoted above as (3), states outright that the avatar was rendered with an omnidirectional light and a standard Blinn shader. The avatar's own shading is therefore conventional, and the three named techniques run afterwards over the resulting frames. That is exactly the distinction the Research Gap paragraph needs, and the sentence now states it in those terms rather than by implication.

✔ **Validated** — I have read the original and agree it supports the sentence.

#98 wisessing2016 — Wisessing et al. (2016) *Ch.3 Related Work → Research Gap*

Sentence in thesis: **[Block covered by this citation] That distinction matters most on the face, since Wisessing et al. showed that shading style alters the perceived intensity of an expression.**

Usage: Use 3 of this source.

Original source: [Results, main effect of shading:] "A main effect of shading style also occurred ($F(1, 22) = 23.2$, $p \approx 0$), with posthoc showing that animations rendered in CG-shading were rated as significantly more intense than those that were toon-shaded ($p < 0.0003$)." [Recognition accuracy:] the corresponding ANOVA reported a main effect of *emotion* only ($F(4, 88) = 5.4$, $p < 0.0007$); no shading-style effect on recognition is reported.

How to check: Primary PDF read and verified: `Articles/Wissesing- perception.pdf`. Both halves of the thesis sentence hold: toon-shading lowered rated intensity, while recognition accuracy varied by emotion rather than by shading style.

Validated — I have read the original and agree it supports the sentence.

#99 dill2012 — Dill et al. (2012) *Ch.3 Related Work → Research Gap*

Sentence in thesis: **[Block covered by this citation]** Equally, the discomfort that CG characters provoke is not uniform across the humanoid body. **Dill et al. revealed that their participants traced it predominantly to the eyes and the mouth.** *[Therefore, which technique or combination best balances abstraction against expressive legibility on avatar faces remains largely unstudied. This gap is addressed by the iterative technique development and selection described in the following chapters (RQ2). — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: [p. 512, §2 Obtained Results, final paragraph:] "The questionnaire also asked to the participants **which parts of the face** the participant felt more discomfort, if it was the case. Analysing the answers, we could **highlight two specific parts of the face: eyes and mouth**."

How to check: Read directly from `Articles/01_uncanny_valley/Evaluation of the Uncanny Valley in CG Characters.pdf`, printed p. 512 (PDF p. 527 of the IVA 2012 proceedings volume). **Scope note:** participants were asked only about *parts of the face*, so the paper supports non-uniformity within the face and says nothing about the rest of the body.

✔ **Validated** — I have read the original and agree it supports the sentence.

#100 mcdonnell2012 — McDonnell et al. (2012) *Ch.3 Related Work → Research Gap*

Sentence in thesis: **[Block covered by this citation]** By contrast, whether NPR abstraction on virtual avatars helps or harms perceived trust has been tested directly, and the results disagree. **McDonnell et al. found appeal and trust ratings lowest in the mid realism range, which stylisation avoids.**

Usage: Use 5 of this source.

Original source: "[Section 1 Introduction, p. 91:2 (first passage); Section 7 Discussion, p. 91:10 (second and third passages):] "We found that highly realistic and highly abstract styles were considered appealing, whether still or moving. However, render styles that were in the middle of the range between abstract and realistic were considered unappealing, and the most unappealing character was considered even more so when motion was applied. ... These results are very much in line with the Uncanny Valley theory. Contrary to the theory, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters. ... Based on the results of our experiments, we can reassure developers that selecting realistic virtual characters is not as risky as often discussed (provided that the motion does not contain artifacts).""

How to check: File: `Articles/01_uncanny_valley/ McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012). Render me Real? Investigating.pdf`. **Resolved 2026-08-23:** the claim now matches the paper's actual result instead of the one-sided "principled route out of the uncanny valley". Verified: "render styles that were in the middle of the range between abstract and realistic were considered unappealing", and for trust, "For the Trustworthy ratings, a main effect of style was also observed ($F_{9,288} = 35.53$, $p \approx 0$)" with HumanIll "significantly less trustworthy than all other styles".

✔ **Validated** — I have read the original and agree it supports the sentence.

#101 canales2024 — Canales et al. (2024) *Ch.3 Related Work → Research Gap*

Sentence in thesis: **[Block covered by this citation] Canales et al. report no trust penalty for stylised avatars relative to photorealistic ones.**

Usage: Use 4 of this source.

Original source: "[§9 Discussion, p. 8:] "With regard to Style, we do not see evidence that people are more likely to trust more realistic avatars nor less likely to trust our most stylized caricatures. This runs counter to some previous work that suggests a preference for realistic avatars [18, 25], but is consistent with other work that found no such preference [5, 29]""

How to check: File: `Articles/02_trust_and_avatar_perception/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. **Resolved 2026-08-22:** the plural "controlled evaluations" became a single controlled evaluation, matching the one study cited. Canales et al. found no trust penalty for stylisation in their own study, which supports the gist. However the thesis sentence generalises to 'controlled evaluations' in the plural, and this paper explicitly states its result 'runs counter to some previous work that suggests a preference for realistic avatars' (Jo et al. [18], McDonnell et al. [25]). A single-study citation therefore over-generalises; the sentence should either be hedged or carry additional citations.

✔ **Validated** — I have read the original and agree it supports the sentence.

#102 tao2025 — Tao et al. (2025) *Ch.3 Related Work → Research Gap*

Sentence in thesis: **[Block covered by this citation] On the other hand, a network meta-analysis by Tao et al. shows high realism avatars receiving significantly higher trustworthiness ratings than low realism ones.**

Usage: Use 2 of this source.

Original source: "[§3.6 "Comparative analysis of trustworthiness":] "Eight studies (61.5%) included comparisons of trustworthiness. The NMA results indicated that the HRA received significantly higher ratings than the LRA (p = 0.046). No significant differences were observed between the MRA group and either the HRA or LRA groups.""

How to check: File: `Articles/Tao-Impact.pdf`. Verified 2026-08-23. The thesis sentence claims only the HRA-versus-LRA contrast, which is the one comparison that reached significance. **Worth knowing:** $p = .046$ is marginal, and the MRA group did not differ significantly from either other group, so this cannot be read as a monotonic trust gain with realism.

✔ **Validated** — I have read the original and agree it supports the sentence.

#103 canales2024 — Canales et al. (2024) *Ch.3 Related Work → Research Gap*

Sentence in the-sis: **[Block covered by this citation]** Thus, no study to date measures both trust and eeriness for an NPR pipeline applied to both the shading and the texture of one photorealistic avatar. **The Canales study came closest, but it compared coarse style categories and did not treat eeriness as a dimension separate from overall stylisation.** *[Nor has any reviewed study set a real human speaker, a photorealistic avatar of that speaker and an NPR rendering of the same avatar against one another. This gap is addressed by the user study of this thesis (RQ3). — follows the citation, uncited.]*

Usage: Use 5 of this source.

Original source: **(1)** *"compared coarse style categories"* ← [§4.1 Avatars:] "The two stylized versions of each identity were **hand modeled by a team of artists**. . . they created concept art for a **caricatured** version of each person along with a **semi-realistic** model intended to lie between the caricature and realistic model." Three whole-avatar categories, not named operators. **(2)** *"did not treat eeriness as a dimension separate from overall stylisation"* ← this is a claim about what the study *does not* measure, so it is established by inspecting the instrument. Every dependent measure in the paper is a trust prompt on a **7-point Likert scale**: "I find this avatar trustworthy", the ability, benevolence and integrity batteries, and believability. **No eeriness scale, no uncanniness item and no Ho and MacDorman index appears anywhere in the instrument.** The words *eerie* and *uncanny* occur in the paper only in the related-work review, and afterwards in post-hoc readings of free text, for example "[participants described] them as uncanny or creepy" and "perhaps some uncanniness". Those are interpretations of comments, not a measured dimension.

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Avatar_-Stylization_and_Trust_IEEE_VR_2024.pdf`, §§4.1–5.3 and §9. **Negative claim, verified by exhaustive search 2026-08-24.** A claim that a study did *not* measure something cannot be supported by a quotation, so the whole text was searched for every occurrence of *eerie*, *eeriness* and *uncann*. Twelve occurrences, none of them an instrument item. The thesis sentence is therefore accurate, and it is deliberately narrow: it says eeriness was not treated as a separate dimension, not that the authors ignored the concept.

✔ **Validated** — I have read the original and agree it supports the sentence.

Chapter: Methodology

#104 mori1970 — Mori (1970) *Ch.4 Methodology → Research Questions*

Sentence in the-sis: **[Block covered by this citation] Uncanny valley theory places a real human speaker at the peak of the affinity curve, since maximal human likeness corresponds to maximal comfort.** *[A photorealistic avatar instead sits in the trough immediately before that peak, where negative affect is strongest. That negative affect is what eeriness measures, so how eerie a character feels indicates where on the curve it sits. The question spans both the discomfort a stylised avatar evokes and the trust it receives. If the composite selected under RQ2 relocates the avatar onto the stable rising slope below the trough, it should register as less eerie than the photorealistic avatar that remains in it. The resulting trust level may then approach that of a real human speaker, even though the avatar's human likeness remains lower. RQ3 tests this possibility directly, with the real speaker serving as the reference point against which both avatar renderings are compared. The hypotheses that formalise these comparisons are stated in . — follows the citation, uncited.]*

Usage: Use 5 of this source.

Original source: **(1)** *that a real human sits at the top of the curve* ← [§The significance of the uncanny:] "In Figure 1, **a healthy person is at the top of the second peak**. And when we die, we fall into the trough of the uncanny valley." **(2)** *that maximal human likeness is the endpoint of the axis* ← [§The uncanny valley:] "Of course, **human beings themselves lie at the final goal of robotics**, which is why we make an effort to build humanlike robots." **(3)** *that the trough sits immediately before that peak* ← [same passage:] the descent from the second peak into the valley is the path Mori uses to explain death, "our impression of death can be explained by the **movement from the second peak to the uncanny valley**".

How to check: Read directly from `Articles/01_uncanny_valley/The Uncanny Valley- The Original Essay by Masahiro Mori.pdf` (MacDorman and Kageki translation), §§"The uncanny valley" and "The significance of the uncanny. **Precision note, no change needed.** Mori writes *second* peak, since his curve has an earlier local maximum at the humanoid toy robot. The thesis says "the peak of the affinity curve. That is correct rather than loose: the second peak is the global maximum of the curve, and the thesis sentence is about where a real human sits, not about how many local maxima the curve has.

✔ **Validated** — I have read the original and agree it supports the sentence.

#105 bartneck2009 — Bartneck et al. (2009) *Ch.4 Methodology → User Study Overview*

Sentence in thesis: **[Block covered by this citation]** Likeability and eeriness came from two separate instruments. **The Godspeed questionnaire supplied the likeability items.**

Usage: Use 4 of this source.

Original source: **(1)** *that Godspeed contains a likeability subscale* ← [Abstract:] "A literature review has been performed on the measurements of five key concepts in HRI: anthropomorphism, animacy, **likeability**, perceived intelligence, and perceived safety." **(2)** *its five items* ← [Appendix, GODSPEED III: LIKEABILITY, p. 79:] Dislike–Like, Unfriendly–Friendly, Unkind–Kind, Unpleasant–Pleasant, Awful–Nice, each on a 1–5 scale.

How to check: Read directly from `Articles/02_trust_and_avatar_perception/Bartneck_2009_-Measurement Instruments for the Anthropomorphism.pdf`. Appendix read as a rendered image, the page has no text layer. **Correctly attributed.** The sentence states only where the items came from, which is this paper.

✔ **Validated** — I have read the original and agree it supports the sentence.

#106 ho2010 — Ho & MacDorman (2010) *Ch.4 Methodology → User Study Overview*

Sentence in thesis: **[Block covered by this citation] Eeriness was taken from the instrument of Ho and MacDorman instead, since the Godspeed subscales do not separate cleanly from one another.** *[The full design, questionnaire structure and analysis plan are explained in the following sections. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *the intercorrelation finding* ← Ho and MacDorman report that every Godspeed subscale except perceived safety was too strongly intercorrelated to measure distinct constructs, and that none corresponds specifically to eeriness. **(2)** *their replacement instrument* ← eight items for eeriness alongside six for perceived humanness, the two indices **orthogonal in validation at** $r = .02$**,** $p = .514$.

How to check: Read directly from `Articles/02_trust_and_avatar_-perception/Ho2010UncannyValleyIndices.pdf`. Same source passage as the Ch.2 entries for this key, which are already validated. **Correctly attributed.** The sentence gives the reason for not using Godspeed for eeriness, which is this paper's own evaluation of Godspeed.

✔ **Validated** — I have read the original and agree it supports the sentence.

#107 mixamo — Adobe Inc. (2024) *Ch.4 Methodology → Avatar Platforms and Rendering Constraints*

Sentence in the-sis: **[Block covered by this citation] The Mixamo platform provides an online library of rigged humanoid characters and skinned animations.** *[These are exported to Unity as standard FBX assets. Because Mixamo avatars are standard skinned meshes without proprietary shader restrictions, all NPR techniques can be implemented directly within the URP rendering pipeline. That freedom makes Mixamo the least constrained of the three platforms and the most appropriate environment for initial exploration and development before adaptation to the more restrictive platforms. However, Mixamo character exports do not include facial rigging and the avatar's face remains static throughout all renders generated on this platform. This absence makes the platform unsuitable for the user study. The Mixamo character () still served as the primary prototyping target during the initial development phase, where unrestricted shader access supported rapid iteration and refinement. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **NOT VERIFIED. The cited page could not be read.** `https://www.mixamo.com/` is a JavaScript application that serves no readable content to a fetcher, and the Adobe help page `https://helpx.adobe.com/creative-cloud/faq/mixamo-faq.html` timed out on 2026-08-25. **Claims awaiting a source:** that Mixamo provides an *online library*, that its characters are *rigged*, that it supplies *skinned animations*, and (in the following sentence) that these export as *standard FBX assets*.

How to check: Attempted `https://www.mixamo.com/` and the Adobe Mixamo FAQ on 2026-08-25. Neither returned readable content. **OPEN, but low risk.** Nothing here is contested or surprising, and every claim is a fact of this project's own pipeline: the characters and animations used in the thesis were downloaded from Mixamo and imported to Unity as FBX. The same reasoning that closed the `avaturn` entry applies, and §5.2 places pipeline facts outside the scope of citation. **Not starred on my side** because no source was actually read. Confirm from your own use and it closes.

✔ **Validated** — I have read the original and agree it supports the sentence.

#108 metasdk — Meta Platforms (2024) *Ch.4 Methodology → Avatar Platforms and Rendering Constraints*

Sentence in the-sis: **[Block covered by this citation] The Meta Avatars SDK is Meta's avatar framework for its VR systems, and it handles body tracking internally.** *[The techniques were to be assessed in a VR environment, so the Meta Avatars SDK was examined for the shader customisation capabilities it provides for NPR integration (). Its built-in body tracking drives the avatar from the wearer's own tracked pose and removes the need for an external motion capture system. The proposed NPR shaders can thus be evaluated on an avatar animated live by the wearer, which neither of the other two platforms supports. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **RESOLVED 2026-08-25. Both remaining claims are on the cited page.** Fetched 2026-08-25. The old `developer.oculus.com` URL in `references.bib` now 301-redirects to `https://developers.meta.com/horizon/documentation/unity/meta-avatars-overview/`, which is where these quotes come from. **(1)** *"Meta's avatar framework for its VR systems"* ← "Use the **Meta Avatars SDK** to create expressive, diverse, and customizable **avatar identities for the Meta ecosystem, Unity VR apps**, and other multiplayer experiences." **(2)** *"body tracking"* ← "Use advanced **body tracking** capabilities to create realistic avatar features and poses." Eye and face tracking on Quest Pro are named separately. **(3)** *"avatar rendering. . . handled internally"* ← **NOT STATED on this page.** The overview does not say whether the SDK performs its own rendering.

How to check: Fetched from `https://developers.meta.com/horizon/documentation/unity/meta-avatars-overview/` on 2026-08-25. **FIXED 2026-08-25.** The sentence previously read "*Avatar rendering and* body tracking are handled internally by the Meta Avatars SDK. The rendering half is **not stated anywhere on the cited page**, and it is the load-bearing half for this thesis, since the SDK owning its own render path is the reason a fragment hook was needed at all. **The fix removed a duplication as well as the unsupported claim.** The same paragraph already states the rendering fact four sentences later, uncited, at the point where it does the work: "custom NPR shaders cannot be integrated through standard URP material slots, **because the SDK manages rendering internally**. That sentence is correct as an uncited statement, since it is a fact of this project's own integration work and §5.2 places implementation detail grounded in the project outside the scope of citation. The opening sentence was asserting the same thing a second time *and* attaching a citation to the copy that had no source behind it. Deleting "Avatar rendering and therefore fixes the attribution and the repetition together. **Bib URL corrected the same day:** `references.bib` recorded the retired `developer.oculus.com` address, which 301-redirects. It now records the canonical `developers.meta.com` address quoted above.

✔ **Validated** — I have read the original and agree it supports the sentence.

#109 avaturn — Avaturn (2024) *Ch.4 Methodology → Avatar Platforms and Rendering Constraints*

Sentence in thesis: **[Block covered by this citation] Avaturn is an avatar creation service that reconstructs a personalised rigged avatar with facial blendshape targets from three uploaded photographs ().**

Usage: First use of this source.

Original source: **THE CITED PAGE DOES NOT SUPPORT THIS SENTENCE.** `https://docs.avaturn.me/docs/what-is-avaturn/` was fetched on 2026-08-25 and checked for each of the four claims. It states none of them. **What the page does say:** Avaturn is "a realistic avatar creator" that lets users "customize body shape, clothes, accessories, hairstyles, shoes and glasses" and returns "a compiled ready-to-use 3D model". **Absent from the page:** any statement of *how many photographs* are required, whether the avatar *comes rigged*, whether it carries *facial blendshapes or morph targets*, and the *export format*.

How to check: Fetched from `https://docs.avaturn.me/docs/what-is-avaturn/` on 2026-08-25, and the site index at `https://docs.avaturn.me/` attempted as well; the documentation is a JavaScript-rendered application and serves no readable content to a fetcher. **RESOLVED 2026-08-25. Nidanur confirmed the three-photograph figure directly, from having built the avatars used in this thesis with the service. No change to the thesis text. What the check established, kept on record.** The cited page states none of the four claims: not the photograph count, not the rig, not the blendshape targets, not the export format. It says only that Avaturn is "a realistic avatar creator returning "a compiled ready-to-use 3D model. **Why the citation is nonetheless correct as it stands.** `avaturn` is a `@misc` reference to the service itself, in the same role as `mixamo` and `metasdk`: it identifies the platform, it is not offered as evidence for the specifications. Those specifications are facts of this project's own pipeline, established by creating the avatars, and §5.2 of the writing rules places implementation detail grounded in the project's own work outside the scope of citation. The blendshape stream is documented directly in the Implementation chapter. **Residual risk, stated rather than hidden:** a reader who follows this citation will not find the photograph count at the other end. If a documentation URL that states it is ever located, pointing the bib entry there would close the gap entirely.

✔ **Validated** — I have read the original and agree it supports the sentence.

#110 gltf2 — Khronos Group (2017) *Ch.4 Methodology → Avatar Platforms and Rendering Constraints*

Sentence in thesis: **[Block covered by this citation] The service exports its output as .glb files, which implement the glTF 2.0 standard and bundle geometry, PBR materials and skeletal rigging within a single self-contained file.** *[Since Unity does not natively support this format at runtime, an additional import package is required to load the avatar. Once loaded, it functions as a standard skinned mesh with PBR materials and unrestricted access to the URP rendering pipeline. The proposed NPR shaders can thus be implemented directly. Accordingly, Avaturn served as the primary platform for the final user study, since it is the only platform among the three that combines unrestricted shader access with personalised facial reconstruction and full blendshape animation. This complemented the prototyping workflow established with Mixamo and the live tracked animation available on the Meta Avatars SDK. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: **NOT READ. The specification could not be retrieved from this machine.** `https://registry.khronos.org/glTF/specs/2.0/glTF-2.0.html` returns **HTTP 403 Forbidden** to a fetcher, and two mirrors in the Khronos GitHub repository (`specification/2.0/README.md` and `specification/2.0/specification.adoc`) return the repository README and **HTTP 404** respectively. No copy is held in the Articles folder. **Claims awaiting a source:** that `.glb` is the binary container of glTF 2.0, that it is *self-contained* in a single file, and that it carries *geometry*, *PBR materials* and *skeletal rigging* together.

How to check: Attempted `https://registry.khronos.org/glTF/specs/2.0/glTF-2.0.html` and two GitHub mirrors on 2026-08-25. All three failed. **Starred on Nidanur's instruction, 2026-08-25, not on a source reading. Recorded so the basis is not mistaken later. Assessment of the risk, which is low.** Every claim is standard, uncontested description of the format and is independently confirmed by this project's own pipeline: the Avaturn avatars used in the thesis were delivered as `.glb` files and imported to Unity carrying exactly these three things, the mesh, the PBR material set and the rig with blendshape targets. The bib entry's bibliographic fields, Khronos Group, glTF 2.0 Specification, 2017, are not in doubt. **To close it properly:** the specification is open and freely readable in a browser even though it refuses automated fetches. Opening §3.2 (glTF Basics), §3.7 (Materials) and §3.10 (Skins), plus the GLB section of §4, would supply the four quotes and let this entry be upgraded from confirmed-by-use to confirmed-by-source.

✔ **Validated** — I have read the original and agree it supports the sentence.

#111 mcdonnell2012 — McDonnell et al. (2012) *Ch.4 Methodology → NPR Technique Design*

Sentence in the-sis: **[Block covered by this citation]** NPR in computer graphics encompasses a wide range of abstraction techniques and these vary in the level of abstraction they produce. **Neither extreme of that range serves an avatar well, since intermediate realism lowers appeal and heavy abstraction weakens perceived expressive intensity.** *[Selecting an appropriate degree therefore requires calibrating the balance between realism and visual expressiveness. — follows the citation, uncited.]*

Usage: Use 6 of this source.

Original source: **(1)** *"intermediate realism lowers appeal", stated outright by the authors* ← [§Discussion:] "highly abstract styles were considered appealing, whether still or moving. However, **render styles that were in the middle of the range between abstract and realistic were considered unappealing**, and the most unappealing character was considered even more so when motion was applied." **(2)** *the same result from the statistics* ← [§6:] "A main effect of render style was found ($F_{9,288} = 26.54$, $p \approx 0$). Post-hoc analysis showed that **four of the most abstract and the two most realistic styles were rated as the most appealing** (ToonPencil, ToonFlat, ToonShaded, ToonCG, HumanHQ1, HumanHQ2) and **significantly more appealing than all other styles** ($p < 0.05$ in all cases)." **(3)** *the other half of the thesis sentence is not this paper's* ← "heavy abstraction weakens perceived expressive intensity" is carried by the co-cited `wallraven2007`, §5: "one should, in general, **avoid high degrees of abstraction** for stylization…as these **adversely influence the perceived intensity and sincerity**."

How to check: Read directly from `Articles/01_uncanny_valley/McDonnell2012_RenderMeReal.pdf`, §6 and Discussion. **EXCERPT UPGRADED 2026-08-25, prompted by Nidanur questioning whether the sentence was supported.** The entry previously carried only quote (2), the post-hoc appeal ranking. That reaches "intermediate realism lowers appeal" only by inference: the reader has to notice that six of the ten styles sit at the two ends and conclude that the remaining four in the middle are the unappealing ones. Quote (1) makes the same claim in the authors' own words and was missed on the first pass. The support was always there, but the entry understated it. **The sentence is a joint citation and each key carries one half.** `mcdonnell2012` carries the appeal half and measured no expressiveness scale at all, its six scales being realism, appeal, re-assuring, familiar, friendly and trustworthy. `wallraven2007` carries the expressive-intensity half. Neither key supports the whole sentence alone, and the pairing is therefore necessary rather than decorative. **Incidental confirmation:** quote (1)'s phrase "whether still or moving" independently corroborates the still-versus-moving correction applied to this key in Related Work on the same day.

✔ **Validated** — I have read the original and agree it supports the sentence.

#112 wallraven2007 — Wallraven et al. (2007) *Ch.4 Methodology → NPR Technique Design*

Sentence in the-sis: **[Block covered by this citation]** NPR in computer graphics encompasses a wide range of abstraction techniques and these vary in the level of abstraction they produce. **Neither extreme of that range serves an avatar well, since intermediate realism lowers appeal and heavy abstraction weakens perceived expressive intensity.** *[Selecting an appropriate degree therefore requires calibrating the balance between realism and visual expressiveness. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *that heavy abstraction costs expressive intensity* ← [§5 Conclusion:] "one should, in general, **avoid high degrees of abstraction for stylization** (this is true both for real-world data and for animated sequences) as these **adversely influence the perceived intensity and sincerity**."

How to check: Read directly from `Articles/03_npr_rendering/wallraven2007.pdf`, §5. **Correctly attributed, and note the scope is explicit.** Unlike their recognition result, which splits by stimulus domain and required a correction in Related Work, the authors state this one as holding "both for real-world data and for animated sequences. It therefore transfers to avatars without qualification.

✔ **Validated** — I have read the original and agree it supports the sentence.

#113 isenberg2003 — Isenberg et al. (2003) *Ch.4 Methodology → NPR Technique Design → V1 — Geometric Silhouette Outline*

Sentence in the-sis: **[Block covered by this citation] Object silhouettes are a defining visual feature of nonphotorealistic rendering.**

Usage: Use 5 of this source.

Original source: **(1)** ← [pull quote and introduction, journal p. 28:] "**Generating object silhouettes lies at the heart of nonphotorealism.**"

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, p. 28. **Correctly attributed**, the survey's own framing sentence.

✔ **Validated** — I have read the original and agree it supports the sentence.

#114 raskar1999 — Raskar & Cohen (1999) *Ch.4 Methodology → NPR Technique Design → V1 — Geometric Silhouette Outline*

Sentence in the-sis: **[Block covered by this citation]** Accordingly, a reliable outline generation technique was required as the foundation for all NPR techniques developed in this thesis. The geometric silhouette method was selected since it produces outlines without an edge detection pass. **It generates an outline shell by enlarging back-facing polygons and relies on GPU face culling to form the outline.** *[The method operates across two dedicated stages: — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *the enlargement, in the authors' own term "fattening"* ← [§3.3 View-dependent Modification of Back-facing Polygons:] "Assuming polygons are convex, **constant width silhouette edges can be drawn using 'fat' back-facing polygons** so that the projection of extra area in the image plane is of constant width. The back-facing polygon is **fattened by pushing outwards each edge** by a distance proportional to $z/(V \cdot N_B)$ normal to the edge." [Figure 4 caption:] "The backfacing polygon in (a) is **fattened by pushing the edges outwards** and creating a triangle fan (b). When rendered in black, they appear as silhouettes (c)." **(2)** *the face culling* ← [§3, algorithm listing:] "**Enable back face culling**, set depth function to 'Less Than'. Render (front facing) polygons in white. **Enable front face culling**, set depth function 'Equal To'. **Draw (back-facing) polygons in black**." **(3)** *that it runs per frame with no preprocessing* ← "**To fatten a polygon at run time involves shifting the vertices of the polygon for every new view.**"

How to check: Read directly from `Articles/03_npr_rendering/Raskar -Image Precision Silhouette Edges.pdf`, §3 and §3.3. **EXCERPT REPLACED 2026-08-25.** This entry had inherited the quote written for the *Related Work* use of the same key, which describes the **translation** variant, "translate back-facing polygons towards the camera. The Methodology sentence describes the **enlargement** variant instead, so the stored excerpt was documenting a different technique from the one the sentence states. The quotes above are the enlargement passages. **Both variants are Raskar and Cohen's, and they rank them.** The paper rejects translation for this purpose in its own words: "It is easy to see why **translation** of back-facing polygons **will not create uniform line width** silhouettes. The width of the resulting silhouette is dependent on the orientation of the back-facing polygon. . . " Enlargement is introduced precisely to fix that. This makes the thesis's split coherent: Related Work presents the simpler translation form to explain the idea, and Methodology adopts the constant-width enlargement form for the implementation. **Naming corrected the same day.** The technique was called *inverted hull* in 20 places across four chapters. That term appears in **no article held for this thesis**, not in Raskar and Cohen, not in Isenberg et al., and not in Real-Time Rendering 4th edition. It is a practitioner term from real-time graphics tooling. A reader following either citation would not have found the technique under that name. It is now called the *geometric silhouette outline* throughout, which matches the framing already used in Related Work, "Silhouette rendering. . . derives line structure from geometry rather than from the lighting term.

✔ **Validated** — I have read the original and agree it supports the sentence.

#115 isenberg2003 — Isenberg et al. (2003) *Ch.4 Methodology → NPR Technique Design → V1 — Geometric Silhouette Outline*

Sentence in thesis: **[Block covered by this citation] This approach is well suited to real-time NPR avatar rendering because it requires no offline mesh preprocessing.** *[It also executes entirely within the GPU vertex transformation stage. As a result, its computational cost scales predictably across assets of varying complexity. However, the expanded shell and the original surface may share similar depth values and this can introduce z-fighting artefacts along the silhouette boundary. A depth bias prevents this by shifting the outline geometry slightly towards the camera in clip space. — follows the citation, uncited.]*

Usage: Use 6 of this source.

Original source: **(1)** *that preprocessing disqualifies a method for real-time animation* ← [Table 1, "Choosing the best silhouette detection algorithm", journal p. 29:] Requirement "**Animate model in real time** beyond a mere flight through the scene" → Recommendation "Real-time technique. **Object space silhouette detection method that uses preprocessing is not appropriate.**" **(2)** *what preprocessing means here* ← [§Precomputation methods:] "To speed up the process of silhouette edge determination, various authors have developed methods that accomplish some type of **preprocessing**. The preprocessing stage sets up data structures used to find silhouette edges more quickly."

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, Table 1 and §Precomputation methods. **Correctly attributed, no change.** This is the survey's own recommendation table rather than a restatement of another paper, so `isenberg2003` is right here even though the neighbouring technique sentence is not. **Worth knowing:** Raskar and Cohen make the same claim about their own method in their abstract, "The method does not require any preprocessing or adjacency information and hence is applicable for dynamic scenes", so either key would be defensible. The survey states it as a general design rule, which is what the thesis sentence does, so it stays.

✔ **Validated** — I have read the original and agree it supports the sentence.

#116 isenberg2003 — Isenberg et al. (2003) *Ch.4 Methodology → NPR Technique Design → V1 — Geometric Silhouette Outline*

Sentence in the-sis: **[Block covered by this citation]** Despite its stability and low rendering cost, the geometric silhouette method has several limitations when applied to avatar rendering. shows the results across all three platforms. Surfaces with degenerate or coincident normals, such as flat planes, yield no visible outline because displacing vertices along identical surface normals does not generate an expanded shell. In addition, the method does not register interior surface boundaries. Consequently, creases between body regions, clothing seams and facial contours receive no edge marking. **The expanded shell yields only the outer contour, which in the terminology of Isenberg et al. excludes the internal silhouettes lying inside it.** *[This limitation is shared by geometry-based outline methods and motivates the interior edge detection techniques introduced in V3. — follows the citation, uncited.]*

Usage: Use 7 of this source.

Original source: **(1)** *the contour versus internal silhouette distinction* ← [§Definitions and terminology, journal p. 28:] “We define a **contour as the subset of the silhouette that separates the object from the background**. Also, the subset of the silhouette, which in 2D divides one portion of the object from another one of the same object, is not part of the contour. Those lines are sometimes called **internal silhouettes**.”

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, p. 28. **Citation placement checked and correct.** The three sentences before this one report limitations observed in this project’s own implementation and reference this thesis’s own figure. They are uncited, which §5.1 rule 5 requires. The citation attaches only to the sentence that uses the survey’s terminology.

✔ **Validated** — I have read the original and agree it supports the sentence.

#117 lake2000 — Lake et al. (2000) *Ch.4 Methodology → NPR Technique Design → V2 — Toon Shading*

Sentence in thesis: **[Block covered by this citation]** Toon shading achieves stylisation by simplifying the tonal representation of a surface rather than reproducing physically accurate illumination. Discretising the illumination response into uniform tonal regions produces the characteristic flat appearance associated with hand-drawn animation. **Correspondingly, the approach is well suited to interactive character rendering.** *[Toon shading was therefore selected as the V2 baseline because it provides a controlled starting point from which each subsequent technique introduces a targeted increase in stylisation. However, implementing this stylisation across the three avatar platforms requires adaptations that account for differences in each platform's rendering architecture. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *that the technique is a real-time one* ← [Abstract, Results:] "In this paper we introduce **new real-time algorithms for cartoon shading**, pencil sketching, and silhouette edge detection and rendering." The paper's own title is *Stylized Rendering Techniques For Scalable **Real-Time** 3D Animation*. **(2)** *that the tonal range runs from flat to smooth* ← [§3:] "single-color flat shading (one texel value) at one end of the spectrum, and **smooth diffuse shading at the other**. By providing the user with just a few parameters, our Painter offers a large variety of stylized and cartoon rendering effects."

How to check: Read directly from `Articles/03_npr_rendering/lake et al Stylized rendering.pdf`, Abstract and §3. **Correctly attributed.** Both halves are the authors' own contribution rather than restated prior work.

✔ **Validated** — I have read the original and agree it supports the sentence.

#118 lake2000 — Lake et al. (2000) *Ch.4 Methodology → NPR Technique Design → V2 — Toon Shading*

Sentence in thesis: **[Block covered by this citation]** where s represents the surface lighting response and n defines the number of shading bands. **This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading.** *[shows the resulting appearance at the default band configuration used in this thesis, with the specific value and its selection criteria given in . The resulting tonal structure consists of highlight, midtone, shadow and deep shadow regions. Furthermore, a rim highlight was added to improve visual separation between the avatar and its background where flattened shading reduces depth cues (). Finally, scene cast shadows were deliberately excluded because they introduce an additional visual variable that could interfere with evaluating the effect of tonal quantisation in isolation. — follows the citation, uncited.]*

Usage: Use 4 of this source.

Original source: **(1)** *the subfigure label "Toon shading"* ← [§3.1 Cartoon Shading:] "Rather than shading the character to give it a three-dimensional appearance, the cartoonist typically uses **solid colors that do not vary across the materials they represent**. . . our hard shading technique finds a transition boundary and shades each side of the boundary with a solid color." The flat bands separated by hard edges visible in the figure are exactly this effect.

How to check: Read directly from `Articles/03_npr_rendering/npar2000_lake_et_al.pdf`, §3.1. The subfigure carries only a two-word label, so the quote supports the technique it names rather than a sentence.

✔ **Validated** — I have read the original and agree it supports the sentence.

#119 barla2006 — Barla et al. (2006) *Ch.4 Methodology → NPR Technique Design → V2.2 — X-Toon Shading*

Sentence in thesis: **[Block covered by this citation] X-Toon shading extends conventional toon shading by replacing the 1D toon ramp with a 2D ramp texture, where the horizontal axis represents the lighting response and the vertical axis controls the abstraction level.** *[This second axis allows the surface appearance to be adjusted independently of the lighting conditions and this provides finer stylisation control. A higher abstraction level reduces local tonal contrast and softens transitions between shading regions towards a flatter, more uniform appearance. illustrates the ramp texture used in this work. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: **(1)** *the 1D to 2D replacement and what the second axis carries* ← [§1:] "we **replace the 1D texture with a 2D texture whose second dimension corresponds to the desired "tone detail"**, which can vary with depth or surface orientation. This supports effects such as **levels-of-abstraction**, aerial perspective, depth-of-field. . . "

How to check: Read directly from `Articles/03_npr_rendering/x-toon.pdf`, §1. **Correctly attributed.** This is the paper's stated contribution, announced in the first person.

✔ **Validated** — I have read the original and agree it supports the sentence.

#120 barla2006 — Barla et al. (2006) *Ch.4 Methodology → NPR Technique Design → V2.2 — X-Toon Shading*

Sentence in thesis: **[Block covered by this citation]** Barla et al. also described a shape detail extension. **The extension blends the surface normals of the mesh with those of a simplified copy of the same shape.** *[This blending was incorporated into the Mixamo and Avaturn implementations. It reduced local surface variation and gave more coherent stylised shading. — follows the citation, uncited.]*

Usage: Use 4 of this source.

Original source: **(1)** *the normal-field blend and its control parameter* ← [§1:] ". . . of normals taken from an **"abstracted shape."** A **global shape detail parameter determines the degree of interpolation between the original and abstracted normal fields**." **(2)** *why it exists* ← [same section:] "in conventional toon shading, every surface location is rendered with full accuracy, so that **even small shape features** are. . . " shown, which the abstracted normal field suppresses.

How to check: Read directly from `Articles/03_npr_rendering/x-toon.pdf`, §1. **Correctly attributed, and checked for an intermediary (§5.3a).** The shape-detail blend carries no inline credit in X-Toon and no corresponding entry appears anywhere in its eleven-item reference list, so it is the authors' own. A suspected Rusinkiewicz or Cignoni origin was looked for and does not exist.

✔ **Validated** — I have read the original and agree it supports the sentence.

#121 barla2006 — Barla et al. (2006) *Ch.4 Methodology → NPR Technique Design → V2.2 — X-Toon Shading*

Sentence in thesis: **[Block covered by this citation] Barla et al. further described a general attribute mapping mechanism that maps a chosen surface attribute onto the abstraction axis and they demonstrated it with depth-based and orientation-based maps.** *[Three driving modes were evaluated here: depth, curvature and manual control. Depth mode and manual mode coincide on the Meta Avatars SDK, since the avatar is framed head and shoulders and its visible surface spans too narrow a depth range for the depth-driven abstraction value to vary. Curvature mode stays distinct there, because it responds to local surface orientation rather than to distance from the camera. The user study used a fixed camera distance much closer to the avatar than the general platform comparison setup. At that distance the depth and curvature modes produce almost no variation, as explains in detail. X-Toon produced a softer and more controllable stylised appearance than V2, since the stylisation level can be adjusted independently of the illumination response. It was consequently retained as the toon shading component in all later technique combinations. — follows the citation, uncited.]*

Usage: Use 5 of this source.

Original source: **(1)** *the mechanism and its generality* ← [§3:] “the designer must choose an **attribute (e.g., orientation or depth)** that will control the tone detail, and provide functions called **attribute maps** that map the attribute to a detail value D at each location on a surface. This **formulation is general in that any attribute can be chosen**, depending on the…”

How to check: Read directly from `Articles/03_npr_rendering/x-toon.pdf`, §3. **Correctly attributed, and the split required by §5.1 rule 5 is in place.** The cited sentence credits Barla et al. with the mechanism and with demonstrating depth and orientation maps. The following sentence, which states that three driving modes were evaluated in this project, carries no citation.

✔ **Validated** — I have read the original and agree it supports the sentence.

#122 isenberg2003 — Isenberg et al. (2003) *Ch.4 Methodology → NPR Technique Design → V3 — Screen Space Normal Edge Detection*

Sentence in the-sis: **[Block covered by this citation] Silhouette extraction techniques can be classified by how they detect edges, into image space, object space and hybrid approaches.** *[Unlike the object-space outline method introduced in V1, image-space techniques analyse per-pixel information generated during rendering to identify visually significant boundaries. One commonly used cue is the surface normal, whose orientation changes abruptly at creases, folds and other geometric features. Access to that cue gives image-space techniques the interior geometric structure that silhouette methods alone leave undetected, as the work reviewed in demonstrates. — follows the citation, uncited.]*

Usage: Use 8 of this source.

Original source: **(1)** ← [§Classification, journal p. 29:] "we classify the approaches according to how they solve each of these problems. . . we distinguish between **image space algorithms** that only operate on image buffers, **hybrid algorithms** that perform manipulations in object space but yield the silhouette in an image buffer, and **object space algorithms** that perform all calculations in. . . "

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, p. 29. **Correctly attributed**, this three-way split is the survey's own organising scheme and is what its Table 1 is built on.

✔ **Validated** — I have read the original and agree it supports the sentence.

#123 roshaan2026 — Roshaan (2026) *Ch.4 Methodology → NPR Technique Design → V5 — Hierarchical Edge Detection with Multiple Cues*

Sentence in the-sis: **[Block covered by this citation] Hierarchical edge detection combines multiple complementary boundary cues into a single, unified representation rather than relying on any one signal alone.**

Usage: Use 3 of this source.

Original source: **(1)** *the three layers and their operators* ← [§2.2, Table 1 "Hierarchical Classification of Algorithms":] **Silhouette** (W_1) → **Depth Difference**; **Structure** (W_2) → **Scharr, Kirsch**; **Texture** (W_3) → **XDoG**. **(2)** *the fusion into one representation* ← [§2.3, Eq. 4:] $\mathrm{Edge}_{\mathrm{final}} = \max\{W_1 \cdot \mathrm{Silhouette}, W_2 \cdot \mathrm{Structure}, W_3 \cdot \mathrm{Texture}\}$, introduced as "To combine the results of all the implemented algorithms, we implement the method of **max-pooling**."

How to check: Read directly from `Articles/03_npr_rendering/Adaptive Hierarchical Edge Detection.pdf`, §§2.2–2.3 and Table 1. **Correctly attributed.**

✔ **Validated** — I have read the original and agree it supports the sentence.

#124 roshaan2026 — Roshaan (2026) *Ch.4 Methodology → NPR Technique Design → V5 — Hierarchical Edge Detection with Multiple Cues*

Sentence in thesis: **[Block covered by this citation]** V5 adopts this approach as a post-processing architecture that operates on the rendered image rather than on the source texture. Depth discontinuities recover object silhouettes and surface normal discontinuities reveal geometric structure. Neither cue reads surface colour, and this reduces the dependence on albedo quality identified in V4.2. Colour boundaries that leave no geometric trace require a third cue, and the colour layer uses the Roberts Cross operator for it. **The original formulation derives its structure layer from gradient operators instead of from surface normals.**

Usage: Use 4 of this source.

Original source: **This citation supports a CONTRAST, not a borrowing. It is attached to the one sentence in the block that describes what AHEAD does differently from V5.** **(1)** *that AHEAD's structure layer uses gradient operators* ← [§2.2, Table 1 "Hierarchical Classification of Algorithms":] **Structure (W_2) → Scharr, Kirsch**. The other two rows are Silhouette (W_1) → Depth Difference and Texture (W_3) → XDoG. **(2)** *that AHEAD has no normal layer at all, which is the stronger form of the claim* ← [Figure 4, pipeline architecture:] the framework's only inputs are a **Color Buffer** and a **Depth Buffer**, feeding an Adaptive Scalar (Luminance Gain), a Structure Layer (Kirsch/Scharr), a Texture Layer (XDoG) and a Silhouette Layer (Depth Difference), combined by Max-Pooling Fusion. **No normal buffer is read anywhere in the pipeline**, so a normal-based layer is not merely absent from Table 1, it is unavailable to the architecture as published.

How to check: Read directly from `Articles/03_npr_rendering/Adaptive Hierarchical Edge Detection.pdf`, §2.2 Table 1 and Figure 4. **Excerpt written 2026-08-25 after Nidanur queried how this entry was supported.** The entry had no source excerpt at all: an earlier override for this key matched the preceding entry and left this one bare, so the checker displayed the whole paragraph as the covered block with nothing to check it against. That is why it read as unsupported. **Citation placement checked and correct.** Five of the six sentences in this block describe this thesis's own design and carry no citation, which is what §5.1 rule 5 requires: the adoption of a post-process architecture, the depth and normal layers of V5, the choice of Roberts Cross for the colour layer, the substitution of the normal cue, and the per-layer weighting. `roshaan2026` attaches only to the sentence reporting AHEAD's own structure layer. **Why the contrast matters, and why it is worth stating.** V5 is not a reimplementation of AHEAD. It keeps the three-layer hierarchy and the max-pooling fusion but replaces the structure cue, and the following sentence gives the reason in the thesis's own voice: baked illumination in the source texture would otherwise register as geometric detail. Without this sentence a reader comparing V5 against the cited paper would find a normal layer that the paper does not have and could take it for a misreading of the source rather than a deliberate departure. **CORRECTION 2026-08-26.** An earlier draft of this entry implied that the Roberts Cross operator does not appear in AHEAD. **It does.** §2.1.5 is a dedicated subsection on it, citing Roberts (1963) as its reference [14], and Figure 1(c) prints the two kernels under the caption "Edge detection kernels used in the Structure Layer. What is accurate is that AHEAD *catalogues and then rejects* it: §2.1.5 warns that "it is very sensitive to fine texture details. It picks up too many details cluttering the screen. . . too much detail might create a harsh look, and §2.2 Selection of Algorithms explains the choice of Scharr and Kirsch instead, which is what Table 1 records. The thesis therefore adopts, for its colour layer, an operator the source considered and set aside for a different application. That is not a contradiction of the source and needs no separate citation, but the distinction should not

✔ **Validated** — I have read the original and agree it supports the sentence.

#125 roshaan2026 — Roshaan (2026) *Ch.4 Methodology → NPR Technique Design → V5 — Hierarchical Edge Detection with Multiple Cues*

Sentence in the-sis: **[Block covered by this citation]** V5 substitutes the normal cue there, because baked illumination in the source texture would otherwise register as geometric detail. Each layer carries an independent weight and the responses are blended through adjustable parameters so that the contribution of each cue can be controlled interactively. **Since a gradient operator produces weak responses in dark regions that fall below the detection threshold, a further mechanism from the same framework scales the edge response inversely with local luminance.** *[V5 applies this gain on the Meta Avatars SDK, where the composited output is darker than the raw albedo the other two platforms read. — follows the citation, uncited.]*

Usage: Use 5 of this source.

Original source: **(1)** *the problem the gain solves* ← [§2.4 Adaptive Feature Enhancement:] "in darker regions of the game, the **standard gradient operator produces a weak gradient response that frequently drops below the binary threshold**." **(2)** *that the scalar is inverse in luminance* ← [same section, Eq. 6:] "The **adaptive scalar $S(x, y)$ is modeled as a modified inverse-linear function of the local pixel luminance** $L(x, y)$", with $S(x, y) = \min\left(\frac{1}{k \cdot L(x,y)+\epsilon}, S_{max}\right)$, $k = 3.0$, $\epsilon = 0.1$, $S_{max} = 4.0$. The section closes "**Lower luminance areas have hi[gher gain]**". **(3)** *that it is a distinct stage of the same framework* ← [Figure 4, pipeline:] the **Adaptive Scalar (Luminance Gain)** block sits alongside the Structure, Texture and Silhouette layers.

How to check: Read directly from `Articles/03_npr_rendering/Adaptive Hierarchical Edge Detection.pdf`, §2.4 and Figure 4. **New citation added 2026-08-26, closing a chapter-ownership gap.** The adaptive gain was credited in the Implementation chapter but never introduced in Methodology, so a mechanism taken from a cited paper appeared only in a chapter of code detail. §6.1 assigns the conceptual description to Methodology. The gap is the same shape as the one found for `blinn1977` on 2026-08-25, but the opposite resolution applies: Blinn-Phong is standard shading vocabulary and its citation was dropped, whereas this gain is genuinely AHEAD's contribution and needed introducing rather than removing. **§5.1 rule 5 split in place.** The cited sentence states only what the framework does. The sentence after it, "V5 applies this gain on the Meta Avatars SDK, where the composited output is darker than the raw albedo the other two platforms read", describes this project's own use and carries no citation. **Origin check (§5.3a), and a deliberate omission.** Roshaan justifies the gain perceptually by the **Weber-Fechner Law**, credited inline to reference [6], and separately credits the AGC framing to "photographic tone reproduction [13]. Neither source is held. The thesis therefore uses Roshaan's own engineering observation instead, that weak gradients in dark regions fall below the detection threshold, which requires no second key. If the perceptual justification is ever wanted in the text, the Weber-Fechner reference must be obtained first.

Validated — I have read the original and agree it supports the sentence.

#126 winnemoller2006 *Ch.4 Methodology → NPR Technique Design → V5 — Hierarchical Edge Detection with Multiple Cues*

Sentence in the-sis: **[Block covered by this citation]** compares the results across the three avatar platforms. V5 detects a broader range of structural boundaries on the Mixamo and Avaturn rendered frames. The depth threshold has little visual influence, since the geometric silhouette outline already provides the silhouette. The normal and colour thresholds instead control the density of interior contours. These contours trace geometric curvature, appearing along the underside of the nose and the outer lip edges, but do not produce clear boundaries between adjacent facial segments. Furthermore, the colour gradient layer occasionally responds to illumination gradients in the rendered image, particularly around the nose bridge where shading transitions are interpreted as texture edges. Although combining the remaining cues still produces richer interior detail than the previous techniques, the absence of screen-space depth prevents silhouette detection on the Meta Avatars SDK. The composited SDK output contains baked lighting gradients absent from the raw albedo, making the texture layer more prone to false responses on this platform. **The colour layer therefore replaces the Roberts Cross computation with the XDoG operator of Winnemöller et al. on this platform.** *[This reduced these false responses, as (b) shows. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *τ and ϕ_e, the parameters the thesis shader implements* ← [§3.2:] "**The parameter τ in Equation 4 controls the amount of center-surround difference required for cell activation, and ϕ_e controls the sharpness of the activation falloff.**" **(2)** *the operator itself* ← [Eq. 4:] $D = 1$ if $(S_{\sigma_e} - \tau \cdot S_{\sigma_r}) > 0$, otherwise $1 + \tanh(\phi_e \cdot (S_{\sigma_e} - \tau \cdot S_{\sigma_r}))$.

How to check: Read directly from `Articles/videoabstraction.pdf`, §3.2 and Eq. 4. **Methodology site of the same correction.** See the Implementation entry for this key for the full account. In short: the operator is the 2006 paper's, *XDoG* is the 2012 paper's name for it, and the thesis shader's `_XDoGTau` and `_XDoGPhi` are τ and ϕ_e from Equation 4 above. Both keys are cited.

Validated — I have read the original and agree it supports the sentence.

#127 winnemoller2012 — Winnemöller et al. (2012) *Ch.4 Methodology → NPR Technique Design → V5 — Hierarchical Edge Detection with Multiple Cues*

Sentence in thesis: **[Block covered by this citation]** compares the results across the three avatar platforms. V5 detects a broader range of structural boundaries on the Mixamo and Avaturn rendered frames. The depth threshold has little visual influence, since the geometric silhouette outline already provides the silhouette. The normal and colour thresholds instead control the density of interior contours. These contours trace geometric curvature, appearing along the underside of the nose and the outer lip edges, but do not produce clear boundaries between adjacent facial segments. Furthermore, the colour gradient layer occasionally responds to illumination gradients in the rendered image, particularly around the nose bridge where shading transitions are interpreted as texture edges. Although combining the remaining cues still produces richer interior detail than the previous techniques, the absence of screen-space depth prevents silhouette detection on the Meta Avatars SDK. The composited SDK output contains baked lighting gradients absent from the raw albedo, making the texture layer more prone to false responses on this platform. **The colour layer therefore replaces the Roberts Cross computation with the XDoG operator of Winnemöller et al. on this platform.** *[This reduced these false responses, as (b) shows. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **(1)** *the name XDoG, which is this paper's* ← the 2012 compendium is the work that names and consolidates the extended difference-of-Gaussians operator as **XDoG** and gives its reparameterisation. **(2)** *its own credit for the underlying operator* ← [§2.4:] "Inspired by those models, **Winnemöller et al. [7]** generated edge images using a DoG variant in which the strength of the inhibitory effect of the larger Gaussian is allowed to vary…" Reference [7] is "Winnemöller, H., Olsen, S.C., Gooch, B.. Real-time video abstraction. ACM Trans Graph 2006;25(3):1221–1226."

How to check: Read directly from `Articles/03_npr_rendering/xdog_winnemoller2012.pdf`, §2.4. **Dual citation, deliberate, applied 2026-08-25.** This key carries the *name* XDoG and the compendium framing, both genuinely its own. It does not carry the operator: the 2012 paper credits that inline to its own reference [7], the 2006 paper, which is now cited alongside it as `winnemoller2006`. The thesis shader settles which parameterisation is in use, since `_XDoGTau` and `_XDoGPhi` are τ and ϕ_e from the 2006 Equation 4, and the 2012 reformulation replaces τ with p.

Validated — I have read the original and agree it supports the sentence.

#128 kyprianidis2009 — Kyprianidis et al. (2009) *Ch.4 Methodology → NPR Technique Design → V6 — Kuwahara Painterly Filter*

Sentence in the-sis: **[Block covered by this citation] The anisotropic Kuwahara formulation aligns an elliptical sampling window with local image structure rather than with the fixed rectangular subregions of the classical filter.** *[V6 adopts this formulation, set out in . Orientation-aligned marks were the property required here, since brushwork ignoring surface direction reads as a uniform blur, not as a painted surface. — follows the citation, uncited.]*

Usage: Use 6 of this source.

Original source: **(1)** ← [Abstract:] "It is based on a generalization of the Kuwahara filter that is **adapted to the local shape of features, derived from the smoothed structure tensor**. Contrary to conventional edge-preserving filters, our approach [produces a] **painting-like flattening effect along the local feature directions** while preserving shape boundaries." **(2)** *that the classical form uses fixed rectangles* ← [§2:] "The **Kuwahara and Tomita-Tsuji filter use rectangular subregions**, while the Nagao-Matsuyama filter uses a rotated elongated bar mask. The use of rectangular subregions results in visible artifacts in the output."

How to check: Read directly from `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`, Abstract and §2. **Correctly attributed.**

Validated — I have read the original and agree it supports the sentence.

#129 isenberg2003 — Isenberg et al. (2003) *Ch.4 Methodology → Avaturn Avatar Preprocessing*

Sentence in the-sis: **[Block covered by this citation]** The head albedo texture was preprocessed in Blender in order to reduce the edge detection artefacts caused by the reconstructed albedo texture identified in . The avatar was then re-exported as a GLB file. Four texture operations were applied to address two distinct problems in the head albedo: baked illumination variation that introduced spurious edge responses at boundaries unrelated to surface structure and the absence of a strong tonal cue at the lip boundary. **The first of these is the characteristic liability of colour-buffer edge detection.**

Usage: Use 9 of this source.

Original source: **(1)** *"the characteristic liability of colour-buffer edge detection"* ← [§Image space algorithms, journal p. 29:] "The easiest way to find significant lines would be to **detect edges in the color buffer**. This, however, **doesn't necessarily detect silhouettes since changes in shading and texturing can cause erroneous edge detection**."

How to check: Read directly from `Articles/03_npr_rendering/Isenberg_2003_ADG.pdf`, journal p. 29. **Added 2026-08-24** so that the preprocessing section names the problem it solves as the one the survey literature already identifies. The corresponding Related Work sentence is the entry for this same passage in §3.3. This is a pointer rather than a re-explanation: the mechanism is not restated here.

Validated — I have read the original and agree it supports the sentence.

#130 demsar2006 — Demšar (2006) *Ch.4 Methodology → Parameter Adjustment Study*

Sentence in thesis: **[Block covered by this citation]** Participants were required to assign a unique rank to every image in the round. Any ballot with a missing rank or a tied rank was excluded. Images were converted to Borda-count points in each round, where the image ranked last received 1 point and the image ranked first received points equal to the number of options. The totals were then summed across participants. **An omnibus Friedman test established whether the ranking within a round differed from chance and a pairwise Wilcoxon signed-rank test compared the configuration ranked first against its closest competitor.** *[The same nonparametric approach is applied to the advisor ranking task in the main study (). reports the outcome of each round and the reasoning that fixed each value. — follows the citation, uncited.]*

Usage: First use of this source.

Original source: [Abstract / Recommendation:] "We recommend Friedman test, with the corresponding post-hoc tests for comparison of more classifiers over multiple data sets." [Sec. 4.1:] "For a two classifier comparison, the Wilcoxon signed ranks test is more appropriate . . . we use average ranks . . . Friedman test . . . followed by post-hoc tests." [Sec. 3.1:] Rank aggregation across datasets: assigns rank 1 to best, rank 2 to second, averages across datasets — equivalent to Borda count aggregation.

How to check: Read directly from `Articles/demsar06a.pdf`. Confirm this passage supports the sentence above.

Validated — I have read the original and agree it supports the sentence.

#131 dubosc2025 — Dubosc et al. (2025) *Ch.4 Methodology → User Study Design → Objective and Hypotheses*

Sentence in thesis: **[Block covered by this citation] Stylising a character lowers perceived eeriness: rendering the same avatar in stylised and semi-realistic form produced significantly lower eeriness ratings than the realistic version.** *[H1 tests whether the NPR composite reproduces that reduction between C2 and C3. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: "[§4.2.3 "Eeriness", Results, p. 11:] "The results show that the stylized character (m = 2.750, SD = 1.256, p = .05), and the semi-realistic character (m = 2.650, SD = 1.331, p = .026) were perceived as significantly less eerie than the realistic character (m = 3.350, SD = 1.703).""

How to check: File: `Articles/02_trust_and_avatar_perception/Effect_of_avatar_stylization_and_facial_expression.pdf`. **Replaced** `mori1970` **on 2026-08-22.** Mori never discusses avatars or rendering, so he could not carry this claim. Dubosc et al. is a within-subjects VR study (n = 40) rendering the *same* actress-driven avatar at three stylisation levels with animation, voice and identity held constant, which mirrors the C2-versus-C3 isolation. **Caveats:** the stylised-versus-realistic comparison sits exactly at $p = .05$ and the effect is small ($\eta^2 = .045$); the trustworthiness advantage is reported only as a trend ($p = .075$), so trust must not be described as a significant gain from this source.

Validated — I have read the original and agree it supports the sentence.

#132 canales2024 — Canales et al. (2024) *Ch.4 Methodology → User Study Design → Objective and Hypotheses*

Sentence in the-sis: **[Block covered by this citation] Prior work varied stylisation level within a fixed advisor role and scenario and found no significant effect on self-reported trust.** *[Benevolence was the one partial exception and there the realistic style trended lowest rather than highest. H2 tests whether this pattern holds when rendering style is the primary variable that changes between conditions. — follows the citation, uncited.]*

Usage: Use 6 of this source.

Original source: "[§10 Conclusion, p. 9:] "This study used a high stakes medical scenario in order to explore how people's tendency to trust was impacted by avatar stylization. Stylization had no significant impact on trust ratings. The Realistic avatars received lower ratings for benevolence on average; however, the differences were not significant.""

How to check: File: `Articles/02_trust_and_avatar_perception/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. Verified against the primary source. The advisor role (doctor) and scenario (kidney lump, second opinion) were fixed for every participant while style varied, and no significant effect on self-reported trust was found. One caveat worth knowing: §6.1 (p. 6) reports that Benevolence was 'the only category that show a significant main effect for Style (chi2(2) = 6.78, p = .034)', although post-hoc tests found no significant style pairs. Avatar identity was also varied simultaneously in the same design.

Validated — I have read the original and agree it supports the sentence.

#133 ho2010 — Ho & MacDorman (2010) *Ch.4 Methodology → User Study Design → Objective and Hypotheses*

Sentence in the-sis: **[Block covered by this citation] Eeriness is a distinct negative affective response rather than general dislike.**

Usage: Use 4 of this source.

Original source: "[Section 1 Plotting emotional responses to humanlike characters, p. 1508:] "Mori (1970) refers to negative shinwakan as bukimi, which translates as eeriness. However, eeriness is not the negative anchor of warmth. A person can be cold and disagreeable without being eerie–at least not eerie in the way that an artificial human being is eerie. In addition, the set of negative emotions that predict eeriness (e.g., fear, anxiety, and disgust) are more specific than coldness (Ho, MacDorman, & Pramono, 2008). Thus, shinwakan and bukimi appear to constitute distinct dimensions.""

How to check: File: `Articles/02_trust_and_avatar_perception/Ho2010UncannyValleyIndices.pdf`. Verified against the primary source. The passage states directly that eeriness is not simply the negative pole of warmth or likeability and is predicted by a more specific set of negative emotions. Reinforced in Section 3, p. 1512: 'Eeriness, which is conceptually distinct from negative warmth (i.e., interpersonal coldness), would need to be included in any set of indices on the uncanny valley.'

Validated — I have read the original and agree it supports the sentence.

#134 alipour2025 — Alipour et al. (2025) *Ch.4 Methodology → User Study Design → Objective and Hypotheses*

Sentence in thesis: **[Block covered by this citation] Its relationship with trust is not uniform, since direct, conditional and absent effects have all been reported.** *[H3 tests whether this relationship is detectable across all three conditions within each participant. — follows the citation, uncited.]*

Usage: Use 4 of this source.

Original source: [§4.4.1 General findings, pp. 10–11:] "A total of **41 studies** provided empirical evidence on the relationship between the UVE and trust or trust-related constructs/intentions. . . . Among the 41 coded studies, **19** were categorized as reporting a direct effect, **24** conditional or indirect effects, and **10** no effect."

How to check: Read directly from the Alipour et al. (2025) systematic review, §4.4.1, pp. 10–11. The thesis sentence states only the three outcome categories; the counts are given here for verification and are deliberately not repeated in the thesis text.

Validated — I have read the original and agree it supports the sentence.

#135 mori1970 — Mori (1970) *Ch.4 Methodology → User Study Design → Objective and Hypotheses*

Sentence in the-sis: **[Block covered by this citation] Uncanny valley theory places a real human speaker at the peak of the affinity curve, where human likeness is maximal and comfort is highest.**

Usage: Use 6 of this source.

Original source: **(1)** *that a real human sits at the top of the curve* ← [§The significance of the uncanny:] "In Figure 1, **a healthy person is at the top of the second peak**. And when we die, we fall into the trough of the uncanny valley." **(2)** *that maximal human likeness is the endpoint of the axis* ← [§The uncanny valley:] "Of course, **human beings themselves lie at the final goal of robotics**, which is why we make an effort to build humanlike robots." **(3)** *that the trough sits immediately before that peak* ← [same passage:] the descent from the second peak into the valley is the path Mori uses to explain death, "our impression of death can be explained by the **movement from the second peak to the uncanny valley**".

How to check: Read directly from `Articles/01_uncanny_valley/The Uncanny Valley- The Original Essay by Masahiro Mori.pdf` (MacDorman and Kageki translation), §§"The uncanny valley" and "The significance of the uncanny. **Precision note, no change needed.** Mori writes *second* peak, since his curve has an earlier local maximum at the humanoid toy robot. The thesis says "the peak of the affinity curve. That is correct rather than loose: the second peak is the global maximum of the curve, and the thesis sentence is about where a real human sits, not about how many local maxima the curve has.

Validated — I have read the original and agree it supports the sentence.

#136 gao2025 — Gao et al. (2025) *Ch.4 Methodology → User Study Design → Objective and Hypotheses*

Sentence in the-sis: **[Block covered by this citation] Gao et al. found a real speaker rated higher than a stylised agent modelled on that same speaker across all three trust dimensions.** *[H4 tests whether the abstraction responsible for H1's predicted reduction in eeriness closes the trust gap enough to reach this reference point, rather than assuming a real human speaker remains categorically more trusted than any avatar. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: "[Abstract:] "Through an experiment with 27 participants, our findings reveal that stylized virtual agents enhance user trust to a degree comparable to real style, while voice has a negligible effect on trust.""

How to check: **CONFIRMED 2026-08-17.** The claim matches the abstract exactly. **Separate bibliography error corrected the same day**: all seven given names in the `references.bib` entry were wrong (Yue/Yang, Yi/Yangbin, Guanyu/Guangtao, Hao/Honglei, Farbod/Fariba, Bo/Binge, Tianshu/Tao). Surnames were correct; given names were not.

Validated — I have read the original and agree it supports the sentence.

#137 canales2024 — Canales et al. (2024) *Ch.4 Methodology → User Study Design → Conditions*

Sentence in thesis: **[Block covered by this citation]** Using the same Avaturn geometry for both C2 and C3 holds avatar identity, facial structure and animation constant across the two avatar conditions. Render style is consequently the primary independent variable between them. **Comparable work crossed stylisation level with speaker identity, and identity produced the stronger effect throughout those results.** *[Holding the speaker constant removes that source of variance from the present comparison. The theoretically critical comparison is C2 versus C3. Since animation is held constant between these two conditions, any observed difference in trust or eeriness ratings is attributable primarily to the change in rendering style, with one material-level exception described in . C1 provides a second point of comparison for H4. This test asks whether C3 reaches the real-person reference point overall rather than isolating rendering style specifically, as animation naturalness and speaker identity also differ between C1 and both avatar conditions. — follows the citation, uncited.]*

Usage: Use 7 of this source.

Original source: "[§5.1 Design, p. 5; §1 Introduction, p. 1; §3.2 Dialogue Testing, p. 3:] "the full experiment uses a 3 (avatar identity) x 3 (avatar style) experimental design, but each participant sees only three avatars, featuring one of each identity and each style ... [§1, p. 1] Each doctor is a different person (varied identity) and each is represented at a different level of stylization ... [§3.2, p. 3] the avatar's voice must match the avatar, which means that a different actor must perform each avatar's dialogue. During the scenario, each avatar must also provide different answers to ensure believability.""

How to check: File: `Articles/02_trust_and_avatar_perception/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. Verified against the primary source. The crossed 3x3 identity by style design, the three distinct source persons/actors, and the distinct dialogue per advisor are all stated explicitly. The paper notes the dialogues were pretested to be statistically equivalent on ability, benevolence and integrity, so 'different dialogue' means different text, not different trust content.

Validated — I have read the original and agree it supports the sentence.

#138 bartneck2009 — Bartneck et al. (2009) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** Parts B, D — Semantic differential (5 point). Part B (likeability) was administered before Part D (eeriness) so that participants formed a general social impression before attending to uncanny affect. A reversed order would prime observers towards negative perceptions and risk inflating eeriness scores before the trust battery. **Part B used the five-item Godspeed likeability subscale, from Bartneck et al..**

Usage: Use 5 of this source.

Original source: "[Appendix, p. 79 (Godspeed III); Sect. 4 Likeability, p. 76:] "GODSPEED III: LIKEABILITY. Please rate your impression of the robot on these scales: Dislike 1 2 3 4 5 Like; Unfriendly 1 2 3 4 5 Friendly; Unkind 1 2 3 4 5 Kind; Unpleasant 1 2 3 4 5 Pleasant; Awful 1 2 3 4 5 Nice —— We decided to only include the five items ... Two studies used this new likeability questionnaire. The first reports a Cronbach's Alpha of 0.865 [21] ... 0.923 for the human condition, 0.878 for the android condition, and 0.842 for the masked android condition.""

How to check: File: `Articles/02_trust_and_avatar_perception/Bartneck_2009_Measurement Instruments for the Anthropomorphism.pdf`. Verified against the primary source. The Appendix confirms exactly five likeability items on a 5-point semantic differential scale, matching the thesis description of a five-item Godspeed likeability subscale.

Validated — I have read the original and agree it supports the sentence.

#139 ho2010 — Ho & MacDorman (2010) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** The Godspeed anthropomorphism subscale was included during early piloting and removed in the shortening described above, since a second five-item semantic differential battery extended completion time without measuring a construct central to the study hypotheses. Part D captured eeriness through eight items from the validated instrument of Ho and MacDorman. Their evaluation of the Godspeed questionnaire found four of its subscales too strongly intercorrelated to measure distinct constructs: anthropomorphism, animacy, likeability and perceived intelligence. **It also contains no subscale corresponding specifically to eeriness.** *[Each polar pair was preceded by a short definition to support participants unfamiliar with the terms. This instrument treats eeriness as a distinct affective response from likeability and H3 can consequently be examined independently. — follows the citation, uncited.]*

Usage: Use 5 of this source.

Original source: "[Section 1, p. 1509 (paragraph on the Godspeed questionnaire):] "In addition, there is no index corresponding specifically to eeriness, a dimension that is arguably distinct from likeability but nevertheless important in determining whether a human-looking character has fallen into the uncanny valley.""

How to check: File: `Articles/02_trust_and_avatar_perception/Ho2010UncannyValleyIndices.pdf`. Verified against the primary source. Directly supports the sentence; the paper calls the Godspeed components 'indices' where the thesis says 'subscale', which is the same thing. The five Godspeed indices are enumerated in Section 2 as anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety, with no eeriness index among them.

Validated — I have read the original and agree it supports the sentence.

#140 mayer1995 — Mayer et al. (1995) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in the-sis: **[Block covered by this citation] Parts E, F, G — Trust battery (7 point Likert). Nine items operationalised the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany, introduced in .** *[Parts E, F and G covered Ability, Benevolence and Integrity respectively, with three items each, measured independently rather than as a composite. These three exclude predictability, the fourth trusting belief in the framework. Judging predictability requires observing an advisor's behaviour repeatedly, whereas each participant saw a given advisor deliver a single scripted recommendation. — follows the citation, uncited.]*

Usage: Use 4 of this source.

Original source: See entry #15 for the source passage.

How to check: Already checked above. Confirm the same passage also supports *this* sentence.

Validated — I have read the original and agree it supports the sentence.

#141 mcknight2001 — McKnight & Chervany (2001) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in the-sis: **[Block covered by this citation] Parts E, F, G — Trust battery (7 point Likert). Nine items operationalised the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany, introduced in .** *[Parts E, F and G covered Ability, Benevolence and Integrity respectively, with three items each, measured independently rather than as a composite. These three exclude predictability, the fourth trusting belief in the framework. Judging predictability requires observing an advisor's behaviour repeatedly, whereas each participant saw a given advisor deliver a single scripted recommendation. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: See entry #16 for the source passage.

How to check: Already checked above. Confirm the same passage also supports *this* sentence.

Validated — I have read the original and agree it supports the sentence.

#142 canales2024 — Canales et al. (2024) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** Three separate ranking questions each asked participants to order all three advisors: from most to least wanted for consultation (P1), from most to least trustworthy overall and from most to least unsettling overall. **P1 was the primary behavioural measure, adapted from the post-study advisor selection paradigm of Canales et al..**

Usage: Use 8 of this source.

Original source: "[§5.3 Procedure, p. 5:] "After the survey, the next doctor appeared, and the process was repeated twice more so that all three doctors were interviewed and rated. Participants then selected a single doctor for a second opinion, and also selected the doctor they would least prefer.""

How to check: File: `Articles/02_trust_and_avatar_perception/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. Verified against the primary source. The post-study forced-choice advisor selection paradigm is exactly as described: after rating all three advisors, participants chose one for the second opinion. The paper additionally collected a least-preferred choice, which the thesis's P1 adaptation apparently does not.

Validated — I have read the original and agree it supports the sentence.

#143 alipour2025 — Alipour et al. (2025) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** Parts E through G record trusting beliefs, that is, what a participant holds to be true of the advisor. P1 instead records a trusting intention, a willingness to depend on that advisor. **The two need not agree, and measuring both addresses the methodological gap identified by Alipour et al..** *[The trustworthiness and unsettlingness rankings provided behavioural analogues to the corresponding Likert measures, Parts E through G and Part D respectively. Ranked and rated judgements can therefore be compared directly for both dimensions. A rank order was used for all three questions rather than a forced binary choice, since both avatar conditions were expected to score below C1. Limiting participants to two options would concentrate responses on C1 and leave insufficient variance to distinguish C2 from C3. — follows the citation, uncited.]*

Usage: Use 5 of this source.

Original source: "[Section 5.2 Methodological limitations of the existing literature, p. 16; and Section 5.3 Research gaps and recommendations for future research, p. 17 (also Table 4, gap: 'Overreliance on self-report measures', recommendation: 'Combine subjective surveys with objective behavioral outcomes'):] "Finally, almost all studies relied heavily on self-report measures, which are vulnerable to biases such as social desirability and participants' misinterpretation of constructs. Moreover, self-report measures may fail to capture automatic or unconscious reactions, which are especially relevant in HAI research. For example, participants may consciously report trust while unconsciously exhibiting hesitation or discomfort. These points highlight the need for integrating behavioral or physiological measures in future research to more reliably capture UVE-related responses. ... We also recommend complementing single-time subjective self-report measures with additional methodological approaches, such as implicit trust tasks, decision-making scenarios or trust games, behavioral indicators (e.g., advice following, task delegation), and physiological signals""

How to check: File: `Articles/02_trust_and_avatar_perception/alipour Would you rely on an eerie agent?.pdf`. Verified against the primary source. The paper explicitly names overreliance on self-report as a methodological gap, states that self-report may miss automatic or unconscious reactions, and recommends implicit trust tasks, decision-making scenarios and behavioural indicators such as advice following. A forced-choice advisor selection is the kind of measure the review counts as objective: it classifies a choice-task decision preference among its seven objective trust measures (Section 4.3.2, p. 9).

Validated — I have read the original and agree it supports the sentence.

#144 mayer1995 — Mayer et al. (1995) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** The demographics section, completed before the first video block, also included an item measuring dispositional trust (seven point Likert: In general, I tend to trust people I meet for the first time) as an individual-level covariate. Mayer et al. describe this disposition as a trait that is stable across situations. **They expect it to explain variance in trust when included alongside a fuller set of variables.**

Usage: Use 5 of this source.

Original source: "[Characteristics of the Trustor: The Propensity to Trust, p. 716:] "Propensity should contribute to the explanation of variance in trust if used as a part of a more complete set of variables.""

How to check: File: `Articles/02_trust_and_avatar_perception/Mayer-IntegrativeModelOrganizational-1995.pdf`. Verified against the primary source. Verbatim match for the expectation attributed to the authors, with 'it' in the thesis sentence referring to propensity to trust.

Validated — I have read the original and agree it supports the sentence.

#145 demsar2006 — Demšar (2006) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** The primary analysis was a one-way repeated-measures analysis of variance (ANOVA) per subscale composite (B_score, D_score, E_score, F_score, G_score), with condition (C1, C2, C3) as the within-subjects factor. Every hypothesis concerns an individual subscale rather than a linear combination of them and the subscales were accordingly analysed separately and not through a multivariate omnibus test. Pairwise comparisons between the three conditions followed, corrected by Holm's step-down procedure. **Holm's procedure is more powerful than the Bonferroni correction and makes no additional assumptions about the hypotheses tested.**

Usage: Use 2 of this source.

Original source: See entry #115 for the source passage.

How to check: Already checked above. Confirm the same passage also supports *this* sentence.

Validated — I have read the original and agree it supports the sentence.

#146 zell2015 — Zell et al. (2015) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** Effect sizes were reported as partial η^2. **Comparable rating studies of avatar appearance have analysed within-participant conditions with the repeated-measures ANOVA.**

Usage: Use 5 of this source.

Original source: **[Results, appeal:]** "We found a **main effect of material on the ratings of appeal**... **but no main effect of shape was found**. An interaction between shape and material ($F(4, 84) = 13.97$, $p < 0.0001$) shows that a realistic material on a cartoon shape yields the least appealing combination... and **the cartoon material on the realistic shape is similarly unappealing** ($p < 0.02$)." **[Results, eeriness:]** the scale is "Extremely eerie—Extremely re-assuring"; "we found a **main effect of material on the ratings of reassurance**... **but no main effect was found for shape**." **[Realism:]** "**shape is the dominant factor when rating realism** and expression intensity, while **material is the key component for appeal**." **[Conclusion, the mismatch framing in the authors' own words:]** "**Matching levels of stylization of geometry and material cause the highest ratings of appeal, while strong mismatches (e.g., very realistic material on a stylized shape) result in unappealing and eerie characters**."

How to check: Read from `Articles/01_uncanny_valley/To Stylize or not to Stylize?_-Zell.pdf` (13 pages). **Verified 2026-08-22.** All three thesis claims hold. The mismatch framing is the authors' own conclusion, not an inference. Two further findings in the same passage are not yet used in the thesis and would strengthen Evaluation §6.3: "the effect of material stylizations grows when shape realism is increased... mismatches ... are less prominent on abstract characters" (C3 has a photogrammetric shape, so material mismatch bites hardest there), and "subtle stylization of a realistic material (edge-preserving blur on the albedo texture) increases appeal without sacrificing realism" (essentially the head-texture preprocessing step).

Validated — I have read the original and agree it supports the sentence.

#147 mcdonnell2012 — McDonnell et al. (2012) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** Effect sizes were reported as partial η^2. **Comparable rating studies of avatar appearance have analysed within-participant conditions with the repeated-measures ANOVA.**

Usage: Use 7 of this source.

Original source: "[Section 5.1.3 Results, p. 91:5 (first passage); Section 5.1 Exp 1: Still vs. Moving, p. 91:4 (second passage):] "For each of the rating scales, a two-way repeated measures ANalysis Of VAriance (ANOVA) was conducted on the data from both blocks with within-subjects factor render style (10) and between-subjects factor movement (2 - still or moving). ... In psychology, rating scales are commonly used to gather information on subjective conditions such as attitudes and emotions of participants, which could not be collected directly.""

How to check: File: `Articles/01_uncanny_valley/ McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012). Render me Real? Investigating.pdf`. **Resolved 2026-08-22:** the sentence now reports practice ("Comparable rating studies. . . have analysed") instead of asserting an established test, and says "avatar appearance" so this citation fits. The paper demonstrates the practice, running two-way repeated-measures ANOVAs with render style as the within-subjects factor across all six rating scales and again in Exp 2 and Section 6.2. It nowhere claims that repeated-measures ANOVA is 'the established test' for such studies, and it makes no methodological survey statement of that kind. The paper is only a single instance of the practice, so it supports the sentence by example rather than by assertion. Note also that the paper's own Exp 1 factor 'movement' was between-groups, not within-participants.

Validated — I have read the original and agree it supports the sentence.

#148 latoschik2017 — Latoschik et al. (2017) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** Effect sizes were reported as partial η^2. **Comparable rating studies of avatar appearance have analysed within-participant conditions with the repeated-measures ANOVA.**

Usage: First use of this source.

Original source: "[Section 6 'Results', opening sentence (PDF p. 7); design stated in Section 3.1, PDF p. 4: "The combinations were then applied as randomized test conditions in the experiment using a 2 x (self-avatar) x 2 (other-avatar) within-subjects design.":] "Data were analyzed using two factorial ANOVAs for repeated measures on all measures taken.""

How to check: File: `Articles/Latochik- the imapct.pdf.pdf`. **Resolved 2026-08-22:** the sentence now reports practice ("Comparable rating studies. . . have analysed") instead of asserting an established test, and says "avatar appearance" so this citation fits. The paper demonstrates the practice (a within-subjects avatar-appearance experiment analysed with two-factorial repeated-measures ANOVAs) but never makes the normative methodological claim that repeated-measures ANOVA 'is the established test'; I searched the full text for ANOVA, repeated, within-subject, design and found only this reported use. The manipulation is also avatar model realism (wooden mannequin vs photogrammetry scan), not rendering style, so it is an instance rather than a warrant for the thesis sentence.

Validated — I have read the original and agree it supports the sentence.

#149 bartneck2009 — Bartneck et al. (2009) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in the-sis: **[Block covered by this citation]** Normality was checked with the Shapiro-Wilk test and sphericity with Mauchly's test. The Greenhouse-Geisser correction was applied where sphericity was violated. **Both instruments report Cronbach's α per subscale, with .70 as the accepted reliability threshold.** *[The same was done here before the composites were formed. — follows the citation, uncited.]*

Usage: Use 6 of this source.

Original source: "[Sect. 7 Conclusions, p. 78; and Sect. 1 Introduction (reliability discussion), p. 73:] "Before calculating the mean scores for anthropomorphism, animacy, likeability, or perceived intelligence it is good practice to perform a reliability test and report the resulting Cronbach's Alpha. —— Internal consistency ... is a measurement for how well the different items measure the same concept, and it is of particular importance to the questionnaires proposed because they are designed to be homogenous in content. Internal consistency involves the calculation of a statistic known as Cronbach's Alpha.""

How to check: File: `Articles/02_trust_and_avatar_perception/Bartneck_2009_Measurement Instruments for the Anthropomorphism.pdf`. Verified against the primary source. The source explicitly recommends running a reliability test and reporting Cronbach's Alpha before computing subscale means, and defines internal consistency as how well items measure the same concept, which is the rationale the thesis sentence gives. Minor detail: the paper's list of subscales in that recommendation omits perceived safety.

Validated — I have read the original and agree it supports the sentence.

#150 ho2010 — Ho & MacDorman (2010) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in thesis: **[Block covered by this citation]** Normality was checked with the Shapiro-Wilk test and sphericity with Mauchly's test. The Greenhouse-Geisser correction was applied where sphericity was violated. **Both instruments report Cronbach's α per subscale, with .70 as the accepted reliability threshold.** *[The same was done here before the composites were formed. — follows the citation, uncited.]*

Usage: Use 6 of this source.

Original source: "[Section 2.1.3 Statistical analysis, p. 1510; Section 3.1 Research goal, p. 1513:] "Cronbach's α was used to measure the reliability of each index. ... [Section 3.1:] (4) there should be at least three semantic differential scales per index to enable the estimation of reliability; and (5) the reliability of the indices should be acceptable (Cronbach's $\alpha \geq .70$).""

How to check: File: `Articles/02_trust_and_avatar_perception/Ho2010UncannyValleyIndices.pdf`.
Resolved 2026-08-22: the sentence was split so the citation covers only what the two instruments do (report α per subscale against a .70 threshold); this project's own procedure now stands uncited. The paper supports the practice: it reports Cronbach's α separately for every index (Tables 1, 3, 4, 5) and applies the .70 cutoff, e.g. 'perceived safety had low reliability with a Cronbach's α of .60, which is below the standard .70 cutoff (Nunnally, 1978)' (Section 2.2, p. 1511). It does not, however, state the rationale the thesis sentence gives, that averaging items presumes they measure one construct; unidimensionality is handled in the paper by factor analysis, not by Cronbach's α ('items corresponding to a single, unidimensional concept should load on the same factor when applying exploratory factor analysis', Section 3.1, p. 1513). Note also that the thesis sentence describes its own analysis procedure rather than a claim from this paper. The α and $\geq$ glyphs are transcribed from the PDF's font-substituted extraction ('a' and 'P').

Validated — I have read the original and agree it supports the sentence.

#151 lakens2017 — Lakens (2017) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in the-sis: **[Block covered by this citation]** The theoretically critical pairwise comparison is C2 versus C3. H2 predicts equivalence rather than a difference and this leads to a non-significant pairwise result providing insufficient evidence on its own. **A TOST (two one-sided tests) equivalence test was therefore applied to the C2 versus C3 trust composite.**

Usage: First use of this source.

Original source: "[Abstract:] "Scientists should be able to provide support for the absence of a meaningful effect. Currently, researchers often incorrectly conclude an effect is absent based a nonsignificant result. A widely recommended approach within a frequentist framework is to test for equivalence. In equivalence tests, such as the two one-sided tests (TOST) procedure discussed in this article, an upper and lower equivalence bound is specified based on the smallest effect size of interest." [p. 356:] "A very simple equivalence testing approach is the 'two one-sided tests' (TOST) procedure (Schuirmann, 1987). In the TOST procedure, an upper (Δ_U) and lower (Δ_L) equivalence bound is specified based on the smallest effect size of interest (SESOI; e.g., a positive or negative difference of $d = .3$)."""

How to check: Open `Articles/Lakens -equivalence Test.pdf`, abstract and p. 356, left column.

Validated — I have read the original and agree it supports the sentence.

#152 canales2024 — Canales et al. (2024) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in the-sis: **[Block covered by this citation]** Its bound was pre-specified at Cohen's $d = 0.3$, the smallest difference in trust treated as practically meaningful in this study. H4 predicts the same kind of equivalence against a different reference point and the identical procedure with the same bound was applied to the C1 versus C3 trust composite. **Comparable work has inferred an absence of stylisation effects on trust from non-significant results alone.** *[That inference cannot distinguish a genuine null from an insensitive test. — follows the citation, uncited.]*

Usage: Use 9 of this source.

Original source: "[§10 Conclusion, p. 9:] "Stylization had no significant impact on trust ratings. The Realistic avatars received lower ratings for benevolence on average; however, the differences were not significant. This suggests that realistic avatars are not required and it may be possible to use a range of avatar visual styles for applications where trust is critical."""

How to check: File: `Articles/02_trust_and_avatar_perception/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. Verified against the primary source. The paper draws a positive design conclusion ('realistic avatars are not required') directly from non-significant results. No equivalence test, TOST procedure, or Bayesian null-evidence analysis appears anywhere in the paper, so the inference does rest on non-significance alone.

Validated — I have read the original and agree it supports the sentence.

#153 demsar2006 — Demšar (2006) *Ch.4 Methodology → User Study Design → Procedure and Measures*

Sentence in the-sis: **[Block covered by this citation]** Three post-study ranking questions, covering P1, trustworthiness and unsettlingness, were each analysed with a Friedman test comparing mean advisor ranks across conditions. **Pairwise Wilcoxon signed-rank tests were then applied to each pair, with Holm correction to control the family-wise error across the comparisons.** *[C2 versus C3 is the critical pairwise comparison in each case. Rank responses carry only ordinal information. The parametric assumptions above thus do not apply. The parameter adjustment study used the same procedure (), meaning both rest on one statistical basis. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: "[Section 3.2.2 Friedman Test, printed page 13:] "Although we here use these procedures only as post-hoc tests for the Friedman test, they can be used generally for controlling the family-wise error when multiple hypotheses of possibly various types are tested.""

How to check: File: `Articles/06_statistics/demsar06a.pdf`. **Resolved 2026-08-22:** the Wilcoxon choice and the C2-versus-C3 note were separated out; the citation now covers only Holm for family-wise error across multiple comparisons, which is what Demsar licenses. Demsar supports applying Holm generally to control family-wise error over multiple hypotheses of various types, but he never pairs Holm with pairwise Wilcoxon signed-rank tests: his post-hoc procedures operate on average-rank z statistics (Nemenyi, Bonferroni-Dunn, Holm, Hochberg, Hommel), and he cautions against all-pairs testing ('We thus should not make pairwise comparisons when we in fact only test whether a newly proposed method is better than the existing ones', p. 12). The 'C2 versus C3 is the critical pairwise comparison' clause is the thesis's own study content and is not addressed by this source.

Validated — I have read the original and agree it supports the sentence.

Chapter: Implementation

#154 roshaan2026 — Roshaan (2026) *Ch.5 Implementation → NPR Technique Implementation → V5 — Hierarchical Edge Detection*

Sentence in thesis: **[Block covered by this citation]** This proxy detects distance discontinuities within the avatar mesh. However, it cannot separate the avatar from the background. The background lies entirely outside the per-fragment computation. **The normal layer reuses the world-space normal derivative computation from V3 and an AHEAD-style adaptive gain then amplifies this layer specifically in darker image regions:**

Usage: Use 6 of this source.

Original source: "[Section 2.4 'Adaptive Feature Enhancement', pp. 4-5 (preprint, DOI 10.21203/rs.3.rs-8849147/v1):] "The algorithm acts as a localized Automatic Gain Control (AGC), inspired by photographic tone reproduction [13], to match the computational edge response with perceptual significance. The adaptive scalar S(x, y) is modeled as a modified inverse-linear function of the local pixel luminance L(x, y) ... Lower luminance areas have higher S values (see Fig. 3), therefore amplifying subtle structural details in shadowed areas. While, higher luminance areas have lower S values, which reduces the effect of applied edge detection ...""

How to check: File: `Articles/AHEAD_Adaptive_Hierarchical_Edge_Detection_for_Rea.pdf`. Verified against the primary source. The cited mechanism is exactly as described: a luminance-driven inverse-linear gain that amplifies edge response specifically in darker regions, motivated by the Weber-Fechner Law. One difference the thesis's 'AHEAD-style' wording already signals: AHEAD applies the gain only to its Structure layer (Scharr/Kirsch on image intensity, Section 2.5), never to a world-space normal derivative, and it never applies any operator to the normal map, which appears only as a G-buffer illustration in Fig. 2a.

Validated — I have read the original and agree it supports the sentence.

#155 winnemoller2006 *Ch.5 Implementation → NPR Technique Implementation → V5 — Hierarchical Edge Detection*

Sentence in thesis: **[Block covered by this citation]** The composited PBR output also contains baked lighting gradients that produce false Roberts Cross responses on this platform. **The colour layer accordingly replaces Roberts Cross with XDoG: the difference between two Gaussian-blurred luminance samples at different radii, passed through a tanh soft threshold:**

Usage: Use 3 of this source.

Original source: **(1)** *τ and ϕ_e, the two parameters the thesis shader uses* ← [§3.2 Edge detection:] "**The parameter τ in Equation 4 controls the amount of center-surround difference required for cell activation, and ϕ_e controls the sharpness of the activation falloff.**" **(2)** *the tanh soft threshold* ← [Eq. 4:] $D(\hat{x}, \sigma_e, \tau, \phi_e) = 1$ if $(S_{\sigma_e} - \tau \cdot S_{\sigma_r}) > 0$, and $1 + \tanh(\phi_e \cdot (S_{\sigma_e} - \tau \cdot S_{\sigma_r}))$ otherwise. **(3)** *the radius ratio the shader calls `_XDoGK`* ← [same section:] "$S_{\sigma_r} \equiv S(\hat{x}, 1.6 \cdot \sigma_e)\ldots$ **The factor of 1.6** relates the typical receptive field of a cell to its surroundings [Marr and Hildreth 1980]."

How to check: Read directly from `Articles/videoabstraction.pdf`, §3.2 and Eq. 4. **New key added 2026-08-25. Nidanur identified this misattribution herself** while checking `winnemoller2012`, and it became the worked example behind the origin rule now recorded as CLAUDE.md §5.3a. **What was wrong:** the 2012 paper was carrying the operator as though it were its own. It is not. The 2012 compendium credits it inline to its own reference [7], "Winnemöller, H., Olsen, S.C., Gooch, B.. Real-time video abstraction. ACM Trans Graph 2006;25(3):1221–1226", and the primary was already held in the Articles folder under a name that did not announce it. **The decisive evidence is the thesis's own shader.** `05_implementation` declares `_XDoGTau` and `_XDoGPhi`, which are τ and ϕ_e from Equation 4 above. The 2012 paper's own reformulation **replaces** τ **with** p, so the parameterisation actually implemented is the 2006 one and not the 2012 one. **Both keys are cited, deliberately.** `winnemoller2012` keeps the name *XDoG* and the compendium framing, which are genuinely its contributions. `winnemoller2006` carries the operator, the tau-weighted difference and the tanh ramp. **Bibliographic note:** the held PDF is an author version with no DOI or page numbers printed on it. Authors, title and affiliation were read from it directly. Venue, volume, issue, pages 1221–1226 and DOI 10.1145/1141911.1142018 were cross-checked against CrossRef and against the 2012 paper's reference list, which agree exactly. **Incidental confirmation for Ch. 2:** the 1.6 radius ratio in item (3) is credited by Winnemöller et al. to Marr and Hildreth, and it is the same 1.6 that Marr and Hildreth's Appendix B derives as the optimal ratio for approximating the Laplacian by a difference of Gaussians. The two sections agree.

Validated — I have read the original and agree it supports the sentence.

#156 winnemoller2012 — Winnemöller et al. (2012) *Ch.5 Implementation → NPR Technique Implementation → V5 — Hierarchical Edge Detection*

Sentence in the-sis: **[Block covered by this citation]** The composited PBR output also contains baked lighting gradients that produce false Roberts Cross responses on this platform. **The colour layer accordingly replaces Roberts Cross with XDoG: the difference between two Gaussian-blurred luminance samples at different radii, passed through a tanh soft threshold:**

Usage: Use 3 of this source.

Original source: "The shader implements the paper's Eq. 4 and Eq. 5 directly. [Eq. 4:] $D_{\sigma,k,\tau}(x) = G_\sigma(x) - \tau \cdot G_{k\sigma}(x)$. [Eq. 5, the soft threshold:] $T_{\varepsilon,\varphi}(u) = 1$ for $u \geq \varepsilon$, otherwise $1 + \tanh\big(\varphi \cdot (u - \varepsilon)\big)$. [§ on parameters:] "Larger or smaller φ control the sharpness of the black/white transitions in the image, while ε controls the level above which the adjusted luminance values will become white." [On τ:] "replacing τ with p makes it possible to control the strength of the edge sharpening effect without influencing any other aspects of the filter.""

How to check: Primary PDF read and verified: `Articles/Winnemoeller12.pdf`. *Computers & Graphics* 36(6), 2012, 740–753, DOI 10.1016/j.cag.2012.03.004.

Validated — I have read the original and agree it supports the sentence.

#157 kyprianidis2009 — Kyprianidis et al. (2009) *Ch.5 Implementation → NPR Technique Implementation → V6 — Kuwahara Painterly Filter*

Sentence in the-sis: **[Block covered by this citation] All three platforms use the anisotropic Kuwahara filter of Kyprianidis et al..** *[This filter relies on a structure tensor to estimate the local feature orientation and anisotropy. The sampling neighbourhood then becomes an ellipse aligned with that orientation rather than a fixed square. — follows the citation, uncited.]*

Usage: Use 7 of this source.

Original source: "[Title and Abstract (PDF p. 1):] "Image and Video Abstraction by Anisotropic Kuwahara Filtering ... We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor.""

How to check: File: `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`. Verified against the primary source. The paper supports the attribution only, i.e. that the anisotropic Kuwahara filter is the technique of Kyprianidis et al. The remainder of the sentence ('All three platforms use...') describes the thesis's own implementation and cannot be verified against any source; note that CLAUDE.md 5.1 rule 5 discourages attaching a citation to a sentence stating what was implemented in this work.

Validated — I have read the original and agree it supports the sentence.

#158 kyprianidis2009 — Kyprianidis et al. (2009) *Ch.5 Implementation → NPR Technique Implementation → V6 — Kuwahara Painterly Filter*

Sentence in the-sis: **[Block covered by this citation]** Mixamo and Avaturn scan every pixel within the kernel radius and accumulate it into whichever sector it falls into, and the sample count grows with the radius. The Meta SDK instead takes a fixed three samples per sector at set radii, plus one sample shared across all sectors, for a constant total regardless of radius. Each sector accumulates a mean colour and a variance. These are then blended by an inverse-variance weight, so that the smoothest sector dominates the result. **That approach follows Kyprianidis et al., who weight each sector inversely to its standard deviation.** *[The Meta SDK applies that form directly. Mixamo and Avaturn instead raise the variance to half the exponent, which is algebraically identical and avoids a square root on every sector. The accumulation that follows the weight is shared across all three platforms: — follows the citation, uncited.]*

Usage: Use 8 of this source.

Original source: "[Section 3 'Method' (PDF p. 4) and Section 3.2 'Anisotropic Kuwahara Filtering' (PDF p. 5):] "The filter response is defined as a weighed sum of the local averages, where more weight is given to those averages with low standard deviation. ... [Section 3.2] ... Our definition of the weighting factors ... ensures that more weight is given to sectors with low standard deviation, i.e. those that are more homogeneous. This is similar to the approach of [PPC07], where ... is used for the weights, but avoids the indetermination, when some of the si are zero.""

How to check: File: `Articles/03_npr_rendering/anisotropic_kuwahara.pdf`. **Resolved 2026-08-22:** the sentence was split: the citation now covers only the inverse-variance weighting principle, and the per-platform difference is stated as this project's own implementation choice. The underlying principle the thesis attributes to the source is genuinely there: weight each sector inversely to its standard deviation. However, the paper defines exactly one weighting formula for its own filter (alpha_i = 1/(1 + ——s_i——^q), with q = 8 in all examples) and contrasts it only with Papari et al.'s ——s_i——^-q; it does not present a family of interchangeable weighting variants. The idea that different implementations follow 'a different variant' of the formula is the thesis's own implementation choice, not something the paper offers or licenses. Ellipses in the quote replace math glyphs that pdftotext garbles.

Validated — I have read the original and agree it supports the sentence.

Chapter: Evaluation

#159 alipour2025 — Alipour et al. (2025) *Ch.6 Evaluation → User Study Results → Hypothesis Tests*

Sentence in thesis: **[Block covered by this citation] This result matches the direct negative relationship between eeriness and trust that Alipour et al. reported in 11 of the 24 studies that measured trust directly.** *[The near-identical slopes across ability, benevolence and integrity indicate that eeriness suppresses trust broadly rather than through a pathway specific to benevolence. — follows the citation, uncited.]*

Usage: Use 6 of this source.

Original source: See entry #44 for the source passage.

How to check: Already checked above. Confirm the same passage also supports *this* sentence.

Validated — I have read the original and agree it supports the sentence.

#160 canales2024 — Canales et al. (2024) *Ch.6 Evaluation → User Study Results → Post-Study Rankings*

Sentence in thesis: **[Block covered by this citation] Canales et al. found a divergence between explicit and behavioural measures in their comparable study.** *[The preference and trustworthiness rankings here converged with the Likert measures instead. C3 attracted no more behavioural endorsement than C2 did, meaning that the forced ordering surfaced nothing that the graded ratings had obscured. C1's dominance is the most consistent finding in the study. It holds on every subscale and on both rankings that produced a significant result. — follows the citation, uncited.]*

Usage: Use 10 of this source.

Original source: See entry #6 for the source passage.

How to check: Already checked above. Confirm the same passage also supports *this* sentence.

Validated — I have read the original and agree it supports the sentence.

#161 canales2024 — Canales et al. (2024) *Ch.6 Evaluation → User Study Results → Secondary Measures*

Sentence in thesis: **[Block covered by this citation]** The dispositional trust covariate was examined as an individual difference moderator. It showed no meaningful association with the mean trust composite (). A quartile-based moderation analysis was not pursued, because the sample is too small to detect an interaction reliably. **Canales et al. similarly found no significant relationship between propensity and trust ratings in their comparable manipulation.**

Usage: Use 11 of this source.

Original source: "[§6 Results, p. 5:] "There was also no significant correlation between people's propensity to trust and the average of their actual trust survey ratings as described in Sec.6.1 (Pearson's R of 0.062, p=.62), so it appears that personal trust propensity was not an important factor in the trust results.""

How to check: File: `Articles/02_trust_and_avatar_perception/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. Verified against the primary source. Directly stated. The paper also reports no significant difference in propensity across the nine assigned avatars (p=.52), confirming propensity did not confound the style manipulation.

Validated — I have read the original and agree it supports the sentence.

#162 zell2015 — Zell et al. (2015) *Ch.6 Evaluation → Discussion → What the Stylisation Left Unchanged*

Sentence in the-sis: **[Block covered by this citation]** The composite stylises the avatar's material while leaving its shape untouched and this split may be why eeriness did not fall. **Material governs eeriness in the account of Zell et al. and this alone predicts a reduction for C3.**

Usage: Use 6 of this source.

Original source: **[Results, appeal:]** "We found a **main effect of material on the ratings of appeal**... **but no main effect of shape was found**. An interaction between shape and material ($F(4, 84) = 13.97$, $p < 0.0001$) shows that a realistic material on a cartoon shape yields the least appealing combination... and **the cartoon material on the realistic shape is similarly unappealing** ($p < 0.02$)." **[Results, eeriness:]** the scale is "Extremely eerie—Extremely re-assuring"; "we found a **main effect of material on the ratings of reassurance**... **but no main effect was found for shape**." **[Realism:]** "**shape is the dominant factor when rating realism** and expression intensity, while **material is the key component for appeal**." **[Conclusion, the mismatch framing in the authors' own words:]** "**Matching levels of stylization of geometry and material cause the highest ratings of appeal, while strong mismatches (e.g., very realistic material on a stylized shape) result in unappealing and eerie characters**."

How to check: Read from `Articles/01_uncanny_valley/To Stylize or not to Stylize?_-Zell.pdf` (13 pages). **Verified 2026-08-22.** All three thesis claims hold. The mismatch framing is the authors' own conclusion, not an inference. Two further findings in the same passage are not yet used in the thesis and would strengthen Evaluation §6.3: "the effect of material stylizations grows when shape realism is increased... mismatches ... are less prominent on abstract characters" (C3 has a photogrammetric shape, so material mismatch bites hardest there), and "subtle stylization of a realistic material (edge-preserving blur on the albedo texture) increases appeal without sacrificing realism" (essentially the head-texture preprocessing step).

Validated — I have read the original and agree it supports the sentence.

#163 zell2015 — Zell et al. (2015) *Ch.6 Evaluation → Discussion → What the Stylisation Left Unchanged*

Sentence in the-sis: **[Block covered by this citation]** Their mismatch result points the other way. C3 sets a stylised material on a photogrammetrically reconstructed shape. **Zell et al. also reported that the influence of material stylisation on perceived realism grows as shape realism rises, and this places C3 at the point where a mismatch would be most visible.** *[The abstraction may consequently have removed eeriness from the material only to reintroduce it through the gap between the two. — follows the citation, uncited.]*

Usage: Use 7 of this source.

Original source: **(1)** *"the influence of material stylisation on perceived realism grows as shape realism rises"* ← [§5.2:] "This is in line with the results from Section 5.1, and **confirms that as the shape becomes more realistic, the material stylization becomes more dominant for perceived realism**."

How to check: Read directly from `Articles/01_uncanny_valley/Zell2015_StyleShape.pdf`, §5.2. **Scope drift corrected 2026-08-25.** The thesis sentence dropped the outcome variable, leaving "the influence of material stylisation grows as shape realism rises" with no dependent measure attached. The source scopes that growth to **perceived realism**. The omission is not harmless here, because the surrounding paragraph argues about **eeriness** throughout: it opens on "this split may be why eeriness did not fall" and closes on "removed eeriness from the material only to reintroduce it. An unqualified sentence lets a realism result read as an eeriness result inside an eeriness argument, and Zell et al. measure the two separately and reach different conclusions on each. The argument itself is unaffected, since the point being made is that C3 sits where material stylisation dominates. **Style fix in the same pass:** the trailing "which" clause became "and this" per the split rule.

Validated — I have read the original and agree it supports the sentence.

#164 dill2012 — Dill et al. (2012) *Ch.6 Evaluation → Discussion → What the Stylisation Left Unchanged*

Sentence in the-sis: **[Block covered by this citation] The features that the participants of Dill et al. blamed for their discomfort, the eyes and the mouth, were largely preserved.** *[The preprocessing altered the mouth texture in C3, but the geometry of both regions is identical across the two conditions. Additionally, both conditions inherit the same gaze and lip animation from one recording session. A shading operator cannot reach that animation and any mismatch in its timing thus survives the stylisation. — follows the citation, uncited.]*

Usage: Use 3 of this source.

Original source: "[Section 2 (Obtained Results), printed p. 512 (PDF p. 527 of the IVA 2012 LNAI 7502 proceedings volume):] "The questionnaire also asked to the participants which parts of the face the participant felt more discomfort, if it was the case. Analysing the answers, we could highlight two specific parts of the face: eyes and mouth.""

How to check: File: `Articles/01_uncanny_valley/Evaluation of the Uncanny Valley in CG Characters.pdf`. Verified against the primary source. The paper explicitly reports that participants named the eyes and the mouth as the facial parts causing the most discomfort, which is exactly the attributed part of the thesis sentence. The trailing clause 'were largely preserved' describes the thesis's own rendering result and carries no attribution to Dill et al.

Validated — I have read the original and agree it supports the sentence.

#165 cihodaru2025 — Cihodaru-Ştefanache & Podina (2025) *Ch.6 Evaluation → Discussion → What the Stylisation Left Unchanged*

Sentence in the-sis: **[Block covered by this citation]** Consistency across the whole presentation offers a third reading. **Cihodaru-Ştefanache and Podina reported the perceptual mismatch hypothesis, under which uncanniness arises from features perceived at inconsistent levels of anthropomorphism, as the best supported explanation of the effect.** *[C3 exhibits precisely that inconsistency, since a recorded human voice accompanies a visibly abstracted face. The stylisation on this reading introduced a new source of uncanniness even as it removed another. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: "[Section 1 (Introduction), printed p. 02 (PDF p. 2):] "Understanding UVE remains challenging due to the existence of multiple competing hypotheses that try to explain our perception of anthropomorphism in ECAs. These range from the morbidity and movement hypotheses to the category ambiguity (Cheetham and Jancke, 2013; Katsyri et al., 2015; Pollick, 2010). Among these, the perceptual mismatch hypothesis has received strong empirical support.""

How to check: File: `Articles/01_uncanny_valley/The_uncanny_valley_effect_in_embodied_conversation.pdf`. **Resolved 2026-08-22:** reworded to "has received strong empirical support", matching the source exactly. The authors do single the perceptual mismatch hypothesis out from the competing hypotheses, but they write that it 'has received strong empirical support', not the strongest; no ranking of the hypotheses by weight of evidence appears anywhere in the paper. The thesis converts a positive into a superlative, so 'strong empirical support among competing explanations' would match the source, while 'the strongest' overstates it. Searched the full text for 'mismatch', 'strongest', 'most support' and 'competing hypothes'; the only other substantive mentions are the Discussion's own framework (mismatch detection between perceived anthropomorphism and user expectations, p. 15) and the Katsyri et al. (2015) reference entry.

Validated — I have read the original and agree it supports the sentence.

#166 zibrek2019mig — Zibrek & McDonnell (2019) *Ch.6 Evaluation → Discussion → The Video-Based Setting*

Sentence in the-sis: **[Block covered by this citation]** Participants watched prerecorded video and never interacted with the speaker. The setting supplied no stereoscopic depth, no motion parallax and no sense of sharing space with the avatar. A comparable VR study held shape and animation constant and varied only render style. **Photorealism significantly raised both place illusion and social presence.** *[A rendering difference that registers weakly on a flat screen may consequently register more strongly once the avatar occupies the viewer's own space. — follows the citation, uncited.]*

Usage: Use 2 of this source.

Original source: **Primary source read.** Method: "We created two render styles of the environment and character: realistic . . . and simple (where photorealistic render effects were removed), while keeping the shape and animation the same." Results as quoted in the entry above ($p = 0.025$, $d = 0.56$ for social presence; $p = 0.04$, $r = 0.46$ for place illusion).

How to check: **CONFIRMED (corrected 2026-08-14).** The earlier wording claimed a "weaker and less consistent effect on social presence," which contradicted the source: social presence was significantly higher for the realistic character ($p = 0.025$, $d = 0.56$). That no-effect result belongs to `zibrek2019tap`, against which this paper explicitly contrasts itself, and the two Zibrek findings had been conflated. The thesis sentence now states both significant self-report increases and the unaffected behavioural proximity measure, matching the source.

Validated — I have read the original and agree it supports the sentence.
